Introduction to Quantum Physics: A Lecture Course

PHYS 214, Spring 2026, ZJUI Institute (Prof. Said Mikki)

Contents

量子物理导论：讲座课程

PHYS 214，2026年春季，浙江工业大学（萨伊德·米基教授）

目录

1 Course Scope and Preliminary General Considerations

Course Roadmap

- **Lecture 1**: Course introduction and overview. Big picture of quantum mechanics: universal scope and multidisciplinary applications. First look at the two approaches to QM: quantization versus the abstract algorithmic view.
- **Lecture 2**: Mathematical foundations for quantum mechanics. Review of waves, complex numbers, probability, and basic linear algebra. Introduction to vectors in Hilbert space and Dirac notation.
- **Lecture 3**: Historical approach to quantum mechanics. Blackbody radiation, photoelectric effect, and Bohr's model. Introduction of de Broglie waves and Heisenberg uncertainty principle.
- **Lecture 4**: Conceptual foundations through the double-slit experiment. Explanation using wave mechanics and discussion of wave-particle duality. Introduction of the probability interpretation and superposition.
- **Lecture 5**: Foundations of wave mechanics I. Postulates of QM in Schrödinger picture: wavefunctions, operators, and the Schrödinger equation. Measurement, expectation values, and the meaning of the wavefunction.
- **Lecture 6**: Foundations of wave mechanics II. Solving the Schrödinger equation in 1D: free particle, infinite potential well, finite well, and particle in a box. Introduction to wave packets.
- **Lecture 7**: Dirac-Hilbert formalism I. Moving from wavefunctions to abstract state vectors in Hilbert space. Introduction of Dirac notation, spin as a qubit system, and quantum measurement in the abstract picture.
- **Lecture 8**: Dirac-Hilbert formalism II. Completeness, basis expansions, and operators in the abstract framework. Unification of wave mechanics and the abstract formalism.
- **Lectures 9-10**: Harmonic oscillator. Both analytic solution (Hermite polynomials) and algebraic method (ladder operators). Importance in physics and quantum field theory.
- **Lectures 11-12**: Atoms, molecules, and solids. Hydrogen atom, angular momentum, and spin. Pauli exclusion principle, multi-electron atoms, and introduction to band theory in solids.
- **Lectures 13-14**: Multi-particle systems, entanglement, and applications. Tensor products, identical particles (bosons and fermions), quantum entanglement, and introduction to quantum information.

1 课程范围与初步一般考虑

课程路线图

- 第一讲：课程介绍和概述。量子力学的宏观图景：普遍范围和多学科应用。首次了解量子力学的两种方法：量子化与抽象算法观。
- 第二讲：量子力学的数学基础。回顾波、复数、概率和基本线性代数。介绍希尔伯特空间中的向量与狄拉克符号。
- 第3讲：量子力学的历史方法。黑体辐射、光电效应和玻尔模型。德布罗意波的引入以及海森堡不确定性原理。
- 第4讲：通过双缝实验理解概念基础。使用波动力学解释，并讨论波粒二象性。概率解释和叠加的引入。
- 第5讲：波动力学基础I。薛定谔图景中的量子力学公设：波函数、算符和薛定谔方程。测量、期望值以及波函数的意义。
- 第6讲：波动力学基础II。求解一维薛定谔方程：自由粒子、无限深势阱、有限势阱和粒子在盒子中。波包的引入。
- 第7讲：狄拉克-希尔伯特形式化I。从波函数到希尔伯特空间中的抽象状态矢量。狄拉克符号的引入、自旋作为量子比特系统以及抽象图景中的量子测量。
- 第八讲：狄拉克-希尔伯特形式主义II。完备性、基展开和抽象框架中的算符。波动力学与抽象形式主义的统一。
- 第九-十讲：谐振子。解析解（厄米多项式）和代数方法（阶梯算符）。在物理学和量子场论中的重要性。
- 第十一-十二讲：原子、分子和固体。氢原子、角动量和自旋。泡利不相容原理、多电子原子以及固体能带理论的介绍。
- 第13-14讲：多粒子系统、纠缠及其应用。张量积、 identical particles（玻色子和费米子）、量子纠缠以及量子信息导论。

1.1 Course Overview and Objectives

Course Overview and Objectives

This course introduces the fundamental principles of quantum mechanics, the theoretical framework that describes physical systems at atomic and subatomic scales. By the end of this course, you will be able to:

- Understand the historical and experimental motivations for quantum theory
- Master the mathematical formalism of quantum mechanics (Hilbert spaces, operators, eigenvalues)
- Solve the Schrödinger equation for key potentials (infinite well, harmonic oscillator, hydrogen atom)
- Apply quantum mechanics to atomic, molecular, and solid-state systems
- Develop intuition for quantum phenomena (superposition, entanglement, measurement)

Prerequisites: Classical Mechanics, Electromagnetism, Linear Algebra, Differential Equations.

1.2 Math background Needed

Key Mathematical Tools You Will Need

- **Differential equations**: Solving the Schrödinger equation
- **Linear algebra**: Vector spaces, eigenvalues, inner products
- **Complex analysis**: Complex number algebra, Euler identity.
- **Probability**: Elementary concepts.
- **Analysis**: Basic function theory, calculus in several dimensions, elementary Fourier transforms.

1.1 课程概述与目标

课程概述与目标

本课程介绍量子力学的基本原理，即描述原子和亚原子尺度物理系统的理论框架。学完本课程后，你将能够：

- 理解量子理论的史实与实验动机
- 掌握量子力学的数学形式化（希尔伯特空间、算子、本征值 ues）
- 求解关键势的薛定谔方程（无限深势阱、谐振子、氢原子）
- 将量子力学应用于原子、分子和固态系统
- 培养对量子现象的直觉（叠加、纠缠、测量）

先修课程：经典力学、电磁学、线性代数、微分方程。

1.2 数学背景要求

所需关键数学工具

- 微分方程：求解薛定谔方程
- 线性代数：向量空间、特征值、内积
- 复分析：复数代数、欧拉公式
- 概率论：基本概念
- 分析学：基本函数理论、多维微积分、初等傅里叶变换

1.3 Engineering Audience Targeted in this Course

Four Undergraduate Disciplines Served

This course is designed to serve students from four major engineering and science disciplines:

Computer Science (CS)
Quantum computing, algorithms

Electrical Engineering (EE)
Quantum devices, nanofabrication

Computer Engineering (CE)
Hardware-software integration

Mechanical Engineering (ME)
Nanomechanics, quantum materials

1.4 Two Major Application Categories

Dual Focus of Quantum Applications

1. Quantum Information Processing
(CE, EE, CS focus)

2. Physical Systems
(ME, CE, EE focus)

a. Materials Science
Quantum properties of solids, fluids, semiconductor materials, metamaterials

b. Quantum Devices
Electron/phonon devices, nanorobots, nanomaterials and systems

Examples of Applications You Can Work on In the Future Opened Up by This Course

Computer Science (CS) Focus

- **Quantum algorithms**: Shor's, Grover's, quantum machine learning
- **Quantum complexity theory**: BQP, QMA complexity classes
- **Quantum programming**: Qiskit, Cirq, quantum circuit design
- **Quantum error correction**: Surface codes, fault tolerance
- **Cryptography**: Post-quantum cryptography, quantum key distribution

1.3 本课程面向的工程受众

服务四个本科专业

本课程旨在服务于来自以下四个主要工程和科学专业的学生：

计算机科学（CS）量子计算、算法

电气工程 (EE) 量子器件，纳米加工

计算机工程 (CE) 硬件-软件集成

机械工程 (ME) 纳米力学，量子材料

1.4 两大主要应用类别

量子应用的两大重点

1. 量子信息处理
(CE, EE, CS 重点)

2. 物理系统
(机械工程、土木工程、电子工程方向)

a. 材料科学
固体、流体、半导体材料、超材料的量子特性

b. 量子器件
电子/声子设备，纳米机器人，纳米材料和系统

本课程为您未来可从事的应用实例

计算机科学（CS）方向

- 量子算法：肖尔算法、格罗弗算法、量子机器学习
- 量子复杂度理论：BQP、QMA复杂度类
- 量子编程：Qiskit、Cirq、量子电路设计
- 量子纠错：表面码、容错性
- 密码学：后量子密码学、量子密钥分发

Electrical & Computer Engineering (EE/CE) Focus

- **Quantum hardware**: Qubit implementation (superconducting, spin, topological)
- **Control systems**: Microwave pulse engineering, cryogenic electronics
- **Nanofabrication**: Josephson junctions, quantum dot devices
- **Photonic integration**: Quantum photonics, integrated optics
- **System architecture**: Quantum-classical interface, readout electronics

Mechanical Engineering (ME) Focus

- **Quantum materials**: 2D materials, topological insulators, superconductors
- **Nanomechanics**: Optomechanical systems, NEMS/MEMS quantum sensors
- **Thermodynamics**: Quantum heat engines, nanoscale thermal transport
- **Fluid dynamics**: Quantum fluids, superfluids, Bose-Einstein condensates
- **Solid mechanics**: Quantum effects in nanomaterials, strain engineering

1.5 Cross-Disciplinary Applications

Application Area	CS	CE	EE	ME
Quantum Computing	$\checkmark$	$\checkmark$	$\checkmark$	
Quantum Sensing		$\checkmark$	$\checkmark$	$\checkmark$
Quantum Materials			$\checkmark$	$\checkmark$
Quantum Communication	$\checkmark$	$\checkmark$	$\checkmark$	
Quantum Simulation	$\checkmark$		$\checkmark$	$\checkmark$
Quantum Metrology		$\checkmark$	$\checkmark$	$\checkmark$

Unifying Theme

Despite different applications, all disciplines share the **same mathematical foundation**:

$$\boxed{i\hbar\frac{\partial}{\partial t}|\psi\rangle = \hat{H}|\psi\rangle} \quad \text{and} \quad \boxed{P(a) = |\langle a|\psi|a|\psi\rangle|^2}$$

This course provides the common quantum mechanics toolkit that enables innovation across computer science, electrical engineering, computer engineering, and mechanical engineering applications.

电气与计算机工程（EE/CE）方向

- 量子硬件：量子比特实现（超导、自旋、拓扑）
- 控制系统：微波脉冲工程、低温电子学
- 纳米加工：约瑟夫森结、量子点器件
- 光子集成：量子光子学，集成光学
- 系统架构：量子-经典接口，读出电子学

机械工程（ME）方向

- 量子材料：二维材料，拓扑绝缘体，超导体
- 纳米力学：光机械系统，NEMS/MEMS量子传感器
- 热力学：量子热机，纳米尺度热传输
- 流体力学：量子流体、超流体、玻色-爱因斯坦凝聚态
- 固体力学：纳米材料的量子效应、应变工程

1.5 跨学科应用

应用领域	CS CE EE			ME
量子计算	$\checkmark$	$\checkmark$	$\checkmark$	
量子传感		$\checkmark$	$\checkmark$	$\checkmark$
量子材料			$\checkmark$	$\checkmark$
量子通信	$\checkmark$	$\checkmark$	$\checkmark$	
量子模拟	$\checkmark$		$\checkmark$	$\checkmark$
量子计量		$\checkmark$	$\checkmark$	$\checkmark$

统一主题

尽管应用领域不同，所有学科都共享相同的基础数学原理：

$$\boxed{i\hbar\frac{\partial}{\partial t}|\psi\rangle = \hat{H}|\psi\rangle} \quad \text{and} \quad \boxed{P(a) = |\langle a|\psi|a|\psi\rangle|^2}$$

本课程提供了通用的量子力学工具包，使计算机科学、电气工程、计算机工程和机械工程应用领域的创新成为可能。

1.6 A Note to Students

Initial Expectations for Lecture 1

This first lecture provides a **high-level overview** of quantum mechanics and its historical development. At this stage:

- **Don't worry if some concepts seem abstract or unfamiliar**—the meaning of mathematical symbols and physical ideas will become clearer as we progress through the course.
- **You may encounter unfamiliar notation** (Dirac notation, operators, Hilbert spaces)—these will be systematically introduced in subsequent lectures.
- **Some historical context may seem disconnected** from the modern formalism—we will show how these pieces fit together mathematically in later lectures.
- **The universal scope of QM** presented here is philosophical and conceptual; the practical mathematical tools for specific problems come later.

Think of this lecture as **laying the conceptual groundwork**—providing motivation and context before we dive into the mathematical details.

1.7 The Universal Scope of Quantum Mechanics

Quantum Mechanics as a Universal Framework

Quantum mechanics is a fundamental theoretical framework for physics that is considered valid across **all physical scales**:

Three Scales Where Quantum Mechanics Applies

Scale 1: Microscopic Scale (fundamental quantum realm):

- Atoms, molecules, subatomic particles (electrons, protons, neutrons)
- Elementary particles (quarks, leptons, gauge bosons)
- Quantum fields and vacuum fluctuations

Scale 2: Intermediate/Mesoscopic Scale (quantum-to-classical transition):

- Macroscopic quantum phenomena (Bose-Einstein condensates, superconductors)
- Nanoscale systems (quantum dots, molecular machines)
- Everyday objects where quantum effects are measurable (interference of large molecules)
- Biological quantum processes (photosynthesis, magnetoreception)

Scale 3: Cosmological Scale (quantum gravity and cosmology):

- Early universe physics (Big Bang, inflation)
- Quantum black holes and Hawking radiation
- Quantum cosmology and the wave function of the universe
- Quantum gravity (string theory, loop quantum gravity)

1.6 致学生

第一讲初始期望

本讲座首先概述了量子力学及其历史发展。现阶段：

- **如果有些概念看起来抽象或不熟悉——随着课程的进行，数学符号和物理概念的意义将会变得更加清晰。** 随着课程的进行，数学符号和物理概念的意义将会变得更加清晰。
- 你可能会遇到不熟悉的符号（狄拉克符号、算符、希尔伯特空间）——这些将在后续讲座中系统介绍。
- **一些历史背景可能与现代形式主义看似脱节——我们将** 在后续讲座中展示这些部分如何从数学上组合在一起。
- 这里呈现的量子力学的普适范围是哲学和概念性的；针对特定问题的实用数学工具将在后面介绍。

将这次讲座视为奠定概念基础——在我们深入数学细节之前，提供动机和背景。

1.7 量子力学的普遍范围

量子力学作为一个通用框架

量子力学是物理学的一个基本理论框架，被认为适用于所有物理尺度：

量子力学适用的三个尺度

尺度1：微观尺度（基本量子领域）：

- 原子、分子、亚原子粒子（电子、质子、中子）
- 基本粒子（夸克、轻子、规范玻色子）
- 量子场和真空涨落

尺度 2：介观尺度（量子到经典的过渡）：

- 宏观量子现象（玻色-爱因斯坦凝聚、超导体）
- 纳米尺度系统（量子点、分子机器）
- 日常物体中可测量的量子效应（大分子的干涉）
- 生物量子过程（光合作用、磁感应）

尺度3：宇宙尺度（量子引力和宇宙学）：

- 早期宇宙物理学（大爆炸、暴胀）
- 量子黑洞和霍金辐射
- 量子宇宙学和宇宙波函数
- 量子引力（弦理论、圈量子引力）

Classical Physics as an Emergent Phenomenon

The Quantum-to-Classical Transition

Classical physics emerges from quantum mechanics through several mechanisms:

- **Decoherence**: Environment-induced suppression of quantum interference
- **Classical limit** ($\hbar \to 0$): Formal mathematical limit
- **Ehrenfest's theorem**: Quantum expectation values follow classical equations

The precise nature of the quantum-to-classical transition remains an active research area in foundations of quantum mechanics.

Quantum Cosmology and the Wave Function of the Universe

Wheeler-DeWitt Equation and Quantum Cosmology

In quantum cosmology, the entire universe is described by a **wave function of the universe**:

$$\Psi_{\text{univ}}[\gamma_{ij}, \phi]$$

where γ_{ij} is the 3-metric of space and ϕ represents matter fields.

- Proposed by Bryce DeWitt and John Wheeler in the 1960s
- Stephen Hawking and James Hartle developed the "no-boundary proposal" in the 1980s
- Attempts to describe the quantum origin of spacetime itself
- The Wheeler-DeWitt equation: $\hat{H}\Psi_{\text{univ}} = 0$ (timeless formulation)

The Universal Reductionist Program

From Quantum Foundations to Complex Phenomena

The reductionist view suggests that all physical phenomena should ultimately be derivable from quantum mechanics:

$$\Psi_{\text{univ}} \xrightarrow{\text{quantum cosmology}} \text{fundamental physics} \xrightarrow{\text{emergence}} \begin{cases} \text{chemistry} \\ \text{biology} \\ \text{neuroscience} \\ \text{sociology} \\ \text{economics} \end{cases}$$

- This represents an ambitious (and controversial) research program
- Practical limitations: Exponential complexity makes direct derivation impossible
- Emergent phenomena often require their own effective theories
- The relationship between quantum foundations and higher-level phenomena remains an open philosophical question

经典物理学作为一种涌现现象

量子到经典的转变

经典物理学通过多种机制从量子力学中涌现：

- 退相干：环境引起的量子干涉抑制
- 经典极限 ($\hbar \to 0$): 形式数学极限
- 埃伦费斯特定理：量子期望值遵循经典方程

量子到经典的转变的精确性质仍然是量子力学基础研究中的一个活跃领域。

量子宇宙学与宇宙波函数

惠勒-德威特方程与量子宇宙学

在量子宇宙学中，整个宇宙由宇宙波函数描述：

$$\Psi_{\text{univ}}[\gamma_{ij}, \phi]$$

其中 γ_{ij} 是空间的三度量， ϕ 表示物质场。

- 由布莱斯·德威特和约翰·惠勒于20世纪60年代提出
- 斯蒂芬·霍金和詹姆斯·哈特雷于20世纪80年代发展了“无边界方案”
- 试图描述时空本身的量子起源
- 惠勒-德威特方程： $H\Psi$univ = 0 ˆ(timeless formulation)

通用还原论计划

从量子基础到复杂现象

还原论观点认为，所有物理现象最终都应可从量子力学推导出来：

$$\Psi_{\text{univ}} \xrightarrow{\text{quantum cosmology}} \text{fundamental physics} \xrightarrow{\text{emergence}} \begin{cases} \text{chemistry} \\ \text{biology} \\ \text{neuroscience} \\ \text{sociology} \\ \text{economics} \end{cases}$$

- 这代表了一个雄心勃勃（且具争议性）的研究计划
- 实际限制：指数级复杂性使得直接推导成为不可能
- 涌现现象往往需要它们自己的有效理论
- 量子基础与高级现象之间的关系仍然是一个开放性的哲学问题

Mathematical Hierarchy of Theories

$$\begin{array}{ccccc} \text{Quantum Gravity} & \supset & \text{Quantum Field Theory} & \supset & \text{Quantum Mechanics} \\ \cap & & \cap & & \cap \\ \text{General Relativity} & \supset & \text{Classical Field Theory} & \supset & \text{Classical Mechanics} \end{array}$$

Interpretation: The more fundamental theory (top row) contains the less fundamental theory (bottom row) as a special case or limiting approximation. For example:

- Quantum Field Theory ⊃ Quantum Mechanics (in the non-relativistic limit)
- Quantum Mechanics ⊃ Classical Mechanics (as $\hbar \to 0$)
- Quantum Gravity ⊃ General Relativity (in the classical limit)

Each theory emerges as an approximation/limit of the more fundamental one above it

1.8 Further Reading

Core Textbooks

Feynman Lectures: The Enduring Classic

- **Complete Reference**: Richard P. Feynman, Robert B. Leighton, and Matthew Sands, *The Feynman Lectures on Physics, Vol. III: Quantum Mechanics.* Basic Books/Caltech, 1963.
- **Key Features**: Unique blend of wavefunction and Hilbert-space formalisms; unparalleled physical insights; includes advanced undergraduate topics like band structures and superconductors.
- **Accessibility**: Freely available online at `https://www.feynmanlectures.caltech.edu/` with high-quality HTML presentation.

Susskind: The Modern Theoretical Minimum

- **Complete Reference**: Leonard Susskind and Art Friedman, *Quantum Mechanics: The Theoretical Minimum.* Basic Books/Penguin, 2014.
- **Key Features**: Firmly based on the modern Dirac-Hilbert space formalism while remaining accessible to undergraduates; includes modern topics like quantum entanglement; inspired by Feynman's pedagogical clarity.
- **Supplementary Materials**: Accompanied by video lectures available on YouTube.

Proposed Reading Methodology

Course Reading Strategy

- **Weekly Assignments**: I will assign specific sections from both books to complement lecture topics.
- **Feynman's Breadth**: Excellent for multidisciplinary audiences and unique applications.
- **Susskind's Modernity**: Provides the abstract mathematical framework essential for modern quantum information science.
- **Hybrid Approach**: Reading both provides both historical insight and modern formalism.

理论数学层级

量子引力 ⊃ 量子场论 ⊃ 量子力学
∩ ∩ ∩广义相对论 ⊃ 经典场论
⊃ 经典力学

解释：更基础的理论（上行）包含更不基础的理论（下行）作为特例或极限近似。例如：

- 量子场论 ⊃ 量子力学（在非相对论极限下）
- 量子力学 ⊃ 经典力学（作为 $\hbar \to 0$）
- 量子引力 ⊃ 广义相对论（在经典极限下）

每种理论都是更根本的上述理论的一种近似或极限。

1.8 深入阅读

核心教材

费曼讲义：不朽的经典

- 完整参考：理查德·费曼、罗伯特·利顿和马修·桑德斯，《费曼物理学讲义》第III卷：量子力学。基础图书/加州理工学院出版社，1963年。
- 主要特点：独特的波函数与希尔伯特空间形式主义的结合；无与伦比的物理洞察力；包含如能带结构和超导体等进阶本科主题。
- **易用性：** 可在 https://www.feynmanlectures.caltech.edu/ 免费在线获取，并提供高质量的HTML演示。

苏斯金德：现代理论最小值

- 完整参考：莱昂纳德·苏斯金德与阿特·弗里德曼，《量子力学：理论最小值》。基础图书/企鹅出版社，2014年。
- 主要特点：基于现代狄拉克-希尔伯特空间形式主义，同时保持对本科生的易读性；包含量子纠缠等现代主题；受费曼教学清晰性的启发。
- 补充材料：配有可在YouTube上观看的视频讲座。

建议阅读方法

课程阅读策略

- 每周作业：我将从两本书中指定章节，以补充讲座主题。
- 费曼的广度：非常适合跨学科受众和独特应用。
- 苏斯金的《现代性》：提供了现代量子信息科学所必需的抽象数学框架
量子信息科学。
- 混合方法：同时阅读既可提供历史洞察，又能体现现代形式主义。

Page 9 of 61

Foundational Texts: Historical and Conceptual

Classical Texts on Quantum Theory

- **David Bohm, *Quantum Theory*. Prentice-Hall, 1951; Dover Reprint.** A classic emphasizing physical intuition and conceptual foundations, with particularly clear discussions of the measurement problem.
- **P. A. M. Dirac, *The Principles of Quantum Mechanics*, 4th ed. Oxford University Press, 1958.** Written by one of QM's founders; most profitable after gaining initial familiarity with the subject. The definitive early statement of the Hilbert space formalism.
- **Roger Penrose, *The Road to Reality: A Complete Guide to the Laws of the Universe*. Jonathan Cape, 2004.** Encyclopedic; quantum physics chapters scattered throughout offer exceptional insight into the mathematical structure of physical theory.

Deeper Second Books on Quantum Mechanics

Conceptual & Structural Texts on Quantum Theory

- **Chris J. Isham, *Lectures on Quantum Theory: Mathematical and Structural Foundations*. Imperial College Press, 1995.** A top-down, mathematically precise approach that builds quantum theory from its abstract Hilbert space foundations. Emphasizes structural clarity over historical narrative.
- **Jakob Schwichtenberg, *A No-Nonsense Introduction to Quantum Mechanics*. Independently published, 2020s.** Strips away historical digressions to present the logical and calculational core of QM in a clean, modern style. Ideal for building a practical understanding of the formalism while introducing deep and advanced topics.

Alternative Interpretations: Bohmian Mechanics

The Causal Interpretation (de Broglie-Bohm Theory)

- **Philosophical Foundation**: An alternative to mainstream interpretations that restores determinism and particle trajectories.
- **David Bohm and B. J. Hiley, *The Undivided Universe: An Ontological Interpretation of Quantum Theory*. Routledge, 1993.** The most complete statement of Bohm's mature interpretation.
- **Peter R. Holland, *The Quantum Theory of Motion: An Account of the de Broglie–Bohm Causal Interpretation*. Cambridge University Press, 1993.** A comprehensive textbook treatment.
- **Detlef Dürr and Stefan Teufel, *Bohmian Mechanics: The Physics and Mathematics of Quantum Theory*. Springer, 2009.** A modern, rigorous mathematical treatment.

Course Relevance: While time constraints prevent detailed coverage, I will reference this important framework throughout the course to highlight interpretive choices in standard QM.

基础文献：历史与概念

量子理论经典文献

- 大卫·玻姆，《量子理论》。普伦蒂斯-霍尔出版社，1951年；多佛重印版。这部经典著作强调物理直觉和概念基础，尤其对测量问题的讨论尤为清晰。
- P. A. M. 狄拉克，《量子力学原理》，第四版，牛津大学出版社，1958年。由量子力学创始人之一撰写；在初步熟悉该学科后最具价值。对希尔伯特空间形式主义的早期权威阐述。
- 罗杰·彭罗斯，《现实之路：宇宙定律的完整指南》，乔纳森·凯普出版社，2004年。百科全书式著作；量子物理学章节散布全书，为物理理论数学结构提供了卓越的见解。

量子力学进阶读物

量子理论的概念与结构文本

- 克里斯·*J*·伊沙姆，《量子理论讲座：数学与结构基础》
基础。帝国理工学院出版社，1995年。一种自上而下、数学精确的方法，从其抽象希尔伯特空间基础构建量子理论。强调结构清晰胜过历史叙述。
- 雅各布·施维钦贝格，《量子力学轻松入门》，独立出版，2020年代。剥离历史旁支，以简洁现代的风格呈现量子力学的逻辑与计算核心。适合在介绍形式主义的同时，深入和高级主题。

替代诠释：玻姆力学

因果解释（德布罗意-玻姆理论）

- 哲学基础：一种替代主流解释、恢复决定论和粒子轨迹的理论。
- 大卫·玻姆与*B. J.* 赫利，《统一宇宙：量子理论的形而上学解释》。*Routledge*出版社，*1993*年。玻姆成熟解释最完整的阐述。
- 彼得·*R*·霍兰，《运动量子理论：德布罗意-玻姆因果解释的说明》。剑桥大学出版社，*1993*年。一本全面的教科书式处理。
- 德特莱夫·*D*¨尔与斯特凡·特菲尔，《玻姆力学：量子理论与数学》。*Springer*出版社，*2009*年。一部现代、严谨的数学处理。

课程相关性：虽然时间限制无法进行详细讲解，但在整个课程中，我会参考这个重要框架，以突出标准QM中的解释性选择。

Foundations and Advanced Undergraduate Perspectives

Foundations and Mathematical Approaches

- **Travis Norsen, *Foundations of Quantum Mechanics: An Exploration of the Physical Meaning of Quantum Theory*. Springer, 2017.** Recommended for undergraduates seeking a clear, modern treatment of interpretational issues.
- **Stephen Bruce Sontz, *An Introductory Path to Quantum Theory: Using Mathematics to Understand the Ideas of Physics*. Springer, 2020.** A mathematically rigorous yet accessible approach, emphasizing how mathematics structures physical understanding.

Physics in Broad Context

Quantum Mechanics in Scientific & Philosophical Context

- **Albert Einstein and Leopold Infeld, *The Evolution of Physics: From Early Concepts to Relativity and Quanta*. Simon & Schuster, 1938.** Situates QM within the historical development of physical concepts.
- **Werner Heisenberg, *Physics and Philosophy: The Revolution in Modern Science*. Harper & Row, 1958.** A founder's perspective on the philosophical implications.
- **Stephen Hawking, *A Brief History of Time: From the Big Bang to Black Holes*. Bantam Books, 1988.** The famous bestseller connecting quantum physics to cosmology.
- **Louis de Broglie, *The Revolution in Physics: A Non-Mathematical Survey of Quanta*. Noondey Press, 1953.** A conceptual overview by the originator of matter waves.

1.9 Mathematical Notation and Conventions

Basic Mathematical Objects

- **Scalar Fields**: $\mathbb{R}$ (real numbers), $\mathbb{C}$ (complex numbers)
- **Complex Unit**: $i = \sqrt{-1}$; complex conjugate: z^* or $\bar{z}$
- **Planck's Constant**: Reduced form $\hbar = h/(2\pi)$
- **Imaginary Unit in QM**: Always i, never j

Vectors and Spaces

- **Cartesian Vectors**: Boldface notation: $\mathbf{r}, \mathbf{p}, \mathbf{x}, \mathbf{q}$
- **Vector Components**: x, y, z or x_i for $i = 1, 2, 3$
- **Hilbert Space**: Abstract state space denoted $\mathcal{H}$
- **Function Space**: $L^2(\mathbb{R}^n) = \{\psi : \int_{\mathbb{R}^n} |\psi(\mathbf{x})|^2 d^n x < \infty\}$

基础与本科生进阶视角

基础与数学方法

- 特拉维斯·诺森，《量子力学基础：对物理的探索》
量子理论的意义。Springer，2017年。推荐给寻求{...}的本科生。
对诠释问题的清晰、现代处理。
- ***Stephen Bruce Sontz***，《量子理论入门：使用数学》
理解物理思想。斯普林格出版社，***2020***年。一本数学严谨但
一种可访问的方法，强调数学如何构建物理理解。

物理学在广义背景下

量子力学在科学与哲学背景中

- 阿尔伯特·爱因斯坦和莱奥波德·英费尔德，《物理学的进化：从早期概念到相对论和量子论》。西蒙与舒斯特出版社，***1938***年。将量子力学置于物理概念历史发展之中。

•维纳·海森堡，《物理学与哲学：现代科学的革命》。哈珀与罗出版公司，***1958***年。一位创始人的哲学意义视角。

•斯蒂芬·霍金，《时间简史：从大爆炸到黑洞》。兰登书屋，1988年。将量子物理学与宇宙学联系起来的著名畅销书。

- 路易·德·布罗意，《物理学革命：量子非数学概述》。正午出版社，***1953***年。物质波创始人的概念概述。

1.9 数学符号与约定

基本数学对象

- 标量场：$\mathbb{R}$ (实数)，$\mathbb{C}$ (复数)
- 复杂单元：$i = \sqrt{-1}$；复共轭：z^* 或 z −
- **普朗克常数：约化形式** $\hbar = h/(2\pi)$
- **量子力学中的虚数单位：始终** i，**绝不** j

向量和空间

- **笛卡尔向量：粗体符号：r,p,x, q**
- **向量分量：** x, y, z **或** x_i **用于** $i = 1,2,3$
- 希尔伯特空间：抽象状态空间，表示为 $\mathcal{H}$
- **函数空间：** $L^2(\mathbb{R}^n) = \{\psi : \int_{\mathbb{R}^n} |\psi(\mathbf{x})|^2 d^n x < \infty\}$

Quantum-Specific Notation

- **Dirac Notation**:
  - State vector (ket): $|\psi\rangle \in \mathcal{H}$
  - Dual vector (bra): $\langle\phi| \in \mathcal{H}^*$
  - Inner product: $\langle\phi|\psi\rangle$
- **Operators**:
  - Hat notation: $\hat{A}, \hat{x}, \hat{p}, \hat{H}$
  - Hermitian conjugate: $\hat{A}^\dagger$
- **Commutator**: $[\hat{A}, \hat{B}] := \hat{A}\hat{B} - \hat{B}\hat{A}$
- **Position Basis**: Wavefunction: $\psi(\mathbf{r}) = \langle\mathbf{r}|\psi\rangle$

1.10 Important Constants

Constant	Symbol	Value
Planck's constant	h	6.626×10^{-34} J·s
Reduced Planck constant	$\hbar = h/2\pi$	1.055×10^{-34} J·s
Speed of light	c	2.998×10^{8} m/s
Electron mass	m_e	9.109×10^{-31} kg
Proton mass	m_p	1.673×10^{-27} kg
Bohr radius	a_0	5.292×10^{-11} m
Rydberg constant	R_H	1.097×10^{7} m^{-1}

量子专用符号

- **狄拉克符号：**
  - 状态向量（括号）： $|\psi\rangle \in \mathcal{H}$
  - 对偶向量（尖括号）： $\langle\phi| \in \mathcal{H}^*$
  - 内积： $\langle\phi|\psi\rangle$
- **算子：**
  - 帽记号： $\hat{A}$, ˆx, ˆp, $\hat{H}$
  - 埃米特共轭：ˆ$A^\dagger$
- **Commutator**: $[\hat{A}, \hat{B}] := \hat{A}\hat{B} - \hat{B}\hat{A}$
- 位置基：波函数： $\psi(\mathrm{r}) = \langle\mathrm{r}|\psi\rangle$

1.10 重要常数

常数	符号	值
普朗克常数	h	6.626×10^{-34} J·s
约化普朗克常数	$\hbar = h/2\pi$	1.055×10^{-34} J·s
光速	c	2.998×10^{8} m/s
电子质量	m_e	9.109×10^{-31} kg
质子质量	m_p	1.673×10^{-27} kg
玻尔半径	a_0	5.292×10^{-11} m
里德伯常数	R_H	1.097×10^{7} m^{-1}

2 Review of Math and Physics Background

2.1 Review of Complex Numbers

The imaginary unit i is defined as a number satisfying $i^2 = -1$. Formally, it can be constructed within set theory as the ordered pair $(0, 1)$, which in turn can be defined as $\{\{0\}, \{0, 1\}\}$. This construction shows that complex numbers exist within rigorous mathematics. Quantum mechanics has demonstrated that the structure of the physical world itself relies fundamentally on complex numbers.

Definition 1 (Complex Number)**.** *A complex number z can be written in two equivalent forms:*

- ***Rectangular form****: $z = x + iy$, where $x, y \in \mathbb{R}$*
- ***Polar form****: $z = re^{i\theta}$, where $r \geq 0$ and $\theta \in [0, 2\pi)$*

Here $r = |z| = \sqrt{x^2 + y^2}$ is the magnitude *and $\theta = \arg(z)$ is the* phase.

Definition 2 (Complex Conjugate)**.** *For $z = x + iy$, the complex conjugate is $z^* = x - iy$. In polar form, if $z = re^{i\theta}$, then $z^* = re^{-i\theta}$.*

Euler's Identity (Fundamental Relation)

$$\boxed{e^{i\theta} = \cos\theta + i\sin\theta}$$

This remarkable identity connects exponential, trigonometric, and complex number theory. Two special cases are particularly important:

$$e^{i\pi} + 1 = 0 \quad \text{and} \quad e^{i\pi/2} = i$$

2.2 Review of Vector and Hilbert Spaces

We provide only the minimum needed to define Hilbert spaces. Think of a Hilbert space as a generalization of the familiar Euclidean vector space $\mathbb{R}^3$.

Example 1 (Euclidean Space $\mathbb{R}^3$)**.** *The space $\mathbb{R}^3$ consists of all vectors $\mathbf{r}$ that can be written as:*

$$\mathbf{r} = \mathbf{i}r_1 + \mathbf{j}r_2 + \mathbf{k}r_3$$

where $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ are three orthonormal basis vectors.

From this example, we extract three key concepts:

1. **Normalization**: $\|\mathbf{i}\| = \|\mathbf{j}\| = \|\mathbf{k}\| = 1$
2. **Orthonormality**: $\mathbf{i} \cdot \mathbf{j} = \mathbf{j} \cdot \mathbf{k} = \mathbf{k} \cdot \mathbf{i} = 0$
3. **Completeness**: Any vector $\mathbf{r} \in \mathbb{R}^3$ can be expressed as a linear combination of the basis vectors

Definition 3 (Hilbert Space)**.** *A Hilbert space $\mathcal{H}$ is a complete inner product space over the complex numbers $\mathbb{C}$.*

- **Vectors**: In Dirac notation, a vector in $\mathcal{H}$ is written as a *ket*: $|\psi\rangle$
- **Basis Expansion**: An arbitrary ket can be expanded in a (possibly countably infinite) basis $\{|\phi_n\rangle\}$:

$$|\psi\rangle = \sum_n c_n |\phi_n\rangle, \quad c_n \in \mathbb{C}$$

2 数学与物理基础回顾

2.1 复数复习

虚数单位 i 定义为一个满足 $i^2 = -1$ 的数。形式上，它可以在集合论中构造为有序对 $(0, 1)$，进而可以定义为 $\{\{0\}, \{0, 1\}\}$。这种构造表明复数存在于严谨的数学之中。量子力学已经证明，物理世界本身的结构根本依赖于复数。

定义*1*（复数）。复数 z 可以写成两种等价形式：

- 矩形形式：$z = x + iy$，其中 x，$y \in \mathbb{R}$
- 极坐标形式：$z = re^{i\theta}$，其中 $r \geq 0$ 和 $\theta \in [0, 2\pi)$

这里 $r = |z| = \sqrt{x^2 + y^2}$ 是幅度，$\theta = \arg(z)$ 是相位。

定义 *2*（复共轭）。对于 $z = x + iy$，复共轭是 $z^* = x - iy$。在极坐标形式中，*if* $z = re^{i\theta}$，则 $z^* = re^{-i\theta}$。

欧拉恒等式（基本关系）

$$\boxed{e^{i\theta} = \cos\theta + i\sin\theta}$$

这个非凡的恒等式连接了指数、三角和复数理论。有两个特殊情况特别重要：

$$e^{i\pi} + 1 = 0 \quad \text{and} \quad e^{i\pi/2} = i$$

2.2 向量空间与希尔伯特空间复习

我们仅提供定义希尔伯特空间所需的最小内容。将希尔伯特空间视为熟悉欧几里得向量空间 $\mathbb{R}^3$ 的一种推广。

示例 *1*（欧几里得空间 $\mathbb{R}^3$）。空间 $\mathbb{R}^3$ 包含所有可以写成以下形式的向量 r：

$$\mathbf{r} = \mathbf{i}r_1 + \mathbf{j}r_2 + \mathbf{k}r_3$$

其中 $\{i, j, k\}$ 是三个正交归一基向量。

从这个示例中，我们提取了三个关键概念：

1. 归一化： $\|\mathbf{i}\| = \|\mathbf{j}\| = \|\mathbf{k}\| = 1$

2. 正交归一性： $\mathbf{i} \cdot \mathbf{j} = \mathbf{j} \cdot \mathbf{k} = \mathbf{k} \cdot \mathbf{i} = 0$

3. 完备性：任何向量 $\mathbf{r} \in \mathbb{R}^3$ 都可以表示为基向量的线性组合

定义 *3*（希尔伯特空间）。希尔伯特空间 $\mathcal{H}$ 是在复数域 $\mathbb{C}$ 上的完备内积空间。

- 向量：在狄拉克符号中，$\mathcal{H}$ 中的向量写作一个bra：$|\psi\rangle$
- 基扩展：任意一个态矢可以在一个（可能是可数无限维的）基 $\{|\phi_n\rangle\}$ 中展开：

$$|\psi\rangle = \sum_n c_n |\phi_n\rangle, \quad c_n \in \mathbb{C}$$

- **Dual Vectors**: The dual of $|\psi\rangle$ is written as a *bra*: $\langle\psi|$. If we think of $|\psi\rangle$ as a column vector, then $\langle\psi|$ is its conjugate transpose:
$$\langle\psi| = |\psi\rangle^\dagger$$
- **Inner Product**: For any two kets $|\phi\rangle, |\psi\rangle \in \mathcal{H}$, the inner product is written as $\langle\phi|\psi\rangle \in \mathbb{C}$.
- **Norm**: The length (norm) of $|\psi\rangle$ is $\| \, |\psi\rangle \, \| = \sqrt{\langle\psi|\psi\rangle}$.

Definition 4 (Operator). *An operator $\hat{A}$ is a linear map $\hat{A} : \mathcal{H} \to \mathcal{H}$.*

Example 2. *In $\mathbb{R}^3$, operators are 3×3 matrices. In Hilbert spaces, operators can be infinite-dimensional matrices.*

Definition 5 (Adjoint Operator). *For an operator $\hat{A}$, its adjoint $\hat{A}^\dagger$ is defined by the relation*
$$\langle\psi|\hat{A}^\dagger|\phi\rangle = \langle\phi|\hat{A}|\psi\rangle^*$$
for all vectors $|\psi\rangle, |\phi\rangle \in \mathcal{H}$, where * denotes complex conjugation. Equivalently,
$$\langle\psi|\hat{A}^\dagger\phi\rangle = \langle\hat{A}\psi|\phi\rangle$$
for all $|\psi\rangle, |\phi\rangle$.

Definition 6 (Algebraic Properties of the Adjoint). *The adjoint operation satisfies the following properties for all operators $\hat{A}$, $\hat{B}$ and scalars $\alpha \in \mathbb{C}$:*

(i) ***Adjoint of an adjoint:*** $(\hat{A}^\dagger)^\dagger = \hat{A}$.

(ii) ***Adjoint of a sum:*** $(\hat{A} + \hat{B})^\dagger = \hat{A}^\dagger + \hat{B}^\dagger$.

(iii) ***Adjoint of a product:*** $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger$ *(reverse order).*

(iv) ***Adjoint of a scalar multiple:*** $(\alpha\hat{A})^\dagger = \alpha^*\hat{A}^\dagger$.

(v) ***Adjoint of the identity:*** $\hat{I}^\dagger = \hat{I}$.

Remark 2.1 (Connection to Matrix Mechanics). *In a finite-dimensional Hilbert space with an orthonormal basis, if $\hat{A}$ is represented by the matrix $\mathbf{A}$, then $\hat{A}^\dagger$ is represented by the conjugate transpose matrix $\mathbf{A}^\dagger = (\mathbf{A}^T)^*$. The product rule $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger$ corresponds to the matrix identity $(\mathbf{AB})^\dagger = \mathbf{B}^\dagger\mathbf{A}^\dagger$.*

Theorem 1 (Eigenvector Relation for $\hat{A}^\dagger$). *Let $\hat{A}$ be an operator on a Hilbert space $\mathcal{H}$ with an eigenvector $|a\rangle$ and corresponding eigenvalue λ, i.e. $\hat{A}|a\rangle = \lambda|a\rangle$. Then the adjoint operator $\hat{A}^\dagger$ satisfies the following:*
$$\langle a|\hat{A}^\dagger = \lambda^*\langle a|$$
That is, $\langle a|$ is a ***left eigenvector*** *of $\hat{A}^\dagger$ with eigenvalue λ^*.*

However, $|a\rangle$ is an eigenvector of $\hat{A}^\dagger$ ***if and only if*** *$\hat{A}$ is a normal operator ($[\hat{A}, \hat{A}^\dagger] = 0$). In that case:*
$$\hat{A}^\dagger|a\rangle = \lambda^*|a\rangle$$

Definition 7 (Hermitian (Self-Adjoint) Operator). *An operator $\hat{A}$ is Hermitian (or self-adjoint) if $\hat{A}^\dagger = \hat{A}$. In terms of inner products, this means*
$$\langle\psi|\hat{A}|\phi\rangle = \langle\phi|\hat{A}|\psi\rangle^*$$
for all $|\psi\rangle, |\phi\rangle \in \mathcal{H}$, or equivalently,
$$\langle\hat{A}\psi|\phi\rangle = \langle\psi|\hat{A}\phi\rangle.$$

- 对偶向量： $|\psi\rangle$ 的对偶用bra表示为 $\langle\psi|$。如果我们把 $|\psi\rangle$ 看作列向量，那么$\langle\psi|$ 就是它的共轭转置：†
$$\langle\psi| = |\psi\rangle$$
- 内积：对于任意两个态矢 $|\phi\rangle$ $|\psi\rangle \in \mathcal{H}$,，内积写作 $\langle\phi|\psi\rangle \in \mathbb{C}$。
- 范数： $|\psi\rangle$ 的长度（范数）是$\| \, |\psi\rangle \, \| = \sqrt{\langle\psi|\psi\rangle}$。

ˆ ˆ 定义*4*（算子）。一个算子 A 是一个线性映射 $A : \mathcal{H} \to \mathcal{H}$。

示例*2*。在 $\mathbb{R}^3$中，算子是 3×3 矩阵。在希尔伯特空间中，算子可以是无限维矩阵。

定义 *5*（伴随算子）。对于算子 ˆA，其伴随算子 ˆ$A^\dagger$ 由以下关系定义
$$\langle\psi|\hat{A}^\dagger|\phi\rangle = \langle\phi|\hat{A}|\psi\rangle^*$$
对于所有向量 $|\psi\rangle$ $|\phi\rangle \in \mathcal{H}$,，其中 * 表示复共轭。等价地，
$$\langle\psi|\hat{A}^\dagger\phi\rangle = \langle\hat{A}\psi|\phi\rangle$$
对于所有 $|\psi\rangle, |\phi\rangle$。

定义 *6*（伴随算子的代数性质）。伴随运算满足以下性质 ˆ ˆ 对于所有算子 A, B 和标量 $\alpha \in \mathbb{C}$:

(i) 伴随的伴随： $(\hat{A}^\dagger)^\dagger = \hat{A}$。

(ii) 和的伴随： $(\hat{A} + \hat{B})^\dagger = \hat{A}^\dagger + \hat{B}^\dagger$.

(iii) 乘积的伴随： $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger$ *(*逆序*)*.

(iv) 数量乘积的伴随： $(\alpha\hat{A})^\dagger = \alpha^*\hat{A}^\dagger$.

(v) 单位元的伴随： $\hat{I}^\dagger = \hat{I}$.

备注 *2.1 (*与矩阵力学的联系*)*。在一个有限维希尔伯特空间中，如果算子 ˆA 用矩阵 $\mathbf{A}$ 表示，那么算子 ˆ $A^\dagger$ 用共轭转置矩阵 $\mathbf{A}^\dagger =(\mathbf{A}^T)^*$ 表示。乘积规则 $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger$ 对应于矩阵恒等式 $(\mathbf{AB})^\dagger = \mathbf{B}^\dagger\mathbf{A}^\dagger$.

定理 *1 (*ˆ$A^\dagger$ 的特征向量关系*)*。设 ˆA 是希尔伯特空间 $\mathcal{H}$ 上的一个算子，具有特征向量$|a\rangle$ 和对应的特征值 λ，即 $\hat{A}|a\rangle = \lambda|a\rangle$。则伴随算子 ˆ$A^\dagger$ 满足以下条件：
$$\langle a|\hat{A}^\dagger = \lambda^*\langle a|$$
也就是说， $\langle a|$ 是ˆ$A^\dagger$ 的一个左特征向量，特征值为 λ^*。

然而， $|a\rangle$ 是ˆ$A^\dagger$ 的特征向量当且仅当ˆA 是一个正规算子 $([\hat{A}, \hat{A}^\dagger] = 0)$。在这种情况下：
$$\hat{A}^\dagger|a\rangle = \lambda^*|a\rangle$$

定义*7*（厄米算子（自伴算子））。一个算子ˆA 是厄米算子（或自伴算子），当且仅当ˆ$A^\dagger =$ ˆA。用内积的语言来说，这意味着。
$$\langle\psi|A|\phi\rangle = \langle\phi|A|\psi\rangle^*$$
对于所有 $|\psi\rangle$、$|\phi\rangle \in \mathcal{H}$，或者等价地，
$$\langle\hat{A}\psi|\phi\rangle = \langle\psi|\hat{A}\phi\rangle.$$

Definition 8 (Eigenvalue Problem). *For an operator $\hat{A}$, the eigenvalue equation is:*

$$\hat{A}|\varphi_n\rangle = \lambda_n|\varphi_n\rangle$$

where λ_n are eigenvalues and $|\varphi_n\rangle$ are eigenvectors.

Theorem 2 (Properties of Hermitian Operators). *For a Hermitian operator $\hat{A}$:*

1. *All eigenvalues are real: $\lambda_n \in \mathbb{R}$*
2. *Eigenvectors corresponding to distinct eigenvalues are orthogonal*
3. *The eigenvectors form a complete orthonormal basis for $\mathcal{H}$ (provided certain technical conditions)*

The Space of Square-Summable Sequences

One of the simplest infinite-dimensional Hilbert spaces is the space ℓ^2 of square-summable sequences:

$$\ell^2 = \left\{(c_1, c_2, c_3, \dots) \in \mathbb{C}^\infty : \sum_{n=1}^{\infty} |c_n|^2 < \infty\right\}.$$

The inner product is defined as:

$$\langle a|b\rangle = \sum_{n=1}^{\infty} a_n^* b_n.$$

The condition $\sum_{n=1}^{\infty} |c_n|^2 < \infty$ ensures that the vectors have finite norm (often interpreted as finite "energy" in physical contexts). This is the natural infinite-dimensional generalization of $\mathbb{C}^n$.

The Space of Square-Integrable Functions

The most important Hilbert space for quantum mechanics is the space $L^2(\mathbb{R}^n)$ of square-integrable functions:

$$L^2(\mathbb{R}^n) = \left\{\psi : \mathbb{R}^n \to \mathbb{C} : \int_{\mathbb{R}^n} |\psi(\mathbf{x})|^2\, d^n x < \infty\right\}.$$

The inner product is:

$$\langle\phi|\psi\rangle = \int_{\mathbb{R}^n} \phi^*(\mathbf{x})\,\psi(\mathbf{x})\, d^n x.$$

Wavefunctions $\psi(\mathbf{r}, t)$ in non-relativistic quantum mechanics are elements of $L^2(\mathbb{R}^3)$ for a single particle, or $L^2(\mathbb{R}^{3N})$ for N particles.

The Relation Between ℓ^2 and L^2

Given an orthonormal basis $\{\varphi_n(\mathbf{x})\}$ of $L^2(\mathbb{R}^n)$, any wavefunction $\psi(\mathbf{x})$ can be expanded as:

$$\psi(\mathbf{x}) = \sum_{n=1}^{\infty} c_n \varphi_n(\mathbf{x}),$$

where the coefficients $c_n = \langle\varphi_n|\psi\rangle$ satisfy:

$$\sum_{n=1}^{\infty} |c_n|^2 = \int_{\mathbb{R}^n} |\psi(\mathbf{x})|^2\, d^n x < \infty.$$

This establishes an isomorphism between $L^2(\mathbb{R}^n)$ and ℓ^2: the square-integrable function ψ corresponds uniquely to the square-summable sequence $\{c_n\}$ of its expansion coefficients.

ˆ定义 8（特征值问题）。对于一个算子 A，特征值方程是：

$$\hat{A}|\varphi_n\rangle = \lambda_n|\varphi_n\rangle$$

特征值 λ_n 在哪里，特征向量 $|\varphi_n\rangle$ 在哪里。

ˆ定理2（厄米算子的性质）。对于厄米算子 A:

1. 所有特征值都是实数： $\lambda_n \in \mathbb{R}$
2. 对应不同特征值的特征向量是正交的
3. 特征向量构成 $\mathcal{H}$ 的完备标准正交基（在满足某些技术条件的情况下）

平方可和序列空间

最简单的无限维希尔伯特空间之一是平方可和序列空间 ℓ^2：

$$\ell^2 = \left\{(c_1, c_2, c_3, \dots) \in \mathbb{C}^\infty : \sum_{n=1}^{\infty} |c_n|^2 < \infty\right\}.$$

内积定义为:

$$\langle a|b\rangle = \sum_{n=1}^{\infty} a_n^* b_n.$$

条件 $\sum_{n=1}^{\infty} |c_n|^2 < \infty$ 确保向量具有有限范数（在物理语境中常被解释为有限的“能量”）。这是 $\mathbb{C}^n$ 的自然无限维推广。

平方可积函数空间

量子力学中最重要的希尔伯特空间是平方可积函数空间 $L^2(\mathbb{R}^n)$：

$$L^2(\mathbb{R}^n) = \left\{\psi : \mathbb{R}^n \to \mathbb{C} : \int_{\mathbb{R}^n} |\psi(\mathbf{x})|^2\, d^n x < \infty\right\}.$$

内积是:

$$\langle\phi|\psi\rangle = \int_{\mathbb{R}^n} \phi^*(\mathbf{x})\,\psi(\mathbf{x})\, d^n x.$$

非相对论量子力学中的波函数 $\psi(\mathbf{r}, t)$ 是单粒子 $L^2(\mathbb{R}^3)$ 或 $L^2(\mathbb{R}^{3N})$ 的元素。

ℓ^2 与 L^2 的关系

给定 $\{\varphi_n(\mathrm{x})\}$ 的正交归一基 $L^2(\mathbb{R}^n)$，任何波函数 $\psi(\mathrm{x})$ 都可以展开为:

$$\psi(\mathbf{x}) = \sum_{n=1}^{\infty} c_n \varphi_n(\mathbf{x}),$$

其中系数 $c_n = \langle\varphi_n|\psi\rangle$ 满足:

$$\sum_{n=1}^{\infty} |c_n|^2 = \int_{\mathbb{R}^n} |\psi(\mathbf{x})|^2\, d^n x < \infty.$$

这建立了 $L^2(\mathbb{R}^n)$ 与 ℓ^2: 平方可积函数 ψ 之间的一一对应关系，平方可和序列 $\{c_n\}$ 唯一对应于其展开系数的序列。

Orthonormal Vectors

A set of vectors $\{|\phi_n\rangle\}$ in a Hilbert space $\mathcal{H}$ is called **orthonormal** if it satisfies:

$$\langle\phi_m|\phi_n\rangle = \delta_{mn} = \begin{cases} 1 & \text{if } m = n, \\ 0 & \text{if } m \neq n. \end{cases}$$

Equivalently:

- **Orthogonality**: $\langle\phi_m|\phi_n\rangle = 0$ for $m \neq n$ (vectors are perpendicular)
- **Normalization**: $\| |\phi_n\rangle \| = \sqrt{\langle\phi_n|\phi_n\rangle} = 1$ (each vector has unit length)

An orthonormal set that is also **complete** (i.e., spans the entire space $\mathcal{H}$) forms an **orthonormal basis**. Any vector $|\psi\rangle \in \mathcal{H}$ can then be expanded as $|\psi\rangle = \sum_n c_n |\phi_n\rangle$ with $c_n = \langle\phi_n|\psi\rangle$.

Here's your revised subsection with a review of statistical independence added:

2.3 Review of Basic Probability Theory

We need only the minimal foundations of probability theory.

Definition 9 (Probability Space)**.** *A probability space consists of:*

- ***Sample space*** Ω*: The set of all possible outcomes*
- ***Events****: A* σ*-algebra* $\mathcal{F}$ *of subsets of* Ω
- ***Probability measure*** $\mathbb{P}$*: A function* $\mathbb{P} : \mathcal{F} \to [0, 1]$ *satisfying:*
  1. $\mathbb{P}(\Omega) = 1$
  2. *Countable additivity: For disjoint events* A_i*,* $\mathbb{P}(\bigcup_i A_i) = \sum_i \mathbb{P}(A_i)$

Definition 10 (Independence of Events)**.** *Two events* A *and* B *are* ***independent*** *if*

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Equivalently, assuming $\mathbb{P}(B) > 0$*,* A *and* B *are independent if* $\mathbb{P}(A \mid B) = \mathbb{P}(A)$*.*

Definition 11 (Conditional Probability)**.** *For events* A *and* B *with* $\mathbb{P}(B) > 0$*, the* ***conditional probability*** *of* A *given* B *is*

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

The concept of independence generalizes to collections of events. Events $A_1, A_2, \ldots, A_n$ are **mutually independent** if for every subset of indices $I \subseteq \{1, \ldots, n\}$,

$$\mathbb{P}\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} \mathbb{P}(A_i).$$

2.4 Review of Classical Mechanics

2.4.1 The Lagrangian Formalism: A Brief Review

The Lagrangian formulation provides a powerful and elegant framework for classical mechanics, operating in **configuration space** rather than phase space.

正交向量

希尔伯特空间 $\mathcal{H}$ 中的一组向量 $\{|\phi_n\rangle\}$，如果满足以下条件，则称为正交归一基：

$$\langle\phi_m|\phi_n\rangle = \delta_{mn} = \begin{cases} 1 & \text{if } m = n, \\ 0 & \text{if } m \neq n. \end{cases}$$

等价地：

- 正交性： $\langle\phi_m|\phi_n\rangle = 0$ 对于 $m \neq n$ (向量互相垂直)
- 归一化：$\| |\phi_n\rangle \| = \sqrt{\langle\phi_n|\phi_n\rangle = 1}$ (每个向量长度为1)

一个既是正交归一集又完备（即张成整个空间 $\mathcal{H}$）的集合构成一个正交归一基。任何向量 $|\psi\rangle \in \mathcal{H}$ 都可以用 $|\psi\rangle = \sum_n c_n |\phi_n\rangle$ 表示为 $c_n = \langle\phi_n|\psi\rangle$。

这是您修订的子章节，增加了统计独立性的说明：

2.3 基础概率论回顾

我们只需要概率论的最小基础。

定义 *9*（概率空间）。一个概率空间包括：

- 样本空间 Ω：所有可能结果的集合
- 事件：一个 σ-代数 $\mathcal{F}\,\Omega$ 的子集
- 概率测度 $\mathbb{P}$：一个满足以下条件的函数 $\mathbb{P} : \mathcal{F} \to [0, 1]$
  1. $\mathbb{P}(\Omega) = 1$
  2. 可数可加性：对于互不相容的事件 A_i、$\mathbb{P}(\bigcup_i A_i) = \sum_i \mathbb{P}(A_i)$

定义 10（事件的独立性）。如果两个事件 A 和 B 满足以下条件，则称它们是独立的

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

等价地，假设 $\mathbb{P}(B) > 0$、A 和 B 是独立的 *if* $\mathbb{P}(A \mid B) = \mathbb{P}(A)$

定义*11*（条件概率）。对于事件 A 和 B，如果 $\mathbb{P}(B) > 0$成立，则 A 在 B 条件下的条件概率是

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

独立性概念可以推广到事件集合。事件 A_1 A_2 … A_n、$I \subseteq \{1, \ldots, n\}$、等如果对于任意指标子集，

$$\mathbb{P}\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} \mathbb{P}(A_i).$$

2.4 经典力学复习

2.4.1 拉格朗日形式：简要回顾

拉格朗日公式为经典力学提供了一个强大而优雅的框架，该框架在构型空间而非相空间中运行。

The Configuration Space Description

For a system of N particles in three-dimensional space, the configuration is specified by $3N$ generalized coordinates. These can be taken as the Cartesian coordinates of each particle:

$$\mathbf{q} = (\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) \in \mathbb{R}^{3N},$$

where each $\mathbf{r}_\alpha = (x_\alpha, y_\alpha, z_\alpha)$ is the position vector of particle α. The full state at a given time also includes the velocities $\dot{\mathbf{q}} = (\dot{\mathbf{r}}_1, \dot{\mathbf{r}}_2, \dots, \dot{\mathbf{r}}_N)$, giving a point in the **tangent bundle** of the configuration space $\mathbb{R}^{3N}$.

The dynamics are governed by the **Lagrangian function**, a map from the tangent bundle of configuration space to the real numbers:

$$L : \mathbb{R}^{6N} \to \mathbb{R}, \quad (\mathbf{q}, \dot{\mathbf{q}}) \mapsto L(\mathbf{q}, \dot{\mathbf{q}}, t),$$

which typically represents the difference between kinetic and potential energy.

The time evolution follows from **Hamilton's principle of stationary action**:

$$\delta S = \delta \int_{t_1}^{t_2} L(\mathbf{q}, \dot{\mathbf{q}}, t)\, dt = 0, \tag{1}$$

which yields the **Euler-Lagrange equations** for each degree of freedom:

$$\boxed{\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0 \qquad (i = 1, \dots, 3N)}. \tag{2}$$

- **Standard form for N particles**: The Lagrangian is typically the difference between total kinetic and potential energy:

$$L(\mathbf{r}_1, \dots, \mathbf{r}_N, \dot{\mathbf{r}}_1, \dots, \dot{\mathbf{r}}_N) = \sum_{\alpha=1}^{N} \frac{1}{2} m_\alpha \dot{\mathbf{r}}_\alpha^2 - V(\mathbf{r}_1, \dots, \mathbf{r}_N).$$

The Euler-Lagrange equations for coordinate x_α then give $m_\alpha \ddot{x}_\alpha = -\partial V / \partial x_\alpha$, recovering Newton's second law.

- **Canonical momentum**: The momentum conjugate to the coordinate q_i (which could be, e.g., x_α) is:

$$p_i := \frac{\partial L}{\partial \dot{q}_i}.$$

For the Cartesian coordinates above, $\mathbf{p}_\alpha = m_\alpha \dot{\mathbf{r}}_\alpha$.

- **Legendre transformation to Hamiltonian**: The Hamiltonian is obtained via:

$$H(\mathbf{q}, \mathbf{p}, t) = \sum_{i=1}^{3N} p_i \dot{q}_i - L(\mathbf{q}, \dot{\mathbf{q}}, t),$$

where the velocities $\dot{q}_i$ are expressed in terms of $(\mathbf{q}, \mathbf{p})$ using the momentum definition. For the standard particle Lagrangian, this yields:

$$H = \sum_{\alpha=1}^{N} \frac{\mathbf{p}_\alpha^2}{2m_\alpha} + V(\mathbf{r}_1, \dots, \mathbf{r}_N).$$

This formulation—based on a single scalar function L and the principle of stationary action—provides a coordinate-independent foundation for mechanics and directly motivates the path integral approach to quantization.

构型空间描述

对于三维空间中的 N 个粒子系统，构型由 $3N$ 广义坐标指定。这些坐标可以是每个粒子的笛卡尔坐标：

$$\mathbf{q} = (\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) \in \mathbb{R}^{3N},$$

其中每个 $\mathbf{r}_\alpha = (x_\alpha, y_\alpha, z_\alpha)$ 是粒子 α 的位置矢量。在给定时间下的完整状态还包括速度 $\dot{\mathbf{q}} = (\dot{\mathbf{r}}_1, \dot{\mathbf{r}}_2, \dots, \dot{\mathbf{r}}_N)$，从而在构型空间的切空间 $\mathbb{R}^{3N}$ 中给出一个点。

动力学由拉格朗日函数支配，该函数是从构型空间的切空间到实数的映射：

$$L : \mathbb{R}^{6N} \to \mathbb{R}, \quad (\mathbf{q}, \dot{\mathbf{q}}) \mapsto L(\mathbf{q}, \dot{\mathbf{q}}, t),$$

它通常表示动能与势能之差。

时间演化遵循哈密顿原理：

$$\delta S = \delta \int_{t_1}^{t_2} L(\mathbf{q}, \dot{\mathbf{q}}, t)\, dt = 0, \tag{1}$$

由此得到每个自由度的欧拉-拉格朗日方程：

$$\boxed{\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0 \qquad (i = 1, \dots, 3N)}. \tag{2}$$

- 标准形式：N个粒子的情形：拉格朗日量通常是总动能与势能之差：

$$L(\mathbf{r}_1, \dots, \mathbf{r}_N, \dot{\mathbf{r}}_1, \dots, \dot{\mathbf{r}}_N) = \sum_{\alpha=1}^{N} \frac{1}{2} m_\alpha \dot{\mathbf{r}}_\alpha^2 - V(\mathbf{r}_1, \dots, \mathbf{r}_N).$$

对于坐标 x_α的欧拉-拉格朗日方程给出 $m_\alpha \ddot{x}_\alpha = -\partial V / \partial x_\alpha$，从而恢复牛顿第二定律。

- 正则动量：与坐标 q_i （例如 x_α）共轭的动量是：

$$p_i := \frac{\partial L}{\partial \dot{q}_i}.$$

对于上述笛卡尔坐标， $\mathbf{p}_\alpha = m_\alpha \dot{\mathbf{r}}_\alpha$。

- **勒让德变换到哈密顿量：哈密顿量通过以下方式获得：**

$$H(\mathbf{q}, \mathbf{p}, t) = \sum_{i=1}^{3N} p_i \dot{q}_i - L(\mathbf{q}, \dot{\mathbf{q}}, t),$$

其中速度 $\dot{q}_i$ 使用动量定义表示为 (q,p)。对于标准粒子的拉格朗日量，这给出了：

$$H = \sum_{\alpha=1}^{N} \frac{\mathbf{p}_\alpha^2}{2m_\alpha} + V(\mathbf{r}_1, \dots, \mathbf{r}_N).$$

这种公式——基于单一标量函数 L 和平稳作用原理——为力学提供了坐标无关的基础，并直接推动了路径积分方法在量子化中的应用。

2.4.2 The Classical Hamiltonian Formalism: A Review

Before examining how quantum mechanics modifies classical mechanics, it is essential to recall the core structure of the Hamiltonian formulation of classical dynamics.

The Phase Space Description

For a system of N particles in three-dimensional space, the complete state is specified by $3N$ generalized position coordinates and their $3N$ conjugate momenta. Using Cartesian coordinates, we write:

$$\mathbf{q} = (\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N) \in \mathbb{R}^{3N}, \quad \mathbf{p} = (\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_N) \in \mathbb{R}^{3N},$$

where $\mathbf{r}_\alpha = (x_\alpha, y_\alpha, z_\alpha)$ is the position of particle α and $\mathbf{p}_\alpha = (p_{\alpha x}, p_{\alpha y}, p_{\alpha z})$ is its conjugate momentum. The instantaneous state is a point $(\mathbf{q}, \mathbf{p})$ in a $6N$-dimensional **phase space** $\mathbb{R}^{6N}$.

The dynamics are governed by the **Hamiltonian function**, a map from phase space to the real numbers:

$$H : \mathbb{R}^{6N} \to \mathbb{R}, \quad (\mathbf{q}, \mathbf{p}) \mapsto H(\mathbf{q}, \mathbf{p}),$$

which typically represents the total energy of the system. The time evolution is given by **Hamilton's canonical equations**:

$$\boxed{\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i} \qquad (i = 1, \ldots, 3N)}. \tag{3}$$

- **Standard form for N particles**: For a system with conservative forces, the Hamiltonian is:

$$H(\mathbf{r}_1, \ldots, \mathbf{r}_N, \mathbf{p}_1, \ldots, \mathbf{p}_N) = \sum_{\alpha=1}^{N} \frac{\mathbf{p}_\alpha^2}{2m_\alpha} + V(\mathbf{r}_1, \ldots, \mathbf{r}_N).$$

Hamilton's equations then yield Newton's second law for each particle. For particle α:

$$\dot{\mathbf{r}}_\alpha = \frac{\partial H}{\partial \mathbf{p}_\alpha} = \frac{\mathbf{p}_\alpha}{m_\alpha}, \quad \dot{\mathbf{p}}_\alpha = -\frac{\partial H}{\partial \mathbf{r}_\alpha} = -\nabla_{\mathbf{r}_\alpha} V \quad \Rightarrow \quad m_\alpha \ddot{\mathbf{r}}_\alpha = -\nabla_{\mathbf{r}_\alpha} V.$$

This elegant structure in phase space—with its clear separation of coordinates and momenta—provides the direct mathematical foundation for the **canonical quantization procedure** that follows.

2.4.2 经典哈密顿正则形式：回顾

在考察量子力学如何修正经典力学之前，有必要回顾经典动力学哈密顿公式的核心结构。

相空间描述

对于三维空间中的 N 个粒子系统，其完整状态由 $3N$ 个广义位置坐标及其 $3N$ 共轭动量指定。使用笛卡尔坐标，我们写：

$$\mathbf{q} = (\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N) \in \mathbb{R}^{3N}, \quad \mathbf{p} = (\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_N) \in \mathbb{R}^{3N},$$

其中 $\mathbf{r}_\alpha = (x_\alpha, y_\alpha, z_\alpha)$ 是粒子 α 的位置， $\mathbf{p}_\alpha = (p_{\alpha x}, p_{\alpha y}, p_{\alpha z})$ 是其共轭动量。瞬时状态是 $6N$维相空间 $\mathbb{R}^{6N}$中的一个点。

动力学由哈密顿函数支配，该函数是从相空间到实数的映射：

$$H : \mathbb{R}^{6N} \to \mathbb{R}, \quad (\mathbf{q}, \mathbf{p}) \mapsto H(\mathbf{q}, \mathbf{p}),$$

它通常表示系统的总能量。时间演化由哈密顿正则方程给出：

$$\boxed{\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i} \qquad (i = 1, \ldots, 3N)}. \tag{3}$$

- N个粒子的标准形式：对于一个有保守力的系统，哈密顿量为：

$$H(\mathbf{r}_1, \ldots, \mathbf{r}_N, \mathbf{p}_1, \ldots, \mathbf{p}_N) = \sum_{\alpha=1}^{N} \frac{\mathbf{p}_\alpha^2}{2m_\alpha} + V(\mathbf{r}_1, \ldots, \mathbf{r}_N).$$

哈密顿方程则给出每个粒子的牛顿第二定律。对于粒子 α：

$$\dot{\mathbf{r}}_\alpha = \frac{\partial H}{\partial \mathbf{p}_\alpha} = \frac{\mathbf{p}_\alpha}{m_\alpha}, \quad \dot{\mathbf{p}}_\alpha = -\frac{\partial H}{\partial \mathbf{r}_\alpha} = -\nabla_{\mathbf{r}_\alpha} V \quad \Rightarrow \quad m_\alpha \ddot{\mathbf{r}}_\alpha = -\nabla_{\mathbf{r}_\alpha} V.$$

这个在相位空间中的优雅结构——其坐标和动量清晰分离——为后续的规范量子化程序提供了直接数学基础。

3 What is Quantum Mechanics: Two Approaches to the Same Topic

3.1 Two Distinct Approaches to Quantum Mechanics

Remarkable Duality in Understanding Quantum Theory

Quantum mechanics can be approached through **two fundamentally different perspectives**: one emphasizing physical intuition and empirical foundations, the other emphasizing abstract mathematical structure and algorithmic principles.

Physical Space Approach (Quantization)
Historically first, physics-driven, not unified

↓ Historical development

Abstract/Algorithmic Approach
Modern, mathematics-driven, structurally clean

The Physical Space Approach (Quantization)

From Classical to Quantum

This approach, historically the first to be developed, emphasizes the **physical content** of quantum mechanics through the process of *quantization*.

- **Historical precedence**: Developed during the quantum revolution (1900-1927)
- **Methodology**: Classical system $\xrightarrow{\text{quantization}}$ Quantum system
- **Core philosophy**: Quantum mechanics as *modification* of classical physics
- **Diverse techniques**: Not a single method but a collection of approaches:
  - Canonical quantization: $q \to \hat{q}$, $p \to \hat{p}$, $[\hat{q}, \hat{p}] = i\hbar$
  - Path integral quantization: Sum over histories (Feynman)
  - Geometric quantization: Symplectic geometry methods
  - Deformation quantization: Star product formalism
  - Stochastic quantization: Random process methods

Physical Approach Flow: $\boxed{\text{Classical Physics}} \xrightarrow[\text{quantization}]{} \boxed{\text{Quantum Physics}}$

The Abstract/Algorithmic Approach

Quantum as Information Processing Framework

This modern approach emphasizes the **mathematical structure** of quantum mechanics, treating it as an abstract computational or information-theoretic framework. It has become dominant in quantum information theory and computer science.

3 什么是量子力学：两种探讨同一主题的方法

3.1 量子力学的两种不同方法

理解量子理论的显著二象性

量子力学可以通过两种根本不同的视角来研究：一种强调物理直觉和经验基础，另一种强调抽象的数学结构和算法原理。

物理空间方法（量子化）历史上首先出现，由物理学驱动，未统一

↓ 历史发展

抽象/算法方法
现代、数学驱动、结构清晰

物理空间方法（量子化）

从经典到量子

这种方法是历史上最早发展起来的，通过量子化过程强调量子力学的物理内容。

- 历史先例：在量子革命期间（1900-1927年）发展起来的
- 方法论：经典系统量子化 ⟶量子系统
- 核心思想：量子力学作为经典物理的修正
- 多样化技术：不是单一方法，而是一系列方法：
  - 正则量子化： $q \to \hat{q}$, $p \to \hat{p}$, $[\hat{q}, \hat{p}] = i\hbar$
  - 路径积分量子化：历史求和（费曼）
  - 几何量子化：辛几何方法
  - 变形量子化：星积形式化
  - 随机量子化：随机过程方法

物理方法流程： $\boxed{\text{经典物理}} \xrightarrow[\text{量子化}]{} \boxed{\text{量子物理}}$

抽象/算法方法

量子作为信息处理框架

这种现代方法强调量子力学的数学结构，将其视为一个抽象的计算或信息论框架。它在量子信息理论和计算机科学中已成为主流。

- **Historical emergence**: Gained prominence in late 20th century with quantum information science
- **Methodology**: Postulates → Mathematical structure → Applications
- **Key insight**: Quantum mechanics as *generalized probability theory*
- **Revolutionary feature**: Can formulate entire quantum theory **without ever mentioning the Schrödinger equation for specific physical systems**
- **Main mathematical tools**:
  - **Linear algebra**: Hilbert spaces, eigenvectors, spectral theorem
  - **Operator theory**: C^*-algebras, von Neumann algebras, completely positive maps
  - **Information theory**: Entropy measures, channel capacities, resource theories
  - **Category theory**: Monoidal categories, string diagrams

Abstract Approach Flow: $\boxed{\text{Mathematical Postulates}}$ → $\boxed{\text{Algebraic Structure}}$ → $\boxed{\text{Physical/Computational Applications}}$

Comparison of Approaches

Aspect	Physical (Quantization)	Abstract/Algorithmic
Starting point	Classical physics	Mathematical postulates
Emphasis	Physical interpretation	Structural properties
Historical period	1900-1927 (development)	1930-present (refinement)
Dominant in	Traditional physics education	Quantum information, computer science
Key equation	Schrödinger equation	Postulates + linear algebra
Main tools	Differential equations, operators	Linear algebra, operator theory, information theory

Pedagogical Strategy for This Course

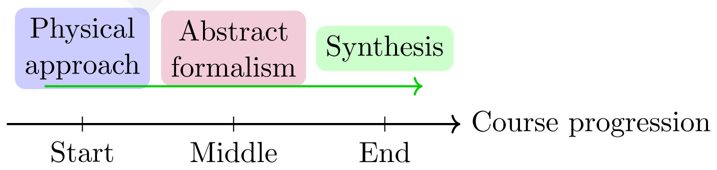
Hybrid Approach: Best of Both Worlds

We adopt a **hybrid pedagogical strategy** that combines the strengths of both approaches:

Phase 1: Build physical intuition: Start with wave mechanics and quantization

Phase 2: Develop abstract formalism: Introduce Hilbert space, Dirac notation, operator methods

Phase 3: Synthesize approaches: Show equivalence and complementary insights



Why This Hybrid Approach?

Balancing Intuition and Rigor

The physical approach provides **concrete intuition** while the abstract approach provides **mathematical clarity**. Together they offer a complete understanding.

Advantages of starting physical:

- 历史出现：在20世纪后期随着量子信息科学的兴起而获得 prominence
- 方法论：公设 → 数学结构 → 应用
- 核心观点：量子力学作为广义概率论
- **革命性特征：可以完全不提及薛定谔方程，就能构建整个量子理论的具体物理系统**
- **主要数学工具：**
  - 线性代数：希尔伯特空间、特征向量、谱定理
  - 算子理论：C^*-代数、冯·诺依曼代数、完全正映射
  - 信息论：熵度量、信道容量、资源理论
  - 同调论：幺半群范畴、字符串图

Abstract Approach Flow: $\boxed{\text{Mathematical Postulates}}$ → $\boxed{\text{Algebraic Structure}}$ → $\boxed{\text{Physical/Computational Applications}}$

方法比较

方面	物理（量化）	抽象/算法
起点	经典物理学	数学公设
重点	物理意义	结构特性
历史时期	1900-1927（发展）	1930-至今（完善）
主导领域	传统物理学教育	量子信息，计算机科学
关键方程	薛定谔¨方程	公设 + 线性代数
主要工具	微分方程，算子	线性代数，算子理论，信息论

本课程的教学策略

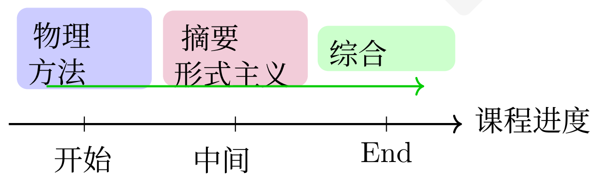
混合方法：取两者之长

我们采用一种混合教学策略，结合了两种方法的优势：

第一阶段：建立物理直觉：从波动力学和量子化开始

第二阶段：开发抽象形式化：引入希尔伯特空间、狄拉克符号、算子方法

第三阶段：综合方法：展示等价性和互补性见解



为何采用这种混合方法?

平衡直觉与严谨

物理方法提供具体直观，而抽象方法提供数学清晰。两者结合，提供完整理解。

开始进行体育活动的优势：

- Connects to experimental physics students already know
- Provides concrete problem-solving methods
- Historical development mirrors conceptual understanding

Advantages of abstract formalism:

- Reveals logical structure independent of specific systems
- Essential for quantum information and computation
- Clean mathematical framework for advanced topics

3.2 The First Worldview: Quantization, From Classical to Quantum

The Physical Space Approach Foundation

To understand the physical space approach to quantum mechanics, we begin with classical mechanics concepts. Quantum theory in this view emerges through quantization of classical systems.

Three Types of Physical Space in Classical Mechanics

In classical mechanics, we distinguish three distinct but related spaces:

Space 1: Physical Spacetime:

- For non-relativistic quantum mechanics: $\mathbb{R}^3 \times \mathbb{R} \cong \mathbb{R}^4$
- Represents actual physical space (x, y, z) and time t
- Where particles and fields physically exist

Space 2: Configuration Space:

- Used in Lagrangian mechanics
- For N particles: $\mathbb{R}^{3N}$ degrees of freedom
- Generalized coordinates: $\mathbf{q}_i$, $i \in \{1, 2, \ldots, N\}$
- Describes all possible positions simultaneously

Space 3: Phase Space:

- Used in Hamiltonian mechanics
- For N particles: $\mathbb{R}^{6N}$ dimensions
- Generalized positions $\mathbf{q}_i$ and conjugate momenta $\mathbf{p}_i$
- Describes both positions and momenta simultaneously

Physical Space	Configuration Space	Phase Space
$\mathbb{R}^3 \times \mathbb{R}$	$\mathbb{R}^{3N}$	$\mathbb{R}^{6N}$
(x, y, z, t)	$\{\mathbf{q}_1, \ldots, \mathbf{q}_N\}$	$\{(\mathbf{q}_i, \mathbf{p}_i)\}$

- 连接到实验物理学生已经知道
- 提供具体的解题方法
- 历史发展反映概念理解

抽象公式的优势：

- 揭示独立于具体系统的逻辑结构
- 对量子信息和计算至关重要
- 为高级主题提供简洁的数学框架

3.2 第一个世界观：量子化，从经典到量子

物理空间方法基础

要理解量子力学的物理空间方法，我们从经典力学概念开始。在这个观点中，量子理论通过经典系统的量子化而出现。

经典力学中的三种物理空间

在经典力学中，我们区分了三个既不同又相关的空间：

空间 1：物理时空：

- 对于非相对论量子力学： $\mathbb{R}^3 \times \mathbb{R} \cong \mathbb{R}^4$
- 表示实际物理空间 (x, y, z) 和时间 t
- 粒子场在此物理存在

空间 2：配置空间：

- 用于拉格朗日力学
- 对于 N 粒子： $\mathbb{R}^{3N}$ 自由度
- 广义坐标：q_i, $i \in \{1, 2, \ldots, N\}$
- 同时描述所有可能的位置

空间3：相空间：

- 用于哈密顿力学
- 对于 N 粒子： $\mathbb{R}^{6N}$ 维度
- 广义坐标 q_i 和共轭动量 p_i
- 同时描述位置和动量

物理空间	配置空间	相空间
$\mathbb{R}^3 \times \mathbb{R}$	$\mathbb{R}^{3N}$	$\mathbb{R}^{6N}$
(x, y, z, t)	$\{q1, \ldots, q_N\}$	$\{(\mathrm{q}_i, \mathrm{p}_i)\}$

The Hamiltonian Foundation of Quantum Mechanics

Hamiltonian Formalism as the Bridge

Conventionally, quantum mechanics in the physical space approach employs the Hamiltonian formalism. A key feature of this worldview is that every quantum system is thought to arise from quantizing some classical system.

- **Philosophical foundation**: Quantum mechanics as *modified* classical mechanics
- **Even for non-classical observables**: When quantum observables lack classical analogues (e.g., spin), one often inserts corresponding terms in the classical Hamiltonian
- **Methodology**: Builds on classical intuition, then introduces fundamentally non-classical features
- **Wave function location**: Despite using Hamiltonian formalism, the wave function ψ lives in configuration space, not phase space

Transition to Wave Mechanics: The Wavefunction as the State

The Radical Change in the State Concept

The physical space approach introduces a fundamental shift: the quantum state replaces the classical phase-space point. This state is not a set of numbers but a complex-valued field—the wavefunction $\psi(\mathbf{r}, t)$.

- **Classical State Space**: For a single particle, the instantaneous state is $(\mathbf{x}, \dot{\mathbf{x}})$ or $(\mathbf{q}, \mathbf{p})$, living in $\mathcal{S} \cong \mathbb{R}^6$. Observables are functions on $\mathcal{S}$.
- **Quantum State Space**: The state is $\psi(\mathbf{r}, t) \in L^2(\mathbb{R}^3)$, a function in configuration space. The collection of all such functions forms a Hilbert space $\mathcal{H}$—a much richer mathematical structure.
- **Information Content**: While a classical state specifies exact values, $\psi(\mathbf{r}, t)$ encodes probabilistic information about all possible measurements.

Physical Interpretation: Born's Rule

From Field to Probability

The wavefunction's physical meaning is provided by the Born rule, which connects the complex field to measurable probabilities.

$$\boxed{P(\text{particle in volume } V) = \int_V |\psi(\mathbf{r}, t)|^2 \, d^3\mathbf{r}}$$

- **Probability Density**: $\rho(\mathbf{r}, t) := |\psi(\mathbf{r}, t)|^2$ acts as a probability density function.
- **Normalization**: States are normalized: $\int_{\mathbb{R}^3} |\psi|^2 \, d^3\mathbf{r} = 1$, ensuring total probability equals 1.
- **Inherent Randomness**: Unlike classical mechanics, QM predictions are fundamentally probabilistic; the wavefunction embodies this indeterminacy.

量子力学的哈密顿基础

哈密顿形式主义作为桥梁

通常情况下，物理空间方法中的量子力学采用哈密顿形式主义。这种世界观的一个关键特征是，每个量子系统被认为是从某个经典系统中量子化而来的。

- 哲学基础：量子力学是修正后的经典力学
- 即使对于非经典可观测量：当量子可观测量缺乏经典对应物（例如自旋）时，人们通常在经典哈密顿量中插入相应的项
- 方法论：基于经典直觉，然后引入根本非经典的特性
- 波函数位置：尽管使用哈密顿形式主义，波函数 ψ 生活在配置空间，而不是相空间

从波动力学到波函数：波函数作为状态

状态概念的根本性转变

物理空间方法引入了一个根本性转变：量子状态取代了经典相空间点。这个状态不是一组数字，而是一个复值场——波函数ψ(r t)。

- 经典状态空间：对于单个粒子，瞬时状态是 (x ˙x) 或 (q, p)，存在于 $\mathcal{S} \cong \mathbb{R}^6$。可观测量是作用在 $\mathcal{S}$ 上的函数。
- 量子态空间：状态是 $\psi(\mathbf{r}, t) \in L^2(\mathbb{R}^3)$，一个在配置空间中的函数。 The 所有这类函数的集合构成希尔伯特空间 $\mathcal{H}$——一种更丰富的数学结构。
- 信息内容：虽然经典状态指定精确值，但 ψ(r t) 编码了关于所有可能测量的概率信息。

物理诠释：玻恩法则

从场到概率

波函数的物理意义由玻恩法则提供，该法则将复数场与可测量的概率联系起来。

$$\boxed{P(\text{particle in volume } V) = \int_V |\psi(\mathbf{r}, t)|^2 \, d^3\mathbf{r}}$$

- 概率密度： $\rho(\mathbf{r}, t) := |\psi(\mathbf{r}, t)|^2$ 充当概率密度函数。
- 归一化：状态被归一化： $\int_{\mathbb{R}^3} |\psi|^2 \, d^3\mathbf{r} = 1$，确保总概率等于1。
- 固有随机性：与经典力学不同，量子力学的预测本质上是概率性的；波函数体现了这种不确定性。

Contrast with Classical Fields

Wavefunction vs. Physical Fields

The wavefunction is *not* analogous to classical fields like the electromagnetic field—it represents a deeper conceptual shift.

Quantum Mechanics (Wavefunction)	Classical Field Theory (e.g., EM)
ψ is the physical state of the particle	Fields are produced by charges/currents
Particle $\Longleftrightarrow$ Wavefunction (dual nature)	Source $\longrightarrow$ Field (causal relationship)
Born rule gives probabilistic interpretation	Field strength gives force/energy density

- No Source Problem: In standard QM, ψ has no known "source" equation like $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$; it is fundamental.
- Quantum Field Theory Perspective: In QFT, this distinction blurs—fields become operator-valued and particles are excitations. Yet in non-relativistic QM, the wavefunction remains the primary description.

Classical Mechanics: States as Measurable Quantities

- The state $(\mathbf{q}, \mathbf{p}) \in \mathcal{S}$ (a phase space point) is directly measurable.
- Observables are functions $f : \mathcal{S} \to \mathbb{R}$; choosing the identity function $f_{\text{id}}(\mathbf{q}, \mathbf{p}) = (\mathbf{q}, \mathbf{p})$ gives the state itself. *Thus in classical physics, states in themselves are directly measurable. This is not the case in quantum physics.*
- State space $\mathcal{S}$ and observable space are essentially the same mathematical object.

Quantum Mechanics: The Radical Separation

- States $\psi \in \mathcal{H}$ (wavefunctions) are not directly measurable.
- Observables are operators $\hat{A} : \mathcal{H} \to \mathcal{H}$ (linear maps from states to states).
- States are abstract vectors; observables are defined independently as Hermitian operators.
- The relationship between state and measurement outcome is probabilistic:

$$P(a|\psi) = \langle \psi | P_a | \psi \rangle$$

where a is an eigenvalue of $\hat{A}$ and P_a is the projection onto the corresponding eigenspace.

Philosophical Implication: The Abstract Construction of QM

- In QM, one must first construct abstract states (the Hilbert space $\mathcal{H}$), then define operators representing observables, and only then combine them via the Born rule to compute probabilities.
- The state-observable relationship becomes mediated by mathematical formalism rather than direct measurement.
- This represents a fundamental epistemological shift: knowledge of a quantum system is inherently inferential and probabilistic, derived from the interaction between abstract states and operator-valued observables.

与经典场的对比

波函数与物理场

波函数不同于电磁场等经典场——它代表了一种更深层次的概念转变。

量子力学（波函数）	经典场论（例如电磁场）
ψ 是粒子的物理状态	场是由电荷/电流产生的
粒子 $\Longleftrightarrow$ 波函数（波粒二象性）	源 $\longrightarrow$ 场（因果关系）
玻恩规则给出概率解释	场强给出力/能量密度

- 没有源问题：在标准量子力学中， ψ 没有已知的“源”方程，就像 $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$; 它一样基本。基本。
- 量子场论视角：在量子场论中，这种区别变得模糊——场成为算子值的，粒子是激发。然而在非相对论量子力学中，波函数仍然是主要的描述。

经典力学：状态作为可测量的量

- 状态（q，p） $\in \mathcal{S}$ （相空间点）是直接可测量的。
- 可观测量是函数 f： $\mathcal{S} \to \mathbb{R}$；选择恒等函数 $f_{\text{id}}(\mathbf{q}, \mathbf{p}) = (\mathbf{q}, \mathbf{p})$ 给出状态本身。因此，在经典物理学中，状态本身是直接可测量的。在量子物理学中则不是这样。
- 状态空间 $\mathcal{S}$ 与可观测空间本质上是一个数学对象。

量子力学：根本的分离

- 状态 $\psi \in \mathcal{H}$ (波函数)不能直接测量。
- 可观测量是算子 A： $\mathcal{H} \to \hat{\mathcal{H}}$ (从状态到状态的线性映射)。
- 状态是抽象向量；可观测量被独立定义为厄米算子。
- 状态与测量结果之间的关系是概率性的：

$$P(a|\psi) = \langle \psi | P_a | \psi \rangle$$

ˆ其中 a 是 A 的一个特征值，而 P_a 是对应特征子空间的投影。

哲学意义：量子力学的抽象构建

- 在量子力学中，必须首先构建抽象态（希尔伯特空间 $\mathcal{H}$），然后定义表示可观测量的算符，最后才通过玻恩法则将它们组合起来计算概率。
- 态-可观测量关系通过数学形式化而不是直接测量而变得中介化。
测量。
- 这代表了一种根本的认识论转变：对量子系统的知识本质上具有推论性和概率性，源于抽象态与算符值可观测量之间的相互作用。

The Quantization Procedure

From Classical to Quantum: The Quantization Recipe

The core of the physical space approach is the quantization procedure that transforms classical systems into quantum ones.

Step 1: Start with classical Hamiltonian:

$$H(\mathbf{q}, \mathbf{p}) = \frac{\mathbf{p}^2}{2m} + V(\mathbf{q})$$

Step 2: Promote variables to operators (canonical quantization):

$$\mathbf{q} \to \hat{\mathbf{q}}, \quad \mathbf{p} \to \hat{\mathbf{p}}, \quad \text{with} \quad [\hat{q}_i, \hat{p}_j] = i\hbar\delta_{ij}$$

Step 3: Construct quantum observables:

- Position: $\hat{x}$ (multiplication operator)
- Momentum: $\hat{p}_x = -i\hbar\frac{\partial}{\partial x}$
- Angular momentum: $\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$
- Hamiltonian: $\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{q}})$

Step 4: Introduce wave function:

$$\psi(\mathbf{q}, t) \in L^2(\mathbb{R}^{3N})$$

defined on configuration space despite Hamiltonian formulation

Quantization Flow:

$$\boxed{\text{Classical } H(\mathbf{q},\mathbf{p})}$$
$$\downarrow \text{quantization}$$
$$\boxed{\text{Quantum } \hat{H}}$$
$$\downarrow$$
$$\boxed{\text{Schrödinger Equation}}$$
$$\downarrow$$
$$\boxed{\text{Wave function } \psi(\mathbf{q}, t)}$$

Beyond Classical Intuition: Intrinsic Quantum Properties

Where Classical Intuition Fails

Despite its classical foundation, quantum mechanics introduces phenomena with no classical counterparts:

- **Spin**: Intrinsic angular momentum (not orbital motion)
- **Isospin**: Internal symmetry of strong interactions
- **Parity**: Discrete spatial inversion symmetry
- **Quantum entanglement**: Non-classical correlations violating Bell inequalities
- **Superposition**: States existing in multiple configurations simultaneously

量子化过程

从经典到量子：量子化配方

物理空间方法的核心是量子化程序，该程序将经典系统转化为量子系统。

步骤1：从经典哈密顿量开始：

$$H(\mathbf{q}, \mathbf{p}) = \frac{\mathbf{p}^2}{2m} + V(\mathbf{q})$$

步骤2：将变量提升为算子（正则量子化）：

$$\mathbf{q} \to \hat{\mathbf{q}}, \quad \mathbf{p} \to \hat{\mathbf{p}}, \quad \text{with} \quad [\hat{q}_i, \hat{p}_j] = i\hbar\delta_{ij}$$

第 3 步：构建量子观测量：

- 位置：ˆx (乘法算符)
- 动量： $\hat{p}_x = -i\hbar\frac{\partial}{\partial x}$ —
- 角动量： $\mathbf{L} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$ ˆ
- 哈密顿量： $\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{q}})$

第 4 步：引入波函数：

$$\psi(\mathbf{q}, t) \in L^2(\mathbb{R}^{3N})$$

在配置空间上定义，尽管采用哈密顿形式化

量化流：

$$\boxed{\text{Classical } H(\mathbf{q},\mathbf{p})}$$
$$\downarrow \text{quantization}$$
$$\boxed{\text{Quantum } \hat{H}}$$
$$\downarrow$$
$$\boxed{\text{Schrödinger Equation}}$$
$$\downarrow$$
$$\boxed{\text{Wave function } \psi(\mathbf{q}, t)}$$

超越经典直觉：内在量子性质

经典直觉失效之处

尽管其基础是经典的，量子力学引入了没有经典对应现象的：

- 自旋：内禀角动量（非轨道运动）
- 同位旋：强相互作用的内部对称性
- 宇称：离散的空间反演对称性
- 量子纠缠：违反贝尔不等式的非经典关联
- 叠加：同时存在于多种配置中的状态

Physical Space Approach: Strengths and Limitations

- **Strengths**:
  - Builds directly on familiar classical concepts
  - Provides concrete calculation methods for specific systems
  - Historical development mirrors conceptual understanding
  - Natural for problems with clear classical limits
- **Quantum Surprises**:
  - Relevant functional space: Wave function lives in configuration space, not physical space
  - Measurement problem: How/when does quantum $\rightarrow$ classical transition occur?
  - Wave function collapse: How/when does ψ collapse?
  - Non-locality: Entanglement suggests connections beyond spacetime
  - Conceptual tension: Quantum systems from classical roots, yet fundamentally different

3.3 The Second Worldview: A Compact Résumé of the Quantum Algorithm

In this section we provide the minimum content of quantum mechanics condensed into four fundamental postulates. This can be done most elegantly using the Dirac-Hilbert space formalism. We will only review below the minimum mathematics required to understand the postulates. A more detailed exposition of the Dirac-Hilbert formulation will be given in later lectures.

Pedagogical Note

This section presents quantum mechanics in its *most abstract form*. Do not be alarmed if some concepts seem unfamiliar—we will build physical intuition gradually. Think of this as the “bare skeleton” of quantum theory; we will add the “flesh” of physical interpretation throughout the course.

3.3.1 Postulates of Quantum Mechanics

The Four-Postulate Algorithm

These four postulates provide a complete recipe for describing quantum systems. Remarkably, they contain no explicit mention of waves, particles, or the Schrödinger equation—these will emerge as consequences.

Postulate I: The Quantum State

Postulate 1 (State Postulate)**.** *Each quantum system S is associated with a Hilbert space $\mathcal{H}_S$. The instantaneous state of the system at time t is completely described by a vector $|\psi(t)\rangle \in \mathcal{H}_S$ with unit norm:*

$$\boxed{\langle\psi(t)|\psi(t)\rangle = 1}$$

Remark 3.1. *The state vector $|\psi(t)\rangle$ contains* all possible information *about the system. Different physical situations correspond to different Hilbert spaces (e.g., spin-1/2 system: $\mathbb{C}^2$, particle in 1D: $L^2(\mathbb{R})$).*

物理空间方法：优势与局限性

- **优势：**
  - 直接基于于熟悉的经典概念
  - 为特定系统提供具体的计算方法
  - 历史发展反映了概念理解
  - 对于具有明确经典极限的问题很自然
- **量子惊喜：**
  - 相关函数空间：波函数存在于构型空间，而非物理空间
  - 测量问题：量子向经典过渡如何/何时发生?
  - 波函数坍缩：ψ 坍缩如何/何时发生?
  - 非局域性：纠缠暗示着超越时空的联系
  - 概念张力：量子系统源于经典，但本质不同

3.3 第二世界观：量子算法的紧凑R´esum´e总结

在本节中，我们提供了量子力学浓缩成的四条基本公设。这最优雅地可以通过狄拉克-希尔伯特空间形式化来完成。我们将仅在本节中回顾理解公设所需的最小数学。狄拉克-希尔伯特形式的更详细阐述将在以后的讲座中给出。

教学提示

本节以最抽象的形式呈现量子力学。如果某些概念看起来陌生，请不要惊慌——我们将逐步建立物理直觉。将此视为量子理论的“赤裸骨架”；在整个课程中，我们将逐步添加物理解释的“血肉”。

3.3.1 量子力学的公设

四公理算法

这四个公理为描述量子系统提供了完整的方案。值得注意的是，它们并未明确提及波、粒子或薛定谔方程——这些将作为推论出现。

公理一：量子态

公理*1*（状态公理）。每个量子系统 S 都与一个希尔伯特空间 $\mathcal{H}_S$相关联。系统在时间 t 的瞬时状态完全由一个单位范数的向量 $|\psi(t)\rangle \in \mathcal{H}_S$ 描述：

$$\boxed{\langle\psi(t)|\psi(t)\rangle = 1}$$

备注 *3.1*。状态向量 $|\psi(t)\rangle$ 包含了系统所有可能的信息。不同的物理情况对应不同的希尔伯特空间（例如，自旋-1/2 系统：$\mathbb{C}^2$，一维粒子：$L^2(\mathbb{R})$）。

Postulate II: The Operators

Postulate 2 (Operator Postulate). *There are two fundamental types of operators in quantum mechanics:*

(a) ***Observables (Measurement Operators)****: Each physically measurable quantity is represented by a unique Hermitian operator $\hat{A}$. The possible measurement outcomes are precisely the real eigenvalues λ_n of $\hat{A}$.*

(b) ***Evolution Operator****: For a closed quantum system S, the time evolution from initial state $|\psi(t_0)\rangle$ to final state $|\psi(t)\rangle$ is described by a unitary operator $\hat{U}(t, t_0)$:*

$$\boxed{|\psi(t)\rangle = \hat{U}(t, t_0)\,|\psi(t_0)\rangle}$$

where $\hat{U}^\dagger \hat{U} = \hat{U}\hat{U}^\dagger = \hat{I}$ (unitarity).

Postulate III: Measurement and Collapse

Postulate 3 (Measurement Postulate - Born Rule). *When measuring an observable $\hat{A}$ on a system in state $|\psi\rangle$, the following occurs:*

1. ***Outcome****: The measurement yields one of the eigenvalues λ_n of $\hat{A}$*
2. ***Probability****: The probability of obtaining λ_n is*

$$\boxed{P(\lambda_n) = |\langle\varphi_n|\psi\rangle|^2}$$

where $|\varphi_n\rangle$ is the eigenvector corresponding to λ_n

3. ***Collapse****: Immediately after measurement, the state collapses to the corresponding eigenstate:*

$$\boxed{|\psi\rangle \xrightarrow{\text{measure } \lambda_n} \frac{|\varphi_n\rangle}{\|\,|\varphi_n\rangle\,\|}}$$

Remark 3.2. *This is the most controversial postulate—it introduces true randomness and discontinuous state collapse. The measurement process itself is not described by unitary evolution.*

Postulate IV: Composite Systems

Postulate 4 (Composite Systems Postulate). *If system S consists of two subsystems S_1 and S_2 with Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$, then the Hilbert space of the composite system is the tensor product:*

$$\boxed{\mathcal{H}_S = \mathcal{H}_1 \otimes \mathcal{H}_2}$$

If $|\psi_1\rangle \in \mathcal{H}_1$ and $|\psi_2\rangle \in \mathcal{H}_2$, then the composite state is $|\psi_1\rangle \otimes |\psi_2\rangle \in \mathcal{H}_1 \otimes \mathcal{H}_2$.

Remark 3.3 (Tensor Product). *We will define the tensor product rigorously in later lectures. For now, think of it as a way to combine systems while preserving the individuality of each subsystem. This postulate enables quantum entanglement—states that cannot be written as simple products $|\psi_1\rangle \otimes |\psi_2\rangle$.*

第二条公设：算符

公设 *2*（算符公设）。量子力学中有两种基本的算符：

(a) 可观测量（测量算符）：每个物理可测量的量都由一个唯一的厄米算符 A表示。可能的测量结果是 λ_n 算符的实特征值 A。

(b) 时间演化算符：对于一个封闭的量子系统 S，从初始状态 $|\psi(t_0)\rangle$ 演化到最终状态 $|\psi(t)\rangle$ 由一个酉算符 $U(t, t_0)$描述：

$$\boxed{|\psi(t)\rangle = \hat{U}(t, t_0)\,|\psi(t_0)\rangle}$$

其中$\hat{U}^\dagger\hat{U} = \hat{U}\hat{U}^\dagger = \hat{I}$（酉性）。

公设三：测量与坍缩

公设*3*（测量公设——玻恩法则）。当测量系统处于状态$|\psi\rangle$时，对其可观测量 A 进行测量，会发生以下情况：

1. 结果：测量结果为 λ_n 的一个本征值。 $\hat{A}$
2. 概率：获得 λ_n 的概率为

$$\boxed{P(\lambda_n) = |\langle\varphi_n|\psi\rangle|^2}$$

其中 $|\varphi_n\rangle$ 是 λ_n对应的本征态。

3. 坍缩：测量完成后，系统立即坍缩到相应的本征态：

$$\boxed{|\psi\rangle \xrightarrow{\text{measure } \lambda_n} \frac{|\varphi_n\rangle}{\|\,|\varphi_n\rangle\,\|}}$$

备注 *3.2*。这是最具争议的假设——它引入了真正的随机性和不连续的状态坍缩。测量过程本身不由幺正演化描述。

假设 IV：复合系统

假设 *4*（复合系统假设）。如果系统 S 由两个子系统 S_1 和 S_2 组成，其希尔伯特空间分别为 $\mathcal{H}_1$ 和 $\mathcal{H}_2$，那么复合系统的希尔伯特空间是张量积：

$$\boxed{\mathcal{H}_S = \mathcal{H}_1 \otimes \mathcal{H}_2}$$

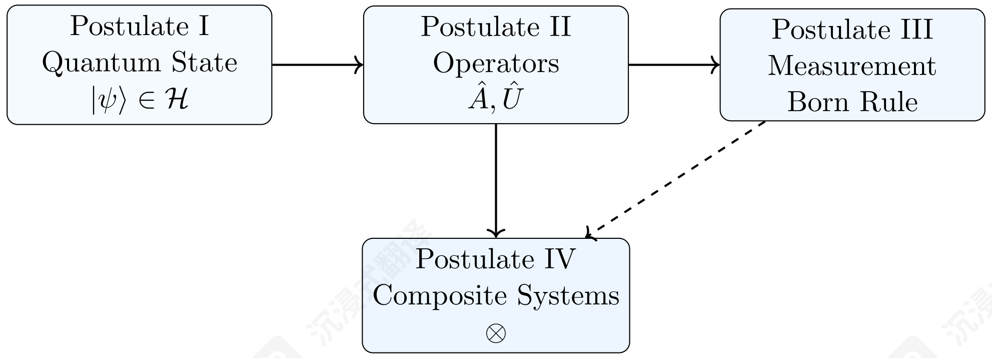
如果 $|\psi_1\rangle \in \mathcal{H}_1$ 和 $|\psi_2\rangle \in \mathcal{H}_2$，那么复合状态是 $|\psi_1\rangle \otimes |\psi_2\rangle \in \mathcal{H}_1 \otimes \mathcal{H}_2$。

备注 *3.3*（张量积）。我们将在后续讲座中严格定义张量积。目前，可以将其视为一种组合系统的方法，同时保留每个子系统的独立性。这一公设使得量子纠缠成为可能——那些不能表示为简单乘积的状态 $|\psi_1\rangle \otimes |\psi_2\rangle$。

3.3.2 Completeness and Emergent Concepts

We may now summarize the quantum algorithm using the following diagrammatic structure:

The Quantum Algorithm



- **Postulate I (The Stage):** The existence of the quantum state and Hilbert space creates the fundamental mathematical "stage" upon which the entire theory is built. Within this structured space, we define **operators**, which encode virtually all physical information about a system—both what can be observed in principle (through Hermitian observables) and how it evolves in time (through unitary evolution operators).
- **The Primacy of Operators:** Some advanced formulations, such as *algebraic quantum mechanics*, highlight the deep foundational role of operators. They demonstrate that one can even dispense with the explicit concept of a quantum state vector and work entirely within the algebra of operators, underscoring their centrality.
- **Postulate II (Dynamics and Observables):** This postulate sets the stage for extracting physical information. It defines what can be measured (observables) and dictates the deterministic evolution of the state (unitary operators), thereby preparing the system for the critical act of **measurement**.
- **Postulate III (Measurement and Collapse):** Governs the actual extraction of information. Through the Born rule, it bridges the abstract formalism with empirical reality by providing probabilities for measurement outcomes. It also describes the state's **discontinuous collapse** into an eigenstate upon observation, completing the connection between theory and experiment.
- **Postulate IV (Building Complexity):** Serves as the architectural or recursive rule for constructing complex systems. It describes how individual subsystems, each with their own Hilbert space, combine via the tensor product $\mathcal{H}_{\text{total}} = \mathcal{H}_1 \otimes \mathcal{H}_2$ to form larger, composite dynamical wholes. This mechanism is essential for enabling quintessentially quantum phenomena like **entanglement**, where the state of the whole cannot be factored into independent states of its parts.

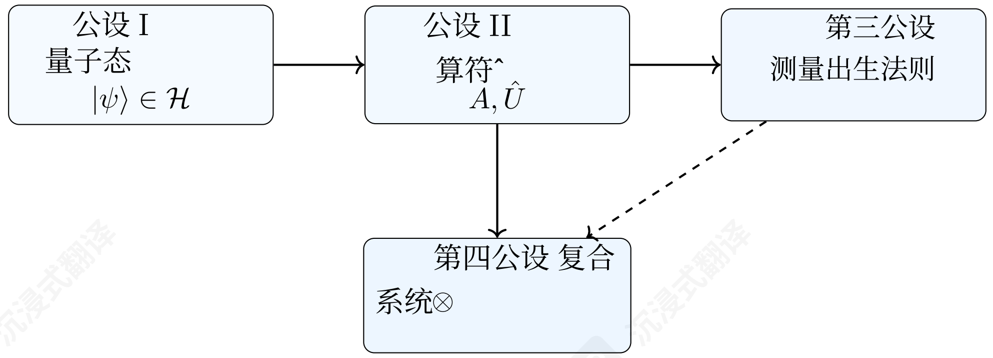
The Historical Debate on Completeness

- **The Algorithm's Claim to Completeness:** The four-postulate quantum algorithm is, in principle, a complete and self-contained logical structure. For over a century, it has provided an astonishingly accurate description of the physical world, from subatomic particles to the chemistry of life.
- **The Core of the Debate: Interpretation of the Quantum State:** The profound question of whether this mathematical description is *complete* has fueled one of the greatest debates in scientific history. This debate crystallized around the physical meaning of the wave function ψ.

3.3.2 完整性与涌现概念

现在，我们可以使用以下图示结构来总结量子算法：

量子算法



- 第一公设（阶段）：量子态和希尔伯特空间的存在构成了整个理论所建立的基础数学“舞台”。在这个结构化的空间中，我们定义算符，它们编码了系统几乎所有物理信息——既包括原则上可以观测的内容（通过厄米算符），也包括系统随时间演变的方式（通过酉演化算符）。

- 算符的首要性：一些高级表述，如代数量子力学，突出了算符的深层基础作用。它们证明，甚至可以舍弃量子态向量的显式概念，完全在算符代数中工作，从而强调了算符的核心地位。

- 第二公设（动力学与可观测量）：该公设为提取物理信息奠定了基础。它定义了可测量的量（可观测量），并规定了状态的决定性演化（幺正算符），从而为关键的测量行为做好了准备。

- 第三公设（测量与坍缩）：规范了信息的实际提取过程。通过玻恩法则，它将抽象的形式主义与经验现实联系起来，为测量结果提供了概率。它还描述了在观测时状态会不连续地坍缩到本征态，从而完成了理论与实验之间的联系。

- 第四公设（构建复杂性）：作为构建复杂系统的架构或递归规则。它描述了具有各自希尔伯特空间的独立子系统，如何通过张量积 $\mathcal{H}_{\text{total}} = \mathcal{H}_1 \otimes \mathcal{H}_2$ 组合形成更大的复合动力学整体。这种机制对于实现本质上量子现象（如纠缠）至关重要，在这些现象中，整体的状态不能分解为其各部分独立状态的总和。

关于完备性的历史争论

- 算法对完备性的主张：四公设量子算法在原则上是一种完整且自洽的逻辑结构。一个多世纪以来，它为物理世界提供了惊人的准确描述，从亚原子粒子到生命的化学。

- 争论的核心：量子态的解释：这个数学描述是否完备的深刻问题，点燃了科学史上最伟大的争论之一。这场争论围绕着波函数 ψ 的物理意义而形成。

Page 27 of 61

– **The Ensemble/Statistical View (Incompleteness):** Physicists like **Einstein, Schrödinger, and de Broglie** argued the algorithm is *incomplete*. They viewed it as a statistical tool for *ensembles* of systems, not a description of individual reality. Einstein famously held that "the wave function cannot be a complete description of reality," believing a deeper, deterministic theory with "hidden variables" must exist.

– **The Copenhagen View (Completeness):** Proponents like **Bohr, Pauli, and von Neumann** maintained the description *is* complete. For them, the wave function represents the full reality of an individual system, and its inherent probabilistic nature is fundamental, not a sign of ignorance.

- **Experimental Verdict (Bell's Inequalities):** Modern experimental evidence, particularly from rigorous tests of **Bell's inequalities**, has strongly vindicated the standard quantum framework. These tests have ruled out broad classes of local hidden variable theories, confirming the non-local correlations predicted by quantum mechanics and thus supporting its completeness at the operational level.

- **Future Directions: Seeking a Deeper Foundation:** Despite its empirical success, the quest for a deeper understanding persists. Many researchers, especially in quantum gravity and foundations, believe quantum mechanics itself may be an **effective theory**—a highly successful approximation emerging from a more fundamental framework (e.g., involving quantum spacetime) yet to be discovered. Therefore, while the quantum algorithm is complete as a *description*, the inquiry into what it implies about the ultimate nature of reality remains a central, driving force in modern physics.

The Algorithmic Completeness

These four postulates provide a complete and self-contained **algorithmic framework** for quantum mechanics. In principle, from these postulates alone, one can derive all standard quantum theory. This framework's power is evident in what it does *not* postulate:

- **The Schrödinger equation**: Emerges from Postulate IIb (unitary evolution) for specific Hamiltonians.
- **The uncertainty principle**: Derivable from Postulates I-III (non-commuting operators).
- **Wave-particle duality**: Contained within the state representation and the measurement postulate.
- **Classical correspondence**: Emerges in the appropriate limits ($\hbar \to 0$, decoherence).

The remainder of this course will be devoted to unpacking these postulates and extracting their physical consequences.

An Important Distinction: It is crucial to distinguish between the *algorithmic completeness* of the framework—its logical self-sufficiency—and the *interpretive completeness* of its central object, the quantum state. Whether the wave function describes an individual system's full reality or merely statistical knowledge of an ensemble has been a central philosophical debate since the theory's inception (Einstein vs. Bohr). While this course will operate within the standard, empirically validated framework, we will remain mindful of this deeper foundational question.

– 系统的集合/统计观点（不完备性）：爱因斯坦、薛定谔和德布罗意等物理学家认为该算法是不完备的。他们认为它只是系统集合的统计工具，而非对个体现实的描述。爱因斯坦曾著名地指出“波函数不能是现实的完整描述”，认为必须存在更深层次的、具有“隐变量”的决定论理论。

– 哥本哈根观点（完备性）：玻尔、泡利和冯·诺依曼等支持者坚持描述是完备的。对他们而言，波函数代表了单个系统的全部现实，其固有的概率性是根本性的，而非无知的表现。

- 实验证据（贝尔不等式）：现代实验证据，特别是贝尔不等式的严格检验，已有力支持了标准量子框架。这些检验排除了广泛的局域隐变量理论，确认了量子力学预测的非局域关联，从而在操作层面支持了其完备性。

- 未来方向：寻求更深的根基：尽管其经验性成功已获认可，但对更深层次的理解的追求仍在持续。许多研究人员，特别是在量子引力和基础理论领域，认为量子力学本身可能只是一个有效理论——一个从更根本的框架（例如涉及量子时空）中产生的极其成功的近似，而这个框架尚未被发现。因此，尽管量子算法作为一个描述是完整的，但探究它对现实终极本质意味着什么的问题仍然是现代物理学中一个核心的、推动性的力量。

算法完备性

这四个公设为量子力学提供了一个完整且自洽的算法框架。原则上，仅从这些公设中，人们就可以推导出所有标准量子理论。这个框架的力量体现在它没有公设的内容：

- 薛定谔方程：源自公设IIb（幺正演化）的特定哈密顿量。
哈密顿量。
- 不确定性原理：可从公设I-III（非交换算符）推导。
- 波粒二象性：包含在状态表示和测量假设中。
- 经典对应性：在适当的极限下出现（$\hbar \to 0$，退相干）。

本课程的其余部分将致力于解析这些假设并提取其物理后果。

一个重要的区别：必须区分框架的算法完备性——其逻辑自足性——与其中心对象，即量子态的解释完备性。波函数是否描述了单个系统的完整现实，还是仅仅是对集合的统计知识，自理论创立以来一直是一个核心的哲学争论（爱因斯坦与玻尔的争论）。尽管本课程将在标准、经验验证的框架内进行，但我们将始终牢记这个更深层次的基础性问题。

4 Foundations of Quantum Mechanics in the Position Space Representation

We begin our formulation of quantum mechanics for a single particle moving in three-dimensional space, $\mathbb{R}^3$. The state of the particle is completely described by a wavefunction, and its evolution and the results of measurements are governed by a set of fundamental postulates.

4.1 Postulate I: The Quantum State

The first postulate establishes the mathematical object that contains all information about a physical system.

Postulate I: The Quantum State

The quantum state of a single particle is completely described by a complex-valued function $\psi(\mathbf{x}, t)$, known as the **wavefunction**. At a fixed time, say $t = 0$, the wavefunction $\psi(\mathbf{x}) \equiv \psi(\mathbf{x}, 0)$ is an element of a complex Hilbert space $\mathcal{H}$.

For a particle in space, the relevant Hilbert space is the space of square-integrable functions, denoted $L^2(\mathbb{R}^3; \mathbb{C})$.

$$\mathcal{H} = L^2(\mathbb{R}^3; \mathbb{C}) = \left\{ \psi : \mathbb{R}^3 \to \mathbb{C} \;\middle|\; \int_{\mathbb{R}^3} |\psi(\mathbf{x})|^2 \mathrm{d}^3x < \infty \right\}. \tag{4}$$

The inner product on this space, which maps two wavefunctions to a complex number, is defined by the integral

$$\langle \phi | \psi \rangle = \int_{\mathbb{R}^3} \phi^*(\mathbf{x}) \psi(\mathbf{x}) \mathrm{d}^3x. \tag{5}$$

The norm of a state is given by

$$||\psi|| = \sqrt{\langle \psi | \psi \rangle}. \tag{6}$$

This norm must be finite. States are typically normalized such that $||\psi|| = 1$, a condition that will be linked to probability by the Born rule.

Remark 4.1 (The Superposition Principle)**.** If $\psi_1(\mathbf{x})$ and $\psi_2(\mathbf{x})$ are possible quantum states of a system, then any linear combination

$$\psi(\mathbf{x}) = c_1\psi_1(\mathbf{x}) + c_2\psi_2(\mathbf{x}) \tag{7}$$

where $c_1, c_2 \in \mathbb{C}$, is also a possible quantum state. The physical justification for this principle will become evident when we discuss the linearity of the Schrödinger equation.

4.2 Postulate II: Operators, Observables, and Time Evolution

This postulate introduces the mathematical objects that represent physical quantities and the law that governs the change of the quantum state over time.

4.2.1 II.a Physical Observables as Hermitian Operators

Physical observables are represented by linear operators acting on the Hilbert space $L^2(\mathbb{R}^3; \mathbb{C})$.

Definition 12 (Linear Operator)**.** A linear operator $\hat{A}$ is a map from $\mathcal{H}$ to $\mathcal{H}$ such that for any $\psi, \phi \in \mathcal{H}$ and $c \in \mathbb{C}$,

$$\hat{A}(c\psi + \phi) = c\hat{A}\psi + \hat{A}\phi. \tag{8}$$

Here, ψ and ϕ are functions of the position variable $\mathbf{x}$, i.e., $\psi \equiv \psi(\mathbf{x})$.

4 位置空间表示中的量子力学基础

我们从单个粒子在三维空间中运动的量子力学表述开始，$\mathbb{R}^3$。粒子的状态完全由一个波函数描述，其演化和测量结果由一组基本公设支配。

4.1 公理I：量子态

第一个公理确立了包含物理系统所有信息的数学对象。

公理I：量子态

单个粒子的量子态完全由一个复值函数 $\psi(\mathbf{x}, t)$ 描述，该函数称为波函数。在固定的时间，比如 $t = 0$，波函数 $\psi(\mathbf{x}) \equiv \psi(\mathbf{x}, 0)$ 是一个复希尔伯特空间 $\mathcal{H}$ 的元素。

对于空间中的粒子，相关的希尔伯特空间是平方可积函数的空间，记作 $L^2(\mathbb{R}^3; \mathbb{C})$。

$$\mathcal{H} = L^2(\mathbb{R}^3; \mathbb{C}) = \left\{ \psi : \mathbb{R}^3 \to \mathbb{C} \;\middle|\; \int_{\mathbb{R}^3} |\psi(\mathbf{x})|^2 \mathrm{d}^3x < \infty \right\}. \tag{4}$$

该空间上的内积，将两个波函数映射为一个复数，由积分定义。

$$\langle \phi | \psi \rangle = \int_{\mathbb{R}^3} \phi^*(\mathbf{x}) \psi(\mathbf{x}) \mathrm{d}^3x. \tag{5}$$

状态范数由下式给出。

$$||\psi|| = \sqrt{\langle \psi | \psi \rangle}. \tag{6}$$

该范数必须是有限的。状态通常被归一化，使得 $||\psi|| = 1$，这一条件将通过玻恩法则与概率联系起来。

注 4.1（叠加原理）。如果 $\psi_1(\mathbf{x})$ 和 $\psi_2(\mathbf{x})$ 是系统可能的量子态，那么任何线性组合

$$\psi(\mathbf{x}) = c_1\psi_1(\mathbf{x}) + c_2\psi_2(\mathbf{x}) \tag{7}$$

c_1 $c_2 \in \mathbb{C}$在哪里，也是一个可能的量子态。当讨论薛定谔方程的线性时，这个原理的物理合理性将变得明显。

4.2 第二个假设：算符、可观测量和时间演化

这个假设引入了代表物理量的数学对象以及支配量子态随时间变化的规律。

4.2.1 第二个假设.a 物理可观测量作为厄米算符

物理可观测量由作用于希尔伯特空间 $L^2(\mathbb{R}^3; \mathbb{C})$的线性算符表示。

ˆ定义12（线性算符）。线性算符 A 是从 $\mathcal{H}$ 到 $\mathcal{H}$ 的映射，使得对于任何 ψ、$\phi \in \mathcal{H}$和 $c \in \mathbb{C}$，

$$\hat{A}(c\psi + \phi) = c\hat{A}\psi + \hat{A}\phi. \tag{8}$$

这里，ψ 和 ϕ 是位置变量x的函数，即 $\psi \equiv \psi(\mathbf{x})$。

Definition 13 (Adjoint Operator)**.** The adjoint of an operator $\hat{A}$, denoted $\hat{A}^\dagger$, is defined by the relation

$$\langle \phi | \hat{A} \psi \rangle = \langle \hat{A}^\dagger \phi | \psi \rangle, \tag{9}$$

which must hold for all $\psi, \phi \in \mathcal{H}$. In the integral form, this is

$$\int_{\mathbb{R}^3} \phi^*(\mathbf{x})(\hat{A}\psi)(\mathbf{x}) \mathrm{d}^3 x = \int_{\mathbb{R}^3} (\hat{A}^\dagger \phi)^*(\mathbf{x}) \psi(\mathbf{x}) \mathrm{d}^3 x. \tag{10}$$

The adjoint operation has the following important properties (for operators $\hat{A}$, $\hat{B}$ and a complex constant c):

- $(\hat{A}^\dagger)^\dagger = \hat{A}$
- $(c\hat{A})^\dagger = c^* \hat{A}^\dagger$
- $(\hat{A} + \hat{B})^\dagger = \hat{A}^\dagger + \hat{B}^\dagger$
- $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger$

Definition 14 (Hermitian Operator)**.** An operator $\hat{A}$ is **Hermitian** (or self-adjoint) if it is equal to its own adjoint: $\hat{A}^\dagger = \hat{A}$. This implies that for all $\psi, \phi \in \mathcal{H}$,

$$\langle \phi | \hat{A} \psi \rangle = \langle \hat{A} \phi | \psi \rangle. \tag{11}$$

The fundamental observables of a single particle are represented by the following Hermitian operators.

Definition 15 (Position Operator)**.** In three dimensions, the position operator $\hat{\mathbf{x}} = (\hat{x}, \hat{y}, \hat{z})$ acts on a wavefunction by multiplication with the coordinate:

$$(\hat{x}\psi)(\mathbf{x}) = x\psi(\mathbf{x}), \tag{12}$$
$$(\hat{y}\psi)(\mathbf{x}) = y\psi(\mathbf{x}), \tag{13}$$
$$(\hat{z}\psi)(\mathbf{x}) = z\psi(\mathbf{x}). \tag{14}$$

In one dimension, this reduces to $(\hat{x}\psi)(x) = x\psi(x)$.

Definition 16 (Momentum Operator)**.** In three dimensions, the momentum operator $\hat{\mathbf{p}} = (\hat{p}_x, \hat{p}_y, \hat{p}_z)$ is defined as a differential operator:

$$(\hat{p}_x\psi)(\mathbf{x}) = -i\hbar \frac{\partial}{\partial x}\psi(\mathbf{x}), \tag{15}$$
$$(\hat{p}_y\psi)(\mathbf{x}) = -i\hbar \frac{\partial}{\partial y}\psi(\mathbf{x}), \tag{16}$$
$$(\hat{p}_z\psi)(\mathbf{x}) = -i\hbar \frac{\partial}{\partial z}\psi(\mathbf{x}). \tag{17}$$

In one dimension, this reduces to

$$(\hat{p}\psi)(x) = -i\hbar \frac{\mathrm{d}}{\mathrm{d}x}\psi(x). \tag{18}$$

The Hermiticity of these operators can be shown using the inner product. For the position operator in 1D:

$$\langle \phi | \hat{x}\psi \rangle = \int_{-\infty}^{\infty} \phi^*(x)\, x\, \psi(x) \mathrm{d}x = \int_{-\infty}^{\infty} (x\phi(x))^* \psi(x) \mathrm{d}x = \langle \hat{x}\phi | \psi \rangle, \tag{19}$$

定义13（伴随算子）。算子ˆA的伴随，记作ˆ$A^\dagger$，由关系定义。

$$\langle \phi | \hat{A} \psi \rangle = \langle \hat{A}^\dagger \phi | \psi \rangle, \tag{9}$$

这对所有 ψ、$\phi \in \mathcal{H}$ 都必须成立。在积分形式中，这是

$$\int_{\mathbb{R}^3} \phi^*(\mathbf{x})(\hat{A}\psi)(\mathbf{x}) \mathrm{d}^3 x = \int_{\mathbb{R}^3} (\hat{A}^\dagger \phi)^*(\mathbf{x}) \psi(\mathbf{x}) \mathrm{d}^3 x. \tag{10}$$

ˆ ˆ 伴随运算具有以下重要性质（对于算子 A、B 和一个复常数 c）：

- $(\hat{A}^\dagger)^\dagger = \hat{A}$
- $(c\hat{A})^\dagger = c^* \hat{A}^\dagger$
- $(\hat{A} + \hat{B})^\dagger = \hat{A}^\dagger + \hat{B}^\dagger$
- $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger$

ˆ 定义 14 (厄米算子). 一个算子 A 是厄米的（或自伴的），如果它等于自己的伴随：$\hat{A}^\dagger = \hat{A}$。这意味着对于所有 $\psi, \phi \in \mathcal{H}$,

$$\langle \phi | \hat{A} \psi \rangle = \langle \hat{A} \phi | \psi \rangle. \tag{11}$$

单个粒子的基本可观测量由以下厄米算子表示。

定义 15 (位置算子). 在三维空间中，位置算子 $\hat{\mathbf{x}} = (\hat{x}, \hat{y}, \hat{z})$ 对波函数的作用是通过与坐标相乘：

$$(\hat{x}\psi)(\mathbf{x}) = x\psi(\mathbf{x}), \tag{12}$$
$$(\hat{y}\psi)(\mathbf{x}) = y\psi(\mathbf{x}), \tag{13}$$
$$(\hat{z}\psi)(\mathbf{x}) = z\psi(\mathbf{x}). \tag{14}$$

在一维空间中，这简化为 $(\hat{x}\psi)(x) = x\psi(x)$。

定义16（动量算符）。在三维空间中，动量算符 $\hat{\mathbf{p}} = (\hat{p}_x, \hat{p}_y, \hat{p}_z)$ 被定义为微分算符：

$$(\hat{p}_x\psi)(\mathbf{x}) = -i\hbar \frac{\partial}{\partial x}\psi(\mathbf{x}), \tag{15}$$
$$(\hat{p}_y\psi)(\mathbf{x}) = -i\hbar \frac{\partial}{\partial y}\psi(\mathbf{x}), \tag{16}$$
$$(\hat{p}_z\psi)(\mathbf{x}) = -i\hbar \frac{\partial}{\partial z}\psi(\mathbf{x}). \tag{17}$$

在一维空间中，这简化为

$$(\hat{p}\psi)(x) = -i\hbar \frac{\mathrm{d}}{\mathrm{d}x}\psi(x). \tag{18}$$

这些算子的厄米性可以通过内积来证明。对于一维中的位置算子：

$$\langle \phi | \hat{x}\psi \rangle = \int_{-\infty}^{\infty} \phi^*(x)\, x\, \psi(x) \mathrm{d}x = \int_{-\infty}^{\infty} (x\phi(x))^* \psi(x) \mathrm{d}x = \langle \hat{x}\phi | \psi \rangle, \tag{19}$$

Page 30 of 61

confirming $\hat{x}^\dagger = \hat{x}$. For the momentum operator in 1D, we require integration by parts:

$$\langle\phi|\hat{p}\psi\rangle = \int_{-\infty}^{\infty} \phi^*(x)\left(-i\hbar\frac{\mathrm{d}\psi}{\mathrm{d}x}\right)\mathrm{d}x \tag{20}$$

$$= \left[-i\hbar\phi^*(x)\psi(x)\right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty}\left(-i\hbar\frac{\mathrm{d}\phi}{\mathrm{d}x}\right)^*\psi(x)\mathrm{d}x. \tag{21}$$

For the operator to be Hermitian, the boundary term must vanish. This imposes a condition on the wavefunctions in our Hilbert space:

$$\lim_{x\to\pm\infty}\psi(x) = 0, \tag{22}$$

which is satisfied for square-integrable functions. With this, we get $\langle\phi|\hat{p}\psi\rangle = \langle\hat{p}\phi|\psi\rangle$, so $\hat{p}^\dagger = \hat{p}$.

4.2.2 II.b Time Evolution: The Schrödinger Equation

The time evolution of a quantum state is governed by the Schrödinger equation.

Postulate II.b: The Schrödinger Equation

The time evolution of the wavefunction $\psi(\mathbf{x},t)$ is determined by the **time-dependent Schrödinger equation**:

$$\boxed{i\hbar\frac{\partial}{\partial t}\psi(\mathbf{x},t) = \hat{H}\psi(\mathbf{x},t)}, \tag{23}$$

where $\hat{H}$ is the Hamiltonian operator representing the total energy of the system.

Illustration 1: The Time-Evolution Operator. For a system with a time-independent Hamiltonian $\hat{H}$, we can solve the Schrödinger equation using the method of separation of variables. This is a standard technique for solving partial differential equations, where we assume the solution can be written as a product of a function of space alone and a function of time alone:

$$\psi(\mathbf{x},t) = \psi(\mathbf{x})T(t). \tag{24}$$

Substituting this into the time-dependent Schrödinger equation gives:

$$i\hbar\psi(\mathbf{x})\frac{\mathrm{d}T}{\mathrm{d}t} = T(t)\hat{H}\psi(\mathbf{x}). \tag{25}$$

Now, divide both sides by $\psi(\mathbf{x})T(t)$ (where the functions are non-zero):

$$i\hbar\frac{1}{T(t)}\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{1}{\psi(\mathbf{x})}\hat{H}\psi(\mathbf{x}). \tag{26}$$

The left-hand side is a function of time only, while the right-hand side is a function of position only. For these to be equal for all $\mathbf{x}$ and t, they must both be equal to a constant, which we call E (which will turn out to be the energy). This yields two separate equations:

$$i\hbar\frac{\mathrm{d}T}{\mathrm{d}t} = ET(t), \tag{27}$$

$$\hat{H}\psi(\mathbf{x}) = E\psi(\mathbf{x}). \tag{28}$$

Equation 27 is an ordinary differential equation for time with the solution

$$T(t) = e^{-iEt/\hbar}. \tag{29}$$

确认 $\hat{x}^\dagger = \hat{x}$。对于一维中的动量算子，我们需要分部积分：

$$\langle\phi|\hat{p}\psi\rangle = \int_{-\infty}^{\infty} \phi^*(x)\left(-i\hbar\frac{\mathrm{d}\psi}{\mathrm{d}x}\right)\mathrm{d}x \tag{20}$$

$$= \left[-i\hbar\phi^*(x)\psi(x)\right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty}\left(-i\hbar\frac{\mathrm{d}\phi}{\mathrm{d}x}\right)^*\psi(x)\mathrm{d}x. \tag{21}$$

为了使算子是厄米算子，边界项必须消失。这给我们的希尔伯特空间中的波函数施加了一个条件：

$$\lim_{x\to\pm\infty}\psi(x) = 0, \tag{22}$$

这在平方可积函数下得到满足。因此，我们得到 $\langle\phi|\hat{p}\psi\rangle = \langle\hat{p}\phi|\psi\rangle$，所以 $\hat{p}^\dagger = \hat{p}$。

4.2.2 II.b 时间演化：薛定谔方程

量子态的时间演化由薛定谔方程支配。

公设 II.b：薛定谔方程

波函数 ψ(x t) 的时间演化由含时薛定谔方程决定：

$$\boxed{i\hbar\frac{\partial}{\partial t}\psi(\mathbf{x},t) = \hat{H}\psi(\mathbf{x},t)}, \tag{23}$$

ˆ其中 H 是代表系统总能量的哈密顿算符。

图解 1：时间演化算符。对于一个哈密顿算符 ˆH 不随时间变化的系统，我们可以用变量分离法求解薛定谔方程。这是求解偏微分方程的标准技巧，我们假设解可以表示为一个仅关于空间和仅关于时间的函数的乘积：

$$\psi(\mathbf{x},t) = \psi(\mathbf{x})T(t). \tag{24}$$

将此式代入含时薛定谔方程得到：

$$i\hbar\psi(\mathbf{x})\frac{\mathrm{d}T}{\mathrm{d}t} = T(t)\hat{H}\psi(\mathbf{x}). \tag{25}$$

现在，将两边除以 ψ(x)$T(t)$（其中函数不为零）：

$$i\hbar\frac{1}{T(t)}\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{1}{\psi(\mathbf{x})}\hat{H}\psi(\mathbf{x}). \tag{26}$$

左侧是仅关于时间的函数，而右侧是仅关于位置的函数。要使它们对所有 x 和 t 都相等，它们必须都等于一个常数，我们称之为 E（这最终将证明是能量）。由此得到两个独立的方程：

$$i\hbar\frac{\mathrm{d}T}{\mathrm{d}t} = ET(t), \tag{27}$$

$$\hat{H}\psi(\mathbf{x}) = E\psi(\mathbf{x}). \tag{28}$$

方程27是一个关于时间的常微分方程，其解为

$$T(t) = e^{-iEt/\hbar}. \tag{29}$$

Page 31 of 61

Equation 28 is the time-independent Schrödinger equation, an eigenvalue equation for the Hamiltonian. The general solution to the original equation can then be written using the **time-evolution operator** $\hat{U}(t, t_0 = 0)$:

$$\psi(\mathbf{x}, t) = e^{-i\hat{H}t/\hbar}\psi(\mathbf{x}, 0) \equiv \hat{U}(t, 0)\psi(\mathbf{x}, 0). \tag{30}$$

The operator $\hat{U}$ is unitary, meaning $\hat{U}^\dagger = \hat{U}^{-1}$, so it preserves the norm of the state. It also satisfies the group properties

$$\hat{U}(t, t) = 1, \tag{31}$$

and

$$\hat{U}(t_2, t_1)\hat{U}(t_1, t_0) = \hat{U}(t_2, t_0). \tag{32}$$

Illustration 2: Time-Independent Schrödinger Equation. As derived above, when the Hamiltonian is time-independent, separating the variables leads directly to the time-independent Schrödinger equation:

$$\boxed{\hat{H}\phi(\mathbf{x}) = E\phi(\mathbf{x})}. \tag{33}$$

Here, E are the allowed energy eigenvalues and $\phi(\mathbf{x})$ are the corresponding energy eigenstates. The full time-dependent solution for an energy eigenstate is then $\phi(\mathbf{x})e^{-iEt/\hbar}$.

4.3 Postulate III: The Measurement Postulate and the Born Rule

Before stating the measurement postulate, we must first understand a fundamental mathematical fact about Hermitian operators and their eigenfunctions.

4.3.1 Mathematical Prelude: Eigenfunctions as Basis Vectors

Recall from linear algebra that in a finite-dimensional vector space, a Hermitian matrix has a complete set of orthonormal eigenvectors that form a basis. Every vector can be uniquely expanded as a linear combination of these eigenvectors. The spectral theorem extends this crucial property to Hermitian operators on the infinite-dimensional Hilbert space $L^2(\mathbb{R}^3; \mathbb{C})$.

Theorem 3 (Spectral Theorem)**.** Let $\hat{A}$ be a Hermitian operator on $\mathcal{H} = L^2(\mathbb{R}^3; \mathbb{C})$ representing a physical observable. Then the eigenfunctions $\{\varphi_n(\mathbf{x})\}$ of $\hat{A}$, corresponding to eigenvalues a_n, form a **complete orthonormal basis** for the Hilbert space.

This theorem has two essential consequences, expressed mathematically in the language of wavefunctions:

1. **Orthonormality**: Different eigenfunctions are orthogonal, and each eigenfunction is normalized. For a discrete spectrum (like bound states in an atom), this is written as:

$$\langle\varphi_m|\varphi_n\rangle = \int_{\mathbb{R}^3} \varphi_m^*(\mathbf{x})\,\varphi_n(\mathbf{x})\,\mathrm{d}^3x = \delta_{mn}, \tag{34}$$

where δ_{mn} is the Kronecker delta ($= 1$ if $m = n$, 0 otherwise).

For operators with a continuous spectrum (such as the momentum operator, which we will study in Application 3), the orthonormality condition involves the Dirac delta function:

$$\int_{\mathbb{R}^3} \varphi_{p'}^*(\mathbf{x})\,\varphi_p(\mathbf{x})\,\mathrm{d}^3x = \delta(p' - p). \tag{35}$$

We will explore the physical meaning of this when we encounter continuous spectra later.

方程28是时间无关的薛定谔方程，是哈密顿算子的本征值方程。然后，原方程的通解可以使用时间演化算子$U(t, t_0 = 0)$来表示：

$$\psi(\mathbf{x}, t) = e^{-i\hat{H}t/\hbar}\psi(\mathbf{x}, 0) \equiv \hat{U}(t, 0)\psi(\mathbf{x}, 0). \tag{30}$$

算符 ˆU是幺正的，这意味着 $\hat{U}^\dagger = \hat{U}^{-1}$，因此它保持态的范数。它还满足群性质。

$$U(t, t) = 1, \tag{31}$$

and

$$\hat{U}(t_2, t_1)\hat{U}(t_1, t_0) = \hat{U}(t_2, t_0). \tag{32}$$

图2：不含时薛定谔方程。如上所述，当哈密顿量不含时，分离变量可直接得到不含时薛定谔方程：

$$\boxed{\hat{H}\phi(\mathbf{x}) = E\phi(\mathbf{x})}. \tag{33}$$

这里， E 是允许的能量本征值， ϕ(x)是对应的能量本征态。然后能量本征态的完整含时解是 $\phi(\mathbf{x})e^{-iEt/\hbar}$。

4.3 第III公设：测量公设和玻恩法则

在阐述测量公设之前，我们必须首先理解关于厄米算符及其本征函数的一个基本数学事实。

4.3.1 数学前奏：本征函数作为基向量

从线性代数中回忆，在有限维向量空间中，厄米矩阵拥有一组完备的正交归一的本征函数，它们构成一个基。每个向量都可以唯一地展开为这些本征函数的线性组合。谱定理将这一关键性质扩展到无限维希尔伯特空间 $L^2(\mathbb{R}^3; \mathbb{C})$上的厄米算符。

定理3（谱定理）。设ˆA 是希尔伯特空间 $\mathcal{H} = L^2(\mathbb{R}^3; \mathbb{C})$ 上的一个厄米算符，表示一个物理ˆ可观测量。那么， A的本征函数 $\{\varphi_n(\mathbf{x})\}$ （对应本征值 a_n）构成希尔伯特空间的一个完备正交归一基。

这个定理有两个基本推论，用波函数的语言在数学上表达如下：

1. 正交归一性：不同的本征函数是正交的，并且每个本征函数都是归一化的。对于离散谱（如原子中的束缚态），这可以写成：

$$\langle\varphi_m|\varphi_n\rangle = \int_{\mathbb{R}^3} \varphi_m^*(\mathbf{x})\,\varphi_n(\mathbf{x})\,\mathrm{d}^3x = \delta_{mn}, \tag{34}$$

δ_{mn} 在哪里是克罗内克δ函数（$= 1$ 如果 $m = n$，否则为0）。

对于具有连续谱的算子（例如我们将要在Application3中研究的动量算子），正交归一性条件涉及狄拉克δ函数：

$$\int_{\mathbb{R}^3} \varphi_{p'}^*(\mathbf{x})\,\varphi_p(\mathbf{x})\,\mathrm{d}^3x = \delta(p' - p). \tag{35}$$

当我们遇到连续光谱时，我们将探讨其物理意义。

2. **Completeness**: Any wavefunction $\psi(\mathbf{x})$ in the Hilbert space can be expanded as a linear combination of the eigenfunctions. For a discrete spectrum:

$$\psi(\mathbf{x}) = \sum_n c_n\,\varphi_n(\mathbf{x}), \tag{36}$$

where the expansion coefficients c_n are complex numbers given by the inner product:

$$c_n = \langle \varphi_n | \psi \rangle = \int_{\mathbb{R}^3} \varphi_n^*(\mathbf{x})\,\psi(\mathbf{x})\,\mathrm{d}^3x. \tag{37}$$

For a continuous spectrum, the sum becomes an integral:

$$\psi(\mathbf{x}) = \int c(p)\,\varphi_p(\mathbf{x})\,\mathrm{d}p, \tag{38}$$

with $c(p) = \langle \varphi_p | \psi \rangle$.

The completeness relation can also be expressed as a condition on the eigenfunctions themselves. In the discrete case:

$$\sum_n \varphi_n^*(\mathbf{x}')\,\varphi_n(\mathbf{x}) = \delta^{(3)}(\mathbf{x} - \mathbf{x}'), \tag{39}$$

which means that any wavefunction can be reconstructed by integrating over its expansion:

$$\psi(\mathbf{x}) = \int_{\mathbb{R}^3} \left(\sum_n \varphi_n^*(\mathbf{x}')\varphi_n(\mathbf{x}) \right) \psi(\mathbf{x}')\,\mathrm{d}^3x'. \tag{40}$$

Remark 4.2 (Looking Ahead)**.** In Applications 3 and 4, we will encounter two important examples of such eigenfunction bases:

- The **momentum eigenfunctions** $\varphi_{\mathbf{p}}(\mathbf{x}) = (2\pi\hbar)^{-3/2} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}$, which form a continuous basis and lead to the momentum space representation via Fourier transforms.
- The **position eigenfunctions** $\varphi_{\mathbf{x}'}(\mathbf{x}) = \delta^{(3)}(\mathbf{x} - \mathbf{x}')$, which are generalized functions (distributions) that form a continuous basis and directly yield the Born rule probability density $|\psi(\mathbf{x})|^2$.

Both are special cases of the general spectral theorem stated above, and their properties will be derived in detail later.

4.3.2 The Measurement Postulate

With this mathematical framework in place, we can now state the measurement postulate. We assume the wavefunction is normalized:

$$\int_{\mathbb{R}^3} |\psi(\mathbf{x})|^2\,\mathrm{d}^3x = 1. \tag{41}$$

Consider an observable represented by a Hermitian operator $\hat{A}$. Let $\{\varphi_n(\mathbf{x})\}$ be the complete set of eigenfunctions of $\hat{A}$ with corresponding eigenvalues a_n, satisfying:

$$\hat{A}\varphi_n = a_n\varphi_n. \tag{42}$$

Since the eigenfunctions form a complete orthonormal basis, any state $\psi(\mathbf{x})$ can be expanded as:

$$\psi(\mathbf{x}) = \sum_n c_n\varphi_n(\mathbf{x}), \tag{43}$$

where the expansion coefficients are given by:

$$c_n = \langle \varphi_n | \psi \rangle = \int_{\mathbb{R}^3} \varphi_n^*(\mathbf{x})\,\psi(\mathbf{x})\,\mathrm{d}^3x. \tag{44}$$

2. 完备性：希尔伯特空间中的任意波函数 $\psi(\mathrm{x})$ 都可以展开为特征函数的线性组合。对于离散谱：

$$\psi(\mathbf{x}) = \sum_n c_n\,\varphi_n(\mathbf{x}), \tag{36}$$

其中展开系数 c_n 是由内积给出的复数：

$$c_n = \langle \varphi_n | \psi \rangle = \int_{\mathbb{R}^3} \varphi_n^*(\mathbf{x})\,\psi(\mathbf{x})\,\mathrm{d}^3x. \tag{37}$$

对于连续谱，求和变为积分：

$$\psi(\mathbf{x}) = \int c(p)\,\varphi_p(\mathbf{x})\,\mathrm{d}p, \tag{38}$$

以及 $c(p) = \langle \varphi_p | \psi \rangle$。

完备关系也可以表示为对本征函数本身的条件。在离散情况下：

$$\sum_n \varphi_n^*(\mathbf{x}')\,\varphi_n(\mathbf{x}) = \delta^{(3)}(\mathbf{x} - \mathbf{x}'), \tag{39}$$

这意味着任何波函数都可以通过对其展开积分来重建：

$$\psi(\mathbf{x}) = \int_{\mathbb{R}^3} \left(\sum_n \varphi_n^*(\mathbf{x}')\varphi_n(\mathbf{x}) \right) \psi(\mathbf{x}')\,\mathrm{d}^3x'. \tag{40}$$

注释 4.2（展望）。在应用 3 和 4 中，我们将遇到两个这样本征函数基的重要例子：

- 动量本征函数 $\varphi_{\mathbf{p}}(\mathbf{x}) = (2\pi\hbar)^{-3/2} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}$，它们构成一个连续基，并通过傅里叶变换导致动量空间表示。
- 位置本征函数 $\varphi_{\mathbf{x}'}(\mathbf{x}) = \delta^{(3)}(\mathbf{x} - \mathbf{x}')$，它们是广义函数（分布），构成一个连续基，并直接产生玻恩规则概率密度 $|\psi(\mathrm{x})|^2$。

它们都是上述一般谱定理的特殊情况，其性质将在后面详细推导。

4.3.2 测量基本假设

在这个数学框架下，我们现在可以陈述测量基本假设。我们假设波函数是归一化的：

$$\int_{\mathbb{R}^3} |\psi(\mathbf{x})|^2\,\mathrm{d}^3x = 1. \tag{41}$$

ˆ 考虑一个由厄米算子 A 表示的可观测量。设 $\{\varphi_n(\mathbf{x})\}$ 是 A 的完备ˆ本征函数集，其对应的本征值为 a_n，满足：

$$\hat{A}\varphi_n = a_n\varphi_n. \tag{42}$$

由于本征函数构成完备正交归一基，任何状态 $\psi(\mathrm{x})$ 都可以展开为：

$$\psi(\mathbf{x}) = \sum_n c_n\varphi_n(\mathbf{x}), \tag{43}$$

展开系数由以下给出：

$$c_n = \langle \varphi_n | \psi \rangle = \int_{\mathbb{R}^3} \varphi_n^*(\mathbf{x})\,\psi(\mathbf{x})\,\mathrm{d}^3x. \tag{44}$$

Page 33 of 61

Postulate III: Measurement

If a quantum system is in a state described by the normalized wavefunction

$$\psi(\mathbf{x}) = \sum_n c_n \varphi_n(\mathbf{x}), \tag{45}$$

then a measurement of the observable corresponding to $\hat{A}$ will yield:

1. **Outcome**: One of the eigenvalues a_n of $\hat{A}$.
2. **Probability**: The probability of obtaining a specific eigenvalue a_n is given by the square of the magnitude of its expansion coefficient,

$$\mathbb{P}(A = a_n) = |c_n|^2. \tag{46}$$

3. **State Collapse**: Immediately after the measurement, the quantum state of the system collapses from $\psi(\mathbf{x})$ to the corresponding eigenfunction $\varphi_n(\mathbf{x})$.

If the measured eigenfunction is not normalized, the post-measurement state must be renormalized.

Remark 4.3. It is crucial to distinguish the three fundamental quantities involved:

- **Quantum State**: $\varphi_n(\mathbf{x})$ (a function in L^2).
- **Measurement Outcome**: a_n (a real number).
- **Probability Amplitude**: $c_n = \langle \varphi_n | \psi \rangle$ (a complex number).
- **Probability**: $\mathbb{P}(A = a_n) = |c_n|^2$ (a real number between 0 and 1).

Remark 4.4 (Continuous Spectra)**.** For operators with a continuous spectrum (such as position or momentum), the expansion becomes an integral and the probability becomes a probability density. For example, the position operator $\hat{\mathbf{x}}$ has eigenfunctions $\varphi_{\mathbf{x}'}(\mathbf{x}) = \delta^{(3)}(\mathbf{x} - \mathbf{x}')$, and the expansion coefficient for a state $\psi(\mathbf{x})$ is simply $c(\mathbf{x}') = \psi(\mathbf{x}')$. This leads naturally to the Born rule for position measurements, which we state next.

Postulate III: Measurement

If a quantum system is in a state described by the normalized wavefunction

$$\psi(\mathbf{x}) = \sum_n c_n \varphi_n(\mathbf{x}), \tag{47}$$

then a measurement of the observable corresponding to $\hat{A}$ will yield:

1. **Outcome**: One of the eigenvalues a_n of $\hat{A}$.
2. **Probability**: The probability of obtaining a specific eigenvalue a_n is given by the square of the magnitude of its expansion coefficient,

$$\mathbb{P}(A = a_n) = |c_n|^2. \tag{48}$$

3. **State Collapse**: Immediately after the measurement, the quantum state of the system collapses from $\psi(\mathbf{x})$ to the corresponding eigenfunction $\varphi_n(\mathbf{x})$.

If the measured eigenfunction is not normalized, the post-measurement state must be renormalized.

第三条公设：测量

如果一个量子系统处于由归一化波函数描述的状态

$$\psi(\mathbf{x}) = \sum_n c_n \varphi_n(\mathbf{x}), \tag{45}$$

ˆ 那么对与 A 对应的可观测量进行测量将得到：

1. 结果： a_n 的一个本征值 $\hat{A}$.
2. 概率：获得特定本征值 a_n 的概率由其展开系数的模平方给出，

$$\mathbb{P}(A = a_n) = |c_n|^2. \tag{46}$$

3. 状态坍缩：测量立即之后，系统的量子态从 $\psi(\mathrm{x})$ 坍缩到相应的本征函数 $\varphi_n(\mathbf{x})$。

如果测量的本征函数未归一化，则测量后的状态必须重新归一化。

备注 4.3。区分所涉及的三个基本量至关重要：

- 量子态： $\varphi_n(\mathbf{x})$(一个在 L^2 中的函数)。
- **测量结果： a_n (一个实数)。**
- **概率幅： $c_n = \langle \varphi_n | \psi \rangle$ (一个复数)。**
- 概率： $\mathbb{P}(A = a_n) = |c_n|^2$ (一个介于0和1之间的实数)。

备注4.4（连续谱）。对于具有连续谱的算子（如位置或动量），展开式变为积分，概率变为概率密度。例如，位置算子ˆx有本征函数 $\varphi_{\mathbf{x}'}(\mathbf{x}) = \delta^{(3)}(\mathbf{x} - \mathbf{x}')$，状态$\psi$(x)的展开系数简单地是 $c(\mathbf{x}') = \psi(\mathbf{x}')$。这自然地引出了位置测量的玻恩法则，我们接下来陈述。

第三条公设：测量

如果一个量子系统处于由归一化波函数描述的状态

$$\psi(\mathbf{x}) = \sum_n c_n \varphi_n(\mathbf{x}), \tag{47}$$

ˆ，那么对与 A 对应的可观测量进行测量将得到：

ˆ 1. 结果： a_n 的一个本征值 a_n 。

2. 概率：获得特定本征值 a_n 的概率由其展开系数的模平方给出，

$$\mathbb{P}(A = a_n) = |c_n|^2. \tag{48}$$

3. 状态坍缩：测量后立即，系统的量子状态从 ψ(x)坍缩到相应的本征函数 $\varphi_n(\mathbf{x})$。

如果测量的本征函数未归一化，则测量后的状态必须重新归一化。

Page 34 of 61

Remark 4.5. It is crucial to distinguish the three fundamental quantities involved:

- **Quantum State**: $\varphi_n(\mathbf{x})$ (a function in L^2).
- **Measurement Outcome**: a_n (a real number).
- **Probability Amplitude**: $c_n = \langle \varphi_n | \psi \rangle$ (a complex number).
- **Probability**: $\mathbb{P}(A = a_n) = |c_n|^2$ (a real number between 0 and 1).

4.3.3 Interlude: A Brief Review of Probability Concepts

To better understand the measurement postulate, let us review some basic ideas from probability theory. Consider a discrete random variable X that can take on values $x_1, x_2, \ldots, x_n$ with probabilities $p_1, p_2, \ldots, p_n$. These probabilities satisfy

$$\sum_{i=1}^{n} p_i = 1. \tag{49}$$

Definition 17 (Average Value (Mean))**.** The average value (or mean) of X, denoted $\langle X \rangle$, is calculated by summing each possible value multiplied by its probability:

$$\langle X \rangle = \sum_{i=1}^{n} x_i p_i. \tag{50}$$

This represents the value we expect to get on average if we perform the measurement many times.

Definition 18 (Variance and Standard Deviation)**.** The variance, σ_X^2, measures the spread of the possible outcomes around the mean. It is defined as the average of the squared deviation from the mean:

$$\sigma_X^2 = \langle (X - \langle X \rangle)^2 \rangle = \sum_{i=1}^{n} (x_i - \langle X \rangle)^2 p_i. \tag{51}$$

The variance can also be expressed in a form that is often easier to compute:

$$\begin{aligned} \sigma_X^2 &= \langle X^2 - 2X\langle X \rangle + \langle X \rangle^2 \rangle & (52)\\ &= \langle X^2 \rangle - 2\langle X \rangle \langle X \rangle + \langle X \rangle^2 & (53)\\ &= \langle X^2 \rangle - \langle X \rangle^2. & (54) \end{aligned}$$

The **standard deviation** σ_X (often denoted ΔX) is the square root of the variance:

$$\Delta X = \sigma_X = \sqrt{\langle X^2 \rangle - \langle X \rangle^2}. \tag{55}$$

The standard deviation gives a measure of the uncertainty in the outcome of a measurement.

We can now apply these concepts directly to quantum mechanics. When we measure an observable $\hat{A}$ on a state ψ, the eigenvalues a_n are the possible outcomes, and the probabilities are $|c_n|^2$.

Definition 19 (Expectation Value in Quantum Mechanics)**.** The expected value (or mean value) of many measurements of the observable $\hat{A}$ on a system in the state $\psi(\mathbf{x})$ is given by

$$\langle \hat{A} \rangle_\psi = \sum_n a_n |c_n|^2. \tag{56}$$

This can be expressed more directly in terms of the wavefunction using the inner product:

$$\boxed{\langle \hat{A} \rangle_\psi = \langle \psi | \hat{A} | \psi \rangle = \int_{\mathbb{R}^3} \psi^*(\mathbf{x}) (\hat{A}\psi)(\mathbf{x}) \mathrm{d}^3 x}. \tag{57}$$

注释4.5。区分所涉及三个基本量至关重要：

- 量子态： $\varphi_n(\mathbf{x})$（一个在 L^2中的函数）。
- **测量结果：** a_n **（一个实数）**。
- **概率幅：** $c_n = \langle \varphi_n | \psi \rangle$ **（一个复数）**。
- 概率： $\mathbb{P}(A = a_n) = |c_n|^2$ （一个介于0和1之间的实数）。

4.3.3 间奏：概率概念简述

为了更好地理解测量公设，让我们回顾概率论中的一些基本概念。考虑一个离散随机变量 X，它可以取值 x_1、 x_2、 …、 x_n，对应的概率为 p_1、 p_2、 …、 p_n。这些概率满足

$$\sum_{i=1}^{n} p_i = 1. \tag{49}$$

定义 17（平均值（均值））。 X 的平均值（或均值）， 记作 $\langle X \rangle$，是通过将每个可能值乘以其概率然后求和来计算的：

$$\langle X \rangle = \sum_{i=1}^{n} x_i p_i. \tag{50}$$

这表示如果我们多次进行测量，期望得到的平均值。

定义 18（方差和标准差）。方差 σ_X^2 衡量可能结果围绕均值的分布程度。它定义为与均值的平方偏差的平均值：

$$\sigma_X^2 = \langle (X - \langle X \rangle)^2 \rangle = \sum_{i=1}^{n} (x_i - \langle X \rangle)^2 p_i. \tag{51}$$

方差也可以表示为一种通常更容易计算的形式：

$$\begin{aligned} \sigma_X^2 &= \langle X^2 - 2X\langle X \rangle + \langle X \rangle^2 \rangle & (52)\\ &= \langle X^2 \rangle - 2\langle X \rangle \langle X \rangle + \langle X \rangle^2 & (53)\\ &= \langle X^2 \rangle - \langle X \rangle^2. & (54) \end{aligned}$$

标准差 σ_X （通常用 ΔX 表示）是方差的平方根：

$$\Delta X = \sigma_X = \sqrt{\langle X^2 \rangle - \langle X \rangle^2}. \tag{55}$$

标准差给出了测量结果不确定性的度量。

ˆ 我们现在可以直接将这些概念应用于量子力学。当我们对一个状态 ψ 测量可观测量 A 时，本征值 a_n 是可能的结果，而概率是 $|c_n|^2$。

定义 19（量子力学中的期望值）。可观测量 A 在系统处于状态 ψ(x) 时的多次测量期望值（或平均值）由下式给出

$$\langle \hat{A} \rangle_\psi = \sum_n a_n |c_n|^2. \tag{56}$$

这可以用波函数的内积更直接地表示：

$$\boxed{\langle \hat{A} \rangle_\psi = \langle \psi | \hat{A} | \psi \rangle = \int_{\mathbb{R}^3} \psi^*(\mathbf{x}) (\hat{A}\psi)(\mathbf{x}) \mathrm{d}^3 x}. \tag{57}$$

Page 35 of 61

Definition 20 (Standard Deviation of an Operator)**.** For a system in a state ψ, the standard deviation (or uncertainty) ΔA of an observable represented by the Hermitian operator $\hat{A}$ is defined, in analogy with the classical definition, as

$$\Delta A = \sqrt{\langle(\hat{A} - \langle\hat{A}\rangle)^2\rangle}. \tag{58}$$

This can be expanded to the equivalent and more practical form:

$$\boxed{\Delta A = \sqrt{\langle\hat{A}^2\rangle_\psi - \langle\hat{A}\rangle_\psi^2}}. \tag{59}$$

This quantity represents the root-mean-square deviation of measurement outcomes from the average value.

Definition 21 (Born Rule)**.** The probability density for finding a particle in a state $\psi(\mathbf{x})$ at position $\mathbf{x}$ is $|\psi(\mathbf{x})|^2$. More formally, the probability of finding the particle in a region $D \subset \mathbb{R}^3$ is

$$\boxed{\mathbb{P}(\mathbf{x} \in D) = \int_D |\psi(\mathbf{x})|^2 \mathrm{d}^3x}. \tag{60}$$

4.4 Applications and Further Topics

4.4.1 Application 1: Commutators and the Uncertainty Principle

The order in which operators act on a state is significant. This is captured by the commutator.

Definition 22 (Commutator)**.** The commutator of two operators $\hat{A}$ and $\hat{B}$ is defined as

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}. \tag{61}$$

The commutator is linear and satisfies important identities such as

$$[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]. \tag{62}$$

Definition 23 (Canonical Commutation Relations (CCR))**.** The fundamental commutation relation between position and momentum in three dimensions is:

$$\boxed{[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}}, \tag{63}$$

where $i, j = x, y, z$. In one dimension, this is

$$[\hat{x}, \hat{p}] = i\hbar. \tag{64}$$

We are now in a position to state the most famous uncertainty principle.

Theorem 4 (Generalized Uncertainty Principle)**.** For any two Hermitian operators $\hat{A}$ and $\hat{B}$ representing physical observables, the product of their standard deviations in any quantum state ψ obeys the Robertson-Schrödinger uncertainty relation:

$$\boxed{(\Delta A)^2(\Delta B)^2 \geq \left|\frac{1}{2i}\langle[\hat{A}, \hat{B}]\rangle\right|^2}. \tag{65}$$

A simpler and more common form, which follows from the above, is:

$$\boxed{\Delta A\, \Delta B \geq \frac{1}{2}\left|\langle[\hat{A}, \hat{B}]\rangle\right|}. \tag{66}$$

This inequality shows that if two operators do not commute ($[\hat{A}, \hat{B}] \neq 0$), there is a fundamental lower bound on the product of their uncertainties. It is impossible to prepare a state in which both observables have arbitrarily precise values simultaneously.

定义20（算符的标准差）。对于一个处于状态 ψ的系统，由厄米算符 A 表示的可观测量（或ˆ不确定性） ΔA 的标准差，类似于经典定义，被定义为

$$\Delta A = \sqrt{\langle(\hat{A} - \langle\hat{A}\rangle)^2\rangle}. \tag{58}$$

这可以扩展为等效且更实用的形式：

$$\boxed{\Delta A = \sqrt{\langle\hat{A}^2\rangle_\psi - \langle\hat{A}\rangle_\psi^2}}. \tag{59}$$

这个量表示测量结果与平均值之间的均方根偏差。

定义21（玻恩法则）。在位置x处找到粒子处于状态 ψ(x)的概率密度为$|\psi(\mathbf{x})|^2$。更正式地说，在区域 $D \subset \mathbb{R}^3$ 中找到粒子的概率为

$$\boxed{\mathbb{P}(\mathbf{x} \in D) = \int_D |\psi(\mathbf{x})|^2 \mathrm{d}^3x}. \tag{60}$$

4.4 应用与进一步主题

4.4.1 应用 1：交换子和不确定性原理

算子作用于态的顺序是重要的。这一点由交换子来描述。

ˆˆ定义 22（交换子）。两个算子 A 和 B 的交换子定义为

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}. \tag{61}$$

交换子是线性的，并满足如下的重要恒等式：

$$[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]. \tag{62}$$

定义 23（正则交换关系（CCR））。在三维空间中，位置和动量之间的基本交换关系是：

$$\boxed{[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}}, \tag{63}$$

其中 i $j = x$ y z, , , 。在一维空间中，这是

$$[\hat{x}, \hat{p}] = i\hbar. \tag{64}$$

我们现在可以陈述最著名的不确定性原理。

ˆˆ定理4（广义不确定性原理）。对于任何两个代表物理量的厄米算子 A 和 B ，在任何量子态 ψ 中，它们的标差乘积遵循罗伯逊-薛定谔不确定性关系：

$$\boxed{(\Delta A)^2(\Delta B)^2 \geq \left|\frac{1}{2i}\langle[\hat{A}, \hat{B}]\rangle\right|^2}. \tag{65}$$

一个更简单和常见的形式，由上述推导得出：

$$\boxed{\Delta A\, \Delta B \geq \frac{1}{2}\left|\langle[\hat{A}, \hat{B}]\rangle\right|}. \tag{66}$$

ˆ ˆ这个不等式表明，如果两个算子不交换 （$[A, B] \neq 0$），它们的不确定性的乘积存在一个基本的下限。不可能制备一个状态，使得两个可观测量同时具有任意精确的值。

We now apply this general principle to the specific case of position and momentum.

Corollary 4.1 (Heisenberg Uncertainty Principle for Position and Momentum)**.** For a single particle in three dimensions, the canonical commutation relations are:

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}, \tag{67}$$

where $i, j = x, y, z$. Applying the generalized uncertainty principle yields the famous Heisenberg uncertainty relations:

$$\Delta x\, \Delta p_x \geq \frac{\hbar}{2}, \tag{68}$$

$$\Delta y\, \Delta p_y \geq \frac{\hbar}{2}, \tag{69}$$

$$\Delta z\, \Delta p_z \geq \frac{\hbar}{2}. \tag{70}$$

It is crucial to note that for different components, e.g., $\hat{x}$ and $\hat{p}_y$, the commutator vanishes:

$$[\hat{x}, \hat{p}_y] = 0. \tag{71}$$

Consequently, there is no fundamental quantum limit preventing the simultaneous precise knowledge of, say, the x-position and the y-momentum of a particle. Their uncertainties can both be zero in principle.

In one dimension, the relations reduce to the single, most commonly quoted form:

$$\boxed{\Delta x\, \Delta p \geq \frac{\hbar}{2}}. \tag{72}$$

Remark 4.6 (Physical Meaning)**.** The Heisenberg uncertainty principle is not a statement about the limitations of our measurement devices, but a fundamental property of quantum states. A particle simply cannot possess both a well-defined position and a well-defined momentum. This is a direct consequence of the wave-like nature of matter: a wave confined to a small region in space (small Δx) is composed of a wide range of wavelengths (large Δp), and vice versa.

Physically, this principle implies that it is impossible to simultaneously know the exact position and exact momentum of a particle. A state cannot be a simultaneous eigenstate of both $\hat{x}$ and $\hat{p}$ because their operators do not commute.

An important example of commuting operators is $[\hat{p}, \hat{H}] = 0$ for a free particle. This indicates that momentum and energy can be measured simultaneously in such a system.

4.4.2 Application 2: Dynamics of a Free Particle

The Hamiltonian for a single particle of mass m in a potential $V(\mathbf{x})$ is

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\mathbf{x}) = -\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{x}). \tag{73}$$

For a **free particle**, $V(\mathbf{x}) = 0$, and the time-dependent Schrödinger equation becomes:

$$i\hbar\frac{\partial}{\partial t}\psi(\mathbf{x}, t) = -\frac{\hbar^2}{2m}\nabla^2\psi(\mathbf{x}, t). \tag{74}$$

We seek to derive the fundamental wave relations from this equation and our operator definitions.

我们现在将这个一般原理应用于位置和动量的特殊情况。

推论4.1（海森堡位置与动量不确定关系）。对于一个三维空间中的单个粒子，正则对易关系为：

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}, \tag{67}$$

其中 i $j = x$ y、 i $j = x$ y、 i $j = x$ y、 z。应用广义不确定关系可以得到著名的海森堡不确定关系：

$$\Delta x\, \Delta p_x \geq \frac{\hbar}{2}, \tag{68}$$

$$\Delta y\, \Delta p_y \geq \frac{\hbar}{2}, \tag{69}$$

$$\Delta z\, \Delta p_z \geq \frac{\hbar}{2}. \tag{70}$$

需要注意的是，对于不同的分量，例如 ˆx 和 ˆp_y，对易子为零：

$$[\hat{x}, \hat{p}_y] = 0. \tag{71}$$

因此，不存在基本的量子限制，阻止同时精确地知道粒子的 x-位置和 y-动量。原则上，它们的测量不确定性都可以为零。

在一维情况下，关系简化为最常见的形式：

$$\boxed{\Delta x\, \Delta p \geq \frac{\hbar}{2}}. \tag{72}$$

备注4.6（物理意义）。海森堡不确定性原理并非关于我们测量设备的限制，而是量子态的基本性质。粒子不可能同时具有明确的位置和明确的动量。这是物质波动性的直接结果：一个被限制在空间小区域（小 Δx）的波，由一系列波长（大 Δp）组成，反之亦然。

从物理上看，这一原理意味着不可能同时知道一个粒子的精确位置和精确动量。一个状态不能同时是ˆx 和ˆp 的纠缠态，因为它们的算子不交换。

ˆ交换算子的一个重要例子是 $[\hat{p}, H] = 0$ 对于自由粒子。这表明在这样系统中，动量和能量可以同时测量。

4.4.2 应用2：自由粒子的动力学

质量为 m 的单个粒子在势能 V(x) 下的哈密顿量为

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\mathbf{x}) = -\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{x}). \tag{73}$$

对于自由粒子， $V(\mathbf{x}) = 0$，含时薛定谔方程变为：

$$i\hbar\frac{\partial}{\partial t}\psi(\mathbf{x}, t) = -\frac{\hbar^2}{2m}\nabla^2\psi(\mathbf{x}, t). \tag{74}$$

我们试图从该方程和我们的算子定义中推导出基本的波动关系。

1. **De Broglie Relation:** First, consider the eigenvalue problem for the momentum operator, $\hat{\mathbf{p}}\varphi_{\mathbf{p}}(\mathbf{x}) = \mathbf{p}\varphi_{\mathbf{p}}(\mathbf{x})$. Using $\hat{\mathbf{p}} = -i\hbar\nabla$, we have:

$$-i\hbar\nabla\varphi_{\mathbf{p}}(\mathbf{x}) = \mathbf{p}\varphi_{\mathbf{p}}(\mathbf{x}). \tag{75}$$

This is a differential equation whose solution is an exponential function. In three dimensions, the solution is:

$$\varphi_{\mathbf{p}}(\mathbf{x}) = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}. \tag{76}$$

By comparing this to a standard plane wave of the form $e^{i\mathbf{k}\cdot\mathbf{x}}$, we identify the wave vector $\mathbf{k}$ as $\mathbf{k} = \mathbf{p}/\hbar$. This yields the de Broglie relation:

$$\boxed{\mathbf{p} = \hbar\mathbf{k}}. \tag{77}$$

2. **Planck Relation and Dispersion:** Now, we look for a solution to the time-dependent equation (74) that is consistent with this. We try a plane wave solution that includes the time dependence we found from separation of variables:

$$\psi(\mathbf{x}, t) = Ae^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}. \tag{78}$$

Substituting this into the left-hand side of (74) gives:

$$i\hbar\frac{\partial}{\partial t}\psi = i\hbar(-i\omega)\psi = \hbar\omega\psi. \tag{79}$$

Substituting into the right-hand side gives:

$$-\frac{\hbar^2}{2m}\nabla^2\psi = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)} \tag{80}$$

$$= -\frac{\hbar^2}{2m}(i^2k_x^2 + i^2k_y^2 + i^2k_z^2)\psi \tag{81}$$

$$= -\frac{\hbar^2}{2m}(-k_x^2 - k_y^2 - k_z^2)\psi \tag{82}$$

$$= \frac{\hbar^2k^2}{2m}\psi. \tag{83}$$

For the equation to hold, the coefficients of ψ on both sides must be equal:

$$\hbar\omega = \frac{\hbar^2k^2}{2m}. \tag{84}$$

Recognizing the right-hand side as the kinetic energy $E = p^2/2m = \hbar^2k^2/2m$, we obtain the Planck relation and the dispersion relation:

$$\boxed{E = \hbar\omega} \quad \text{and} \quad \boxed{E = \frac{\hbar^2k^2}{2m}}. \tag{85}$$

We see that the definitions of the position and momentum operators, combined with the Schrödinger equation, directly yield the fundamental relations of wave-particle duality.

4.4.3 Application 3: Momentum Space Representation

The momentum eigenfunctions $\varphi_{\mathbf{p}}(\mathbf{x})$ form a complete basis. This means that any state $\psi(\mathbf{x})$ can be expressed as a linear combination of these basis functions. Because the momentum eigenvalues $\mathbf{p}$ are continuous, this expansion takes the form of an integral rather than a sum.

1. 德布罗意关系：首先，考虑动量算符的本征值问题，$\hat{\mathbf{p}}\varphi_{\mathbf{p}}(\mathbf{x}) = \mathbf{p}\varphi_{\mathbf{p}}(\mathbf{x})$. 使用 $\hat{\mathbf{p}} = -i\hbar\nabla$，我们得到：

$$-i\hbar\nabla\varphi_{\mathbf{p}}(\mathbf{x}) = \mathbf{p}\varphi_{\mathbf{p}}(\mathbf{x}). \tag{75}$$

这是一个其解为指数函数的微分方程。在三维空间中，解为：

$$\varphi_{\mathbf{p}}(\mathbf{x}) = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}. \tag{76}$$

通过将其与标准平面波形式 $e^{i\mathbf{k}\cdot\mathbf{x}}$ 进行比较，我们确定了波矢 k 为 $\mathbf{k} = \mathbf{p}/\hbar$。这给出了德布罗意关系：

$$\boxed{\mathbf{p} = \hbar\mathbf{k}}. \tag{77}$$

2. 普朗克关系和色散：现在，我们寻找与这个方程（74）一致的时间相关方程的解。我们尝试一个包含变量分离中找到的时间依赖性的平面波解：$i(\mathbf{k}\cdot$ $t)$

$$\psi(\mathbf{x}, t) = Ae \quad \mathbf{x}-\omega \quad . \tag{78}$$

将此代入(74)的左侧得到：

$$i\hbar\frac{\partial}{\partial t}\psi = i\hbar(-i\omega)\psi = \hbar\omega\psi. \tag{79}$$

代入右侧得到：

$$-\frac{\hbar^2}{2m}\nabla^2\psi = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)} \tag{80}$$

$$= -\frac{\hbar^2}{2m}(i^2k_x^2 + i^2k_y^2 + i^2k_z^2)\psi \tag{81}$$

$$= -\frac{\hbar^2}{2m}(-k_x^2 - k_y^2 - k_z^2)\psi \tag{82}$$

$$= \frac{\hbar^2k^2}{2m}\psi. \tag{83}$$

为使该方程成立，两边的 ψ 系数必须相等：

$$\hbar\omega = \frac{\hbar^2k^2}{2m}. \tag{84}$$

将右侧识别为动能 $E = p^2/2m = \hbar^2k^2/2m$，我们得到普朗克关系和色散关系：

$$\boxed{E = \hbar\omega} \quad \text{and} \quad \boxed{E = \frac{\hbar^2k^2}{2m}}. \tag{85}$$

我们看到，位置和动量算符的定义，结合薛定谔方程，直接给出了波粒二象性的基本关系。

4.4.3 应用3：动量空间表示

动量本征态函数 $\varphi_{\mathbf{p}}(\mathbf{x})$ 构成一个完备基。这意味着任何态 $\psi(\mathbf{x})$ 都可以表示为这些基函数的线性组合。由于动量本征值 p 是连续的，这个展开形式表现为一个积分而不是求和。

We begin by writing the expansion as

$$\psi(\mathbf{x}) = \int_{\mathbb{R}^3} c(\mathbf{p})\, \varphi_{\mathbf{p}}(\mathbf{x})\, \mathrm{d}^3 p. \tag{86}$$

Here, $c(\mathbf{p})$ is a complex coefficient that tells us how much of the momentum eigenstate $\mathbf{p}$ is contained in $\psi(\mathbf{x})$. Substituting the explicit form of the momentum eigenfunction, we get:

$$\psi(\mathbf{x}) = \int_{\mathbb{R}^3} c(\mathbf{p})\, \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}\, \mathrm{d}^3 p. \tag{87}$$

It is convenient to define a new function $\tilde{\psi}(\mathbf{p})$, called the **momentum space wavefunction**, which is essentially the density of the coefficient $c(\mathbf{p})$. We set

$$\tilde{\psi}(\mathbf{p}) = (2\pi\hbar)^{3/2}\, c(\mathbf{p}). \tag{88}$$

Substituting this definition into equation (87), the factors of $(2\pi\hbar)^{3/2}$ cancel, yielding:

$$\boxed{\psi(\mathbf{x}) = \frac{1}{(2\pi\hbar)^{3/2}} \int_{\mathbb{R}^3} \tilde{\psi}(\mathbf{p})\, e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}\, \mathrm{d}^3 p}. \tag{89}$$

This expression is precisely the **inverse Fourier transform** of $\tilde{\psi}(\mathbf{p})$.

To find the function $\tilde{\psi}(\mathbf{p})$ from $\psi(\mathbf{x})$, we need the inverse relation. This is obtained from the orthonormality of the momentum eigenfunctions:

$$\langle \varphi_{\mathbf{p}'} | \varphi_{\mathbf{p}} \rangle = \int_{\mathbb{R}^3} \varphi^*_{\mathbf{p}'}(\mathbf{x}) \varphi_{\mathbf{p}}(\mathbf{x}) \mathrm{d}^3 x = \delta^{(3)}(\mathbf{p}' - \mathbf{p}). \tag{90}$$

Now, take the inner product of both sides of the expansion in equation (87) with a specific eigenfunction $\varphi_{\mathbf{p}'}(\mathbf{x})$:

$$\begin{aligned} \langle \varphi_{\mathbf{p}'} | \psi \rangle &= \int_{\mathbb{R}^3} \varphi^*_{\mathbf{p}'}(\mathbf{x}) \psi(\mathbf{x}) \mathrm{d}^3 x & (91)\\ &= \int_{\mathbb{R}^3} \left(\frac{1}{(2\pi\hbar)^{3/2}} e^{-i\mathbf{p}'\cdot\mathbf{x}/\hbar} \right) \left(\frac{1}{(2\pi\hbar)^{3/2}} \int_{\mathbb{R}^3} \tilde{\psi}(\mathbf{p}) e^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \mathrm{d}^3 p \right) \mathrm{d}^3 x. & (92) \end{aligned}$$

Interchanging the order of integration and using the integral representation of the delta function,

$$\int_{\mathbb{R}^3} e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}/\hbar} \mathrm{d}^3 x = (2\pi\hbar)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}'), \tag{93}$$

we obtain:

$$\begin{aligned} \langle \varphi_{\mathbf{p}'} | \psi \rangle &= \frac{1}{(2\pi\hbar)^3} \int_{\mathbb{R}^3} \tilde{\psi}(\mathbf{p}) \left(\int_{\mathbb{R}^3} e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}/\hbar} \mathrm{d}^3 x \right) \mathrm{d}^3 p & (94)\\ &= \frac{1}{(2\pi\hbar)^3} \int_{\mathbb{R}^3} \tilde{\psi}(\mathbf{p}) (2\pi\hbar)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}') \mathrm{d}^3 p & (95)\\ &= \tilde{\psi}(\mathbf{p}'). & (96) \end{aligned}$$

Since the left-hand side is just the inner product, we arrive at the **forward Fourier transform** relation:

$$\boxed{\tilde{\psi}(\mathbf{p}) = \frac{1}{(2\pi\hbar)^{3/2}} \int_{\mathbb{R}^3} \psi(\mathbf{x})\, e^{-i\mathbf{p}\cdot\mathbf{x}/\hbar}\, \mathrm{d}^3 x}. \tag{97}$$

我们首先将展开式写成

$$\psi(\mathbf{x}) = \int_{\mathbb{R}^3} c(\mathbf{p})\, \varphi_{\mathbf{p}}(\mathbf{x})\, \mathrm{d}^3 p. \tag{86}$$

在这里， c(p) 是一个复系数，它告诉我们动量本征态 p 中包含了多少 ψ(x)。代入动量本征函数的具体形式，我们得到：

$$\psi(\mathbf{x}) = \int_{\mathbb{R}^3} c(\mathbf{p})\, \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}\, \mathrm{d}^3 p. \tag{87}$$

˜定义一个新的函数 $\psi(\mathbf{p})$，称为动量空间波函数，它本质上就是系数 c(p) 的密度。我们设

$$\tilde{\psi}(\mathbf{p}) = (2\pi\hbar)^{3/2}\, c(\mathbf{p}). \tag{88}$$

将这个定义代入方程 (87)， $(2\pi\hbar)^{3/2}$ 的因子相互抵消，得到：

$$\boxed{\psi(\mathbf{x}) = \frac{1}{(2\pi\hbar)^{3/2}} \int_{\mathbb{R}^3} \tilde{\psi}(\mathbf{p})\, e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}\, \mathrm{d}^3 p}. \tag{89}$$

˜这个表达式正是 $\psi(\mathbf{p})$ 的逆傅里叶变换。

˜要从 ψ(x)中找到函数 ψ(p)，我们需要逆关系。这可以从动量本征态函数的正交归一性得到：

$$\langle \varphi_{\mathbf{p}'} | \varphi_{\mathbf{p}} \rangle = \int_{\mathbb{R}^3} \varphi^*_{\mathbf{p}'}(\mathbf{x}) \varphi_{\mathbf{p}}(\mathbf{x}) \mathrm{d}^3 x = \delta^{(3)}(\mathbf{p}' - \mathbf{p}). \tag{90}$$

现在，用特定的本征态函数$\varphi_{\mathbf{p}'}(\mathbf{x})$对公式(87)展开式的两边取内积：

$$\begin{aligned} \langle \varphi_{\mathbf{p}'} | \psi \rangle &= \int_{\mathbb{R}^3} \varphi^*_{\mathbf{p}'}(\mathbf{x}) \psi(\mathbf{x}) \mathrm{d}^3 x & (91)\\ &= \int_{\mathbb{R}^3} \left(\frac{1}{(2\pi\hbar)^{3/2}} e^{-i\mathbf{p}'\cdot\mathbf{x}/\hbar} \right) \left(\frac{1}{(2\pi\hbar)^{3/2}} \int_{\mathbb{R}^3} \tilde{\psi}(\mathbf{p}) e^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \mathrm{d}^3 p \right) \mathrm{d}^3 x. & (92) \end{aligned}$$

交换积分顺序，并使用δ函数的积分表示，

$$\int_{\mathbb{R}^3} e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}/\hbar} \mathrm{d}^3 x = (2\pi\hbar)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}'), \tag{93}$$

我们得到：

$$\begin{aligned} \langle \varphi_{\mathbf{p}'} | \psi \rangle &= \frac{1}{(2\pi\hbar)^3} \int_{\mathbb{R}^3} \tilde{\psi}(\mathbf{p}) \left(\int_{\mathbb{R}^3} e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}/\hbar} \mathrm{d}^3 x \right) \mathrm{d}^3 p & (94)\\ &= \frac{1}{(2\pi\hbar)^3} \int_{\mathbb{R}^3} \tilde{\psi}(\mathbf{p}) (2\pi\hbar)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}') \mathrm{d}^3 p & (95)\\ &= \tilde{\psi}(\mathbf{p}'). & (96) \end{aligned}$$

由于左侧只是内积，我们得到正向傅里叶变换关系：

$$\boxed{\tilde{\psi}(\mathbf{p}) = \frac{1}{(2\pi\hbar)^{3/2}} \int_{\mathbb{R}^3} \psi(\mathbf{x})\, e^{-i\mathbf{p}\cdot\mathbf{x}/\hbar}\, \mathrm{d}^3 x}. \tag{97}$$

A key property of the Fourier transform is that it preserves the norm of the function (Plancherel's theorem):

$$\int_{\mathbb{R}^3} |\psi(\mathbf{x})|^2 \mathrm{d}^3x = \int_{\mathbb{R}^3} \left| \frac{1}{(2\pi\hbar)^{3/2}} \int_{\mathbb{R}^3} \tilde{\psi}(\mathbf{p}) e^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \mathrm{d}^3p \right|^2 \mathrm{d}^3x \tag{98}$$

$$= \int_{\mathbb{R}^3} |\tilde{\psi}(\mathbf{p})|^2 \mathrm{d}^3p. \tag{99}$$

This confirms that if $\psi(\mathbf{x})$ is normalized, then $\tilde{\psi}(\mathbf{p})$ is also normalized. Therefore, $|\tilde{\psi}(\mathbf{p})|^2$ can be interpreted as a probability density in momentum space. The probability of measuring the particle's momentum in a region D of momentum space is:

$$\boxed{\mathbb{P}(\mathbf{p} \in D) = \int_D |\tilde{\psi}(\mathbf{p})|^2 \mathrm{d}^3p}. \tag{100}$$

Remark 4.7 (Normalization of Plane Waves)**.** The momentum eigenfunction $\varphi_{\mathbf{p}}(\mathbf{x}) = (2\pi\hbar)^{-3/2} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}$ is not square-integrable, meaning it is not a member of L^2. Its normalization is to a Dirac delta function, as we used above:

$$\langle \varphi_{\mathbf{p}'} | \varphi_{\mathbf{p}} \rangle = \delta^{(3)}(\mathbf{p}' - \mathbf{p}). \tag{101}$$

4.4.4 Application 4: Position Eigenstates

We now seek the eigenfunctions of the position operator $\hat{\mathbf{x}}$. Let $\varphi_{\mathbf{x}'}(\mathbf{x})$ be an eigenstate with eigenvalue $\mathbf{x}'$:

$$\hat{\mathbf{x}} \varphi_{\mathbf{x}'}(\mathbf{x}) = \mathbf{x}' \varphi_{\mathbf{x}'}(\mathbf{x}). \tag{102}$$

Recall the sifting property of the Dirac delta function, which allows any function to be expanded as:

$$\psi(\mathbf{x}) = \int_{\mathbb{R}^3} \psi(\mathbf{x}') \delta^{(3)}(\mathbf{x} - \mathbf{x}') \mathrm{d}^3x'. \tag{103}$$

This expansion suggests that the delta functions $\delta^{(3)}(\mathbf{x} - \mathbf{x}')$ act as a basis for expanding wavefunctions. To verify they are eigenfunctions, we substitute $\varphi_{\mathbf{x}'}(\mathbf{x}) = \delta^{(3)}(\mathbf{x} - \mathbf{x}')$ into the eigenvalue equation:

$$\hat{\mathbf{x}} \delta^{(3)}(\mathbf{x} - \mathbf{x}') = \mathbf{x} \delta^{(3)}(\mathbf{x} - \mathbf{x}') \tag{104}$$

$$= \mathbf{x}' \delta^{(3)}(\mathbf{x} - \mathbf{x}'), \tag{105}$$

where the last equality uses the defining property of the delta function: $f(\mathbf{x})\delta(\mathbf{x} - \mathbf{x}') = f(\mathbf{x}')\delta(\mathbf{x} - \mathbf{x}')$. Thus, the (generalized) eigenfunctions of the position operator are

$$\varphi_{\mathbf{x}'}(\mathbf{x}) = \delta^{(3)}(\mathbf{x} - \mathbf{x}'). \tag{106}$$

Remark 4.8 (Normalization of Delta Functions)**.** These position eigenfunctions are also not square-integrable. They are normalized to a delta function in their eigenvalues:

$$\langle \varphi_{\mathbf{x}'} | \varphi_{\mathbf{x}''} \rangle = \int_{\mathbb{R}^3} \delta^{(3)}(\mathbf{x} - \mathbf{x}') \, \delta^{(3)}(\mathbf{x} - \mathbf{x}'') \, \mathrm{d}^3x = \delta^{(3)}(\mathbf{x}' - \mathbf{x}''). \tag{107}$$

This means that while the position eigenfunctions are not normalizable in the usual sense (their norm is infinite), they satisfy a weaker form of orthonormality: the inner product of two such eigenfunctions is zero unless their eigenvalues are identical, in which case it is infinite in a controlled way represented by the Dirac delta function.

Remark 4.9 (Comparison with Momentum Eigenfunctions)**.** It is instructive to compare the normalization of position and momentum eigenfunctions:

傅里叶变换的一个关键性质是它保持函数的范数（普朗歇尔定理）：

$$\int_{\mathbb{R}^3} |\psi(\mathbf{x})|^2 \mathrm{d}^3x = \int_{\mathbb{R}^3} \left| \frac{1}{(2\pi\hbar)^{3/2}} \int_{\mathbb{R}^3} \tilde{\psi}(\mathbf{p}) e^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \mathrm{d}^3p \right|^2 \mathrm{d}^3x \tag{98}$$

$$= \int_{\mathbb{R}^3} |\tilde{\psi}(\mathbf{p})|^2 \mathrm{d}^3p. \tag{99}$$

这证实了如果 ψ(x) 是归一化的，那么 $\tilde{\psi}(\mathbf{p})$ 也是归一化的。因此，$|\tilde{\psi}(\mathbf{p})|^2$ 可以解释为动量空间中的概率密度。在动量空间的一个区域 D 中测量粒子动量的概率为：

$$\boxed{\mathbb{P}(\mathbf{p} \in D) = \int_D |\tilde{\psi}(\mathbf{p})|^2 \mathrm{d}^3p}. \tag{100}$$

备注 4.7（平面波的归一化）。动量本征函数 $\varphi_{\mathbf{p}}(\mathbf{x}) = (2\pi\hbar)^{-3/2} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}$ 不是平方可积的，这意味着它不是 L^2 的成员。它的归一化是一个狄拉克δ函数，正如我们上面所用的：

$$\langle \varphi_{\mathbf{p}'} | \varphi_{\mathbf{p}} \rangle = \delta^{(3)}(\mathbf{p}' - \mathbf{p}). \tag{101}$$

4.4.4 应用 4：位置本征态

我们现在寻求位置算符 ˆx 的本征函数。设 $\varphi_{\mathbf{x}'}(\mathbf{x})$ 是一个本征态，其本征值为 x′：

$$\hat{\mathbf{x}} \varphi_{\mathbf{x}'}(\mathbf{x}) = \mathbf{x}' \varphi_{\mathbf{x}'}(\mathbf{x}). \tag{102}$$

回顾狄拉克δ函数的筛选性质，它允许任何函数展开为：

$$\psi(\mathbf{x}) = \int_{\mathbb{R}^3} \psi(\mathbf{x}') \delta^{(3)}(\mathbf{x} - \mathbf{x}') \mathrm{d}^3x'. \tag{103}$$

这种展开表明，δ函数 $\delta^{(3)}(\mathbf{x} - \mathbf{x}')$ 充当展开波函数的基。为了验证它们是本征态，我们将 $\varphi_{\mathbf{x}'}(\mathbf{x}) = \delta^{(3)}(\mathbf{x} - \mathbf{x}')$ 代入本征值方程：

$$\hat{\mathbf{x}} \delta^{(3)}(\mathbf{x} - \mathbf{x}') = \mathbf{x} \delta^{(3)}(\mathbf{x} - \mathbf{x}') \tag{104}$$

$$= \mathbf{x}' \delta^{(3)}(\mathbf{x} - \mathbf{x}'), \tag{105}$$

在最后一个等式中使用了δ函数的定义性质：$f(\mathbf{x})\delta(\mathbf{x} - \mathbf{x}') = f(\mathbf{x}')\delta(\mathbf{x} - \mathbf{x}')$。因此，位置算符的（广义）本征函数是

$$\varphi_{\mathbf{x}'}(\mathbf{x}) = \delta^{(3)}(\mathbf{x} - \mathbf{x}'). \tag{106}$$

注4.8（δ函数的归一化）。这些位置本征函数也不是平方可积的。它们的本征值归一化到δ函数：

$$\langle \varphi_{\mathbf{x}'} | \varphi_{\mathbf{x}''} \rangle = \int_{\mathbb{R}^3} \delta^{(3)}(\mathbf{x} - \mathbf{x}') \, \delta^{(3)}(\mathbf{x} - \mathbf{x}'') \, \mathrm{d}^3x = \delta^{(3)}(\mathbf{x}' - \mathbf{x}''). \tag{107}$$

这意味着，虽然位置本征函数在通常意义上不可归一化（它们的范数是无穷大），但它们满足一种较弱的正交归一性：两个此类本征函数的内积为零，除非它们的本征值相同，在这种情况下，它以由狄拉克δ函数表示的受控方式无限大。

备注4.9（与动量本征函数的比较）。比较位置和动量本征函数的归一化是有益的：

- **Position eigenfunctions**: $\varphi_{\mathbf{x}'}(\mathbf{x}) = \delta^{(3)}(\mathbf{x} - \mathbf{x}')$ with normalization

$$\langle \varphi_{\mathbf{x}'} | \varphi_{\mathbf{x}''} \rangle = \delta^{(3)}(\mathbf{x}' - \mathbf{x}''). \tag{108}$$

- **Momentum eigenfunctions**: $\varphi_{\mathbf{p}}(\mathbf{x}) = \dfrac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}$ with normalization

$$\langle \varphi_{\mathbf{p}'} | \varphi_{\mathbf{p}} \rangle = \delta^{(3)}(\mathbf{p}' - \mathbf{p}). \tag{109}$$

In both cases, the eigenfunctions are not elements of L^2 (they are not square-integrable), but they form a complete basis in the sense of Dirac delta function normalization. This is why they are often called **generalized eigenfunctions**.

- **位置本征函数：** $\varphi_{\mathbf{x}'}(\mathbf{x}) = \delta^{(3)}(\mathbf{x} - \mathbf{x}')$ **具有归一化**

$$\langle \varphi_{\mathbf{x}'} | \varphi_{\mathbf{x}''} \rangle = \delta^{(3)}(\mathbf{x}' - \mathbf{x}''). \tag{108}$$

- **动量本征函数：** $\varphi_{\mathbf{p}}(\mathbf{x}) = \dfrac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}$ 具有归一化

$$\langle \varphi_{\mathbf{p}'} | \varphi_{\mathbf{p}} \rangle = \delta^{(3)}(\mathbf{p}' - \mathbf{p}). \tag{109}$$

在两种情况下，本征函数都不是 L^2 的元素（它们不是平方可积的），但在狄拉克δ函数规范的意义下，它们构成一个完备基。这就是为什么它们通常被称为广义本征函数。

Key Differences from Classical Physics

Aspect	Classical Physics	Quantum Mechanics
State Description	Position & momentum (x, p)	Wave function $\Psi(x, t)$
Determinism	Completely deterministic	Probabilistic predictions
Observables	Continuous values	Quantized/discrete values
Measurement	Passive observation	Active disturbance
Superposition	Not allowed	Fundamental principle

Quantum Operators Dictionary

Observable	Classical	Quantum Operator	Eigenvalue Eqn
Position	x	$\hat{x} = x$	$\hat{x}\psi = x_0\psi$
Momentum	p_x	$\hat{p}_x = -i\hbar\frac{\partial}{\partial x}$	$\hat{p}_x\psi = p\psi$
Kinetic Energy	$\frac{p^2}{2m}$	$\hat{T} = -\frac{\hbar^2}{2m}\nabla^2$	$\hat{T}\psi = E\psi$
Potential Energy	$V(x)$	$\hat{V} = V(x)$	$\hat{V}\psi = V(x)\psi$
Total Energy	$H = T + V$	$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V(x)$	$\hat{H}\psi = E\psi$
Angular Momentum	$\mathbf{L} = \mathbf{r} \times \mathbf{p}$	$\hat{L}_z = -i\hbar\frac{\partial}{\partial\phi}$	$\hat{L}_z\psi = m\hbar\psi$

与经典物理学的主要区别

方面	经典物理学	量子力学
状态描述	位置与动量 (x, p)	波函数 $\Psi(x, t)$
决定论	完全决定论	概率预测
可观察量	连续值	量化/离散值
测量	被动观察	主动扰动
叠加	不允许	基本原理

量子算子词典

可观察的	经典	量子算符	本征值方程
位置	x	$\hat{x} = x$	$\hat{x}\psi = x_0\psi$
动量	p_x	$\hat{p}_x = -i\hbar\frac{\partial}{\partial x}$	$\hat{p}_x\psi = p\psi$
动能	$\frac{p^2}{2m}$	$\hat{T} = -\frac{\hbar^2}{2m}\nabla^2$	$\hat{T}\psi = E\psi$
势能	$V(x)$	$\hat{V} = V(x)$	$\hat{V}\psi = V(x)\psi$
总能量	$H = T + V$	$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V(x)$	$\hat{H}\psi = E\psi$
角动量	$\mathbf{L} = \mathbf{r} \times \mathbf{p}$	$\hat{L}_z = -i\hbar\frac{\partial}{\partial\phi}$	$\hat{L}_z\psi = m\hbar\psi$

5 The Historical Approach to Quantum Mechanics

Quantum mechanics emerged from experimental observations that classical physics couldn't explain. The development occurred in two main phases:

Old Quantum Theory (1900-1925)
Planck, Einstein, Bohr, Sommerfeld

Matrix & Wave Mechanics

Modern Quantum Mechanics (1925-1930)
Heisenberg, Schrödinger, Dirac, Born, Jordan, Pauli

5.1 The Crisis of Classical Physics (Late 19th Century)

By the late 1800s, classical physics (Newtonian mechanics, Maxwell's electromagnetism, thermodynamics) appeared complete. However, several phenomena defied classical explanation:

Unsolved Problems in 1900

1. **Blackbody Radiation**: Spectrum couldn't be explained by classical electrodynamics
2. **Photoelectric Effect**: Light behaving as particles
3. **Atomic Spectra**: Discrete emission/absorption lines
4. **Specific Heat of Solids**: Deviations from Dulong-Petit law at low temperatures
5. **Stability of Atoms**: Why don't electrons spiral into the nucleus?

In what follows, we will discuss some of these historically importan problems in details.

5.1.1 Blackbody Radiation and Planck's Quantum Hypothesis (1900)

Definition 24 (Blackbody)**.** *An ideal object that absorbs all radiation incident upon it and emits radiation with a characteristic spectrum that depends only on its temperature.*

The classical Rayleigh-Jeans law predicted the **ultraviolet catastrophe**:

$$\rho_{\text{classical}}(\nu, T) = \frac{8\pi\nu^2}{c^3} k_B T \xrightarrow{\nu\to\infty} \infty!$$

Planck's Radical Assumption (December 1900)

Energy of electromagnetic oscillators is **quantized**:

$$E = nh\nu, \quad n = 0, 1, 2, \ldots$$

where $h = 6.626 \times 10^{-34}$ J·s is **Planck's constant**.

This led to Planck's radiation law:

$$\boxed{\rho_{\text{Planck}}(\nu, T) = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/k_B T} - 1}}$$

Significance: First introduction of quantization, though Planck considered it a mathematical trick.

5 量子力学的历史方法

量子力学源于经典物理学无法解释的实验观察。其发展经历了两个主要阶段：

Old Quantum Theory (1900-1925)
Planck, Einstein, Bohr, Sommerfeld

矩阵与波力学

现代量子力学 (1925-1930)
海森堡、薛定谔、狄拉克、玻恩、约当、泡利

5.1 经典物理学的危机（19世纪末）

到19世纪末，经典物理学（牛顿力学、麦克斯韦电磁学、热力学）似乎已经完善。然而，一些现象无法用经典理论解释：

1900年的未解之谜

1. 黑体辐射：经典电磁学无法解释其光谱
2. 光电效应：光表现出粒子性
3. 原子光谱：离散的发射/吸收线
4. 固体的比热容：低温下偏离杜隆-佩蒂定律
5. 原子的稳定性：为什么电子不会螺旋式坠入原子核?

在接下来的内容中，我们将详细讨论其中一些具有历史意义的重要问题。

5.1.1 黑体辐射与普朗克的量子假说（1900年）

定义*24*（黑体）。一种能吸收所有入射辐射的理想物体，并发出仅取决于其温度的特征谱辐射。

经典瑞利-金斯定律预言了紫外灾难：

$$\rho_{\text{classical}}(\nu, T) = \frac{8\pi\nu^2}{c^3} k_B T \xrightarrow{\nu\to\infty} \infty!$$

普朗克的革命性假设（1900年12月）

电磁振子的能量是量子化的：

$$E = nh\nu, \quad n = 0, 1, 2, \ldots$$

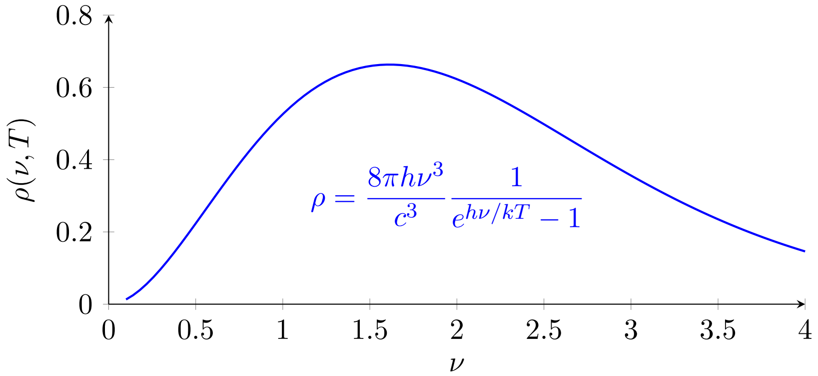
其中 $h = 6.626 \times 10^{-34}$ **J·s 是普朗克常数。**

这导致了普朗克辐射定律：

$$\boxed{\rho_{\text{Planck}}(\nu, T) = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/k_B T} - 1}}$$

意义：首次引入了量子化概念，尽管普朗克认为它是一种数学技巧。

Page 43 of 61



Key Features of Planck's Spectrum

- **Ultraviolet catastrophe**: Rayleigh-Jeans law diverges as ν^2 (orange curve)
- **Quantum resolution**: Planck's law remains finite due to $e^{h\nu/k_BT} - 1$ denominator
- **Wien's displacement law**: Peak frequency $\nu_{\text{max}} \propto T$
- **Stefan-Boltzmann law**: Total power $\int_0^\infty \rho(\nu, T)d\nu \propto T^4$
- **Temperature dependence**: Higher T (red dashed) $\rightarrow$ higher peak, shifts to higher ν

Planck's Radiation Low and the Big Bang: Quantum Mechanics Across Cosmic Scales

Remarkable connection: The cosmic microwave background (CMB)—the afterglow of the Big Bang—exhibits a perfect Planck blackbody spectrum at $T = 2.7255$ K. This provides:

- **Cosmological validation**: Quantum mechanics accurately describes the universe's oldest radiation
- **Experimental triumph**: COBE satellite measurements (1990) showed $< 0.03\%$ deviation from Planck's formula
- **Historical irony**: Planck's "mathematical convenience" describes radiation from 13.8 billion years ago!
- **Universal scope**: Same quantum principles apply from atomic to cosmological scales

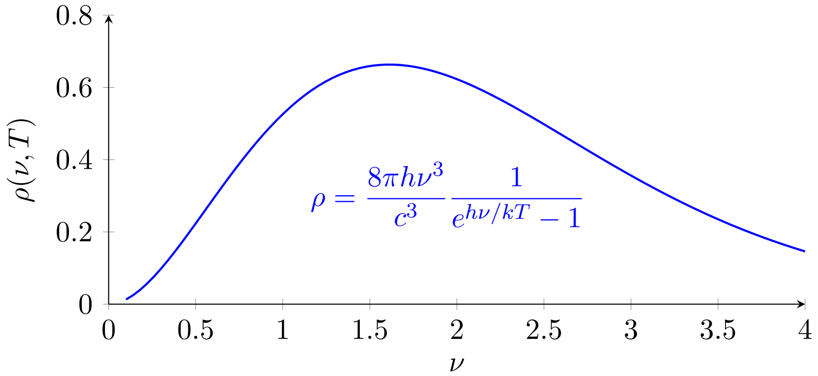
The CMB spectrum represents the most precise blackbody measurement in nature, confirming quantum theory's cosmic validity.

Photoelectric Effect and Einstein's Light Quanta (1905)

Definition 25 (Photoelectric Effect)**.** *Emission of electrons from a metal surface when illuminated by light of sufficiently high frequency.*

Classical predictions vs. experimental observations:

Aspect	Classical Prediction	Experimental Fact
Dependence on intensity	Higher intensity $\Rightarrow$ higher KE	KE independent of intensity
Time delay	Energy accumulates over time	Emission instantaneous ($< 10^{-9}$ s)
Frequency threshold	No minimum frequency	Threshold frequency ν_0 exists



普朗克谱的关键特征

- 紫外灾难：瑞利-金斯定律在 ν^2 （橙色曲线）处发散
- 量子分辨率：普朗克定律由于 $e^{h\nu/k_BT} - 1$ 分母 而保持有限
- 维恩位移定律：峰值频率 ν最大 $\propto T$
- **斯特藩-玻尔兹曼定律：总功率** $\int_0^\infty \rho(\nu, T)d\nu \propto T^4$
- 温度依赖性：更高的 T (红色虚线) $\rightarrow$ 峰值更高，向更高 ν移动

普朗克辐射定律与大爆炸：跨宇宙尺度的量子力学

显著关联：宇宙微波背景辐射（CMB）——大爆炸的余晖——在 $T = 2.7255$ K 时表现出完美的普朗克黑体谱。这提供了：

- 宇宙学验证：量子力学精确描述了宇宙最古老的辐射
  辐射
- 实验胜利：COBE 卫星测量（1990 年）显示 $< 0.03\%$ 偏差
  源自普朗克的公式
- 历史讽刺：普朗克所谓的“数学便利”描述了138亿年前辐射
  的景象！
- 普适性范围：相同的量子原理适用于从原子到宇宙学的各个尺度

宇宙微波背景辐射谱代表了自然界中最精确的黑体测量，证实了量子理论在宇宙中的有效性。

光电效应与爱因斯坦的光量子（1905年）

定义*25*（光电效应）。当金属表面被频率足够高的光照射时，会发射电子。

经典预测与实验观察：

方面	经典预测	实验事实
依赖于强度	强度越高 $\Rightarrow$ 动能越高	KE与强度无关
时间延迟	能量随时间累积	瞬时发射 ($< 10^{-9}$ s)
频率阈值	无最小频率	阈值频率 ν_0 存在

Einstein's Explanation (1905)

Light consists of discrete packets (photons) with energy:

$$E_{\text{photon}} = h\nu$$

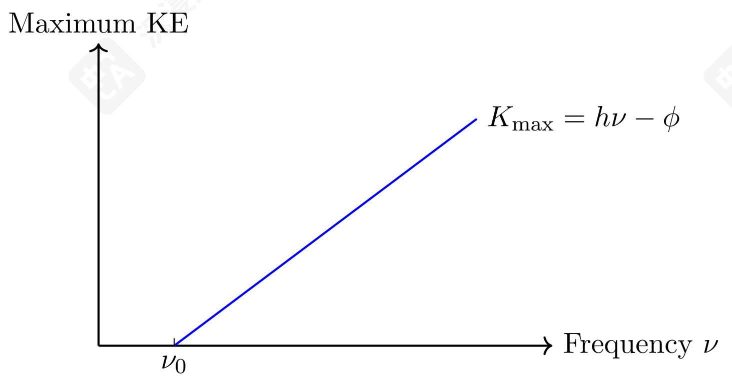
The maximum kinetic energy of emitted electrons:

$$K_{\max} = h\nu - \phi$$

where ϕ is the work function (minimum energy to remove electron).

Mathematical Formulation:

$$\frac{1}{2} m_e v_{\max}^2 = h\nu - \phi$$



Photoelectric Effect: Linear dependence of maximum kinetic energy on frequency

Significance: Light has particle-like properties; wave-particle duality begins.

5.1.2 Atomic Spectra and Bohr's Model (1913)

Atoms emit/absorb light at specific frequencies (spectral lines), not continuous spectra.

Example 3 (Hydrogen Spectrum)**.** *Balmer formula (1885) for visible lines:*

$$\frac{1}{\lambda} = R_H \left(\frac{1}{2^2} - \frac{1}{n^2} \right), \quad n = 3, 4, 5, \ldots$$

where $R_H = 1.097 \times 10^7\ m^{-1}$ *is the Rydberg constant.*

Bohr's Postulates (1913)

1. Electrons orbit nucleus in **stationary states** with fixed energy (no radiation)
2. Radiation occurs only during **quantum jumps** between stationary states:

$$h\nu = E_i - E_f$$

3. Angular momentum is quantized:

$$L = n\hbar, \quad n = 1, 2, 3, \ldots$$

where $\hbar = h/2\pi$.

爱因斯坦的解释（1905年）

光由离散的包（光子）组成，其能量为：

$$E_{\text{photon}} = h\nu$$

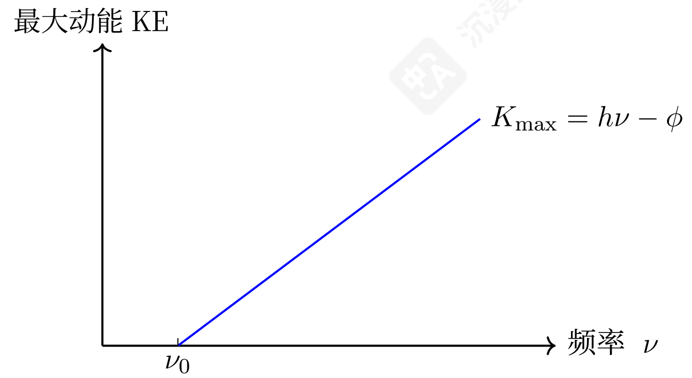
发射出的电子的最大动能：

$$K_{\max} = h\nu - \phi$$

其中 ϕ 是功函数（移除电子所需的最小能量）。

数学公式：

$$\frac{1}{2} m_e v_{\max}^2 = h\nu - \phi$$



光电效应：最大动能与频率的线性关系

意义：光具有粒子性；波粒二象性开始出现。

5.1.2 原子光谱与玻尔模型 (1913)

原子在特定频率处发射/吸收光（形成光谱线），而非连续光谱。

示例 *3*（氢谱）。巴尔末公式（*1885*年）用于可见光线：

$$\frac{1}{\lambda} = R_H \left(\frac{1}{2^2} - \frac{1}{n^2} \right), \quad n = 3, 4, 5, \ldots$$

其中 $R_H = 1.097 \times 10^7\ m^{-1}$ 是里德伯常数。

玻尔假设（1913年）

1. 电子在定态（固定能量）绕原子核运动（不辐射）
2. 辐射仅在定态之间的量子跃迁时发生：

$$h\nu = E_i - E_f$$

3. 角动量是量子化的：

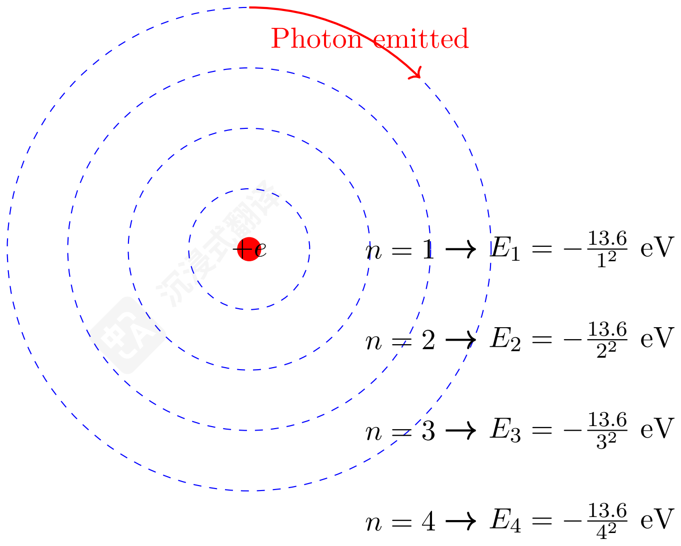
$$L = n\hbar, \quad n = 1, 2, 3, \ldots$$

在 $\hbar = h/2\pi$处。

From these postulates, Bohr derived for hydrogen atom:

$$E_n = -\frac{m_e e^4}{8\epsilon_0^2 h^2}\frac{1}{n^2} = -\frac{13.6 \text{ eV}}{n^2}$$

$$r_n = \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2} n^2 = a_0 n^2 \quad (a_0 = 0.529 \text{ Å})$$



Bohr Model: Quantized Orbits

Successes: Explained hydrogen spectrum, predicted Rydberg constant.

Limitations: Only worked for hydrogen-like atoms; no explanation for line intensities; ad hoc mixture of classical and quantum ideas.

5.2 Two Formulations of Quantum Mechanics

Two Equivalent Approaches to Quantum Mechanics

Historically, quantum mechanics developed through two parallel formulations that were later shown to be mathematically equivalent:

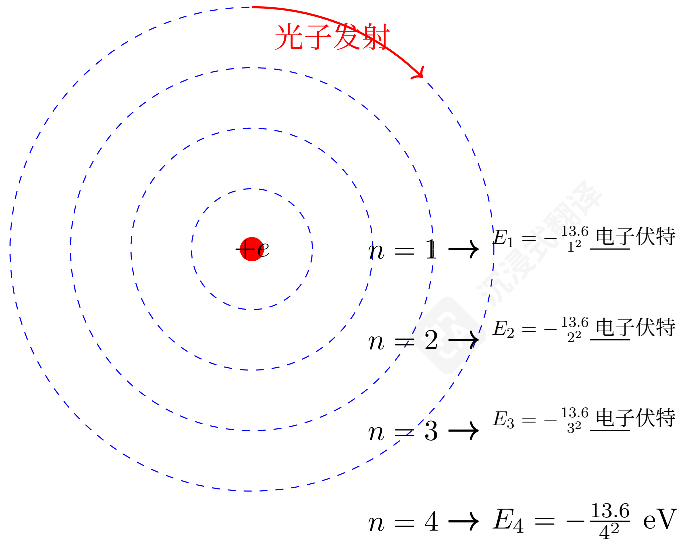
Aspect	**Wave Mechanics (Schrödinger)**	**Matrix Mechanics (Heisenberg)**
Key Figures	de Broglie, Schrödinger	Heisenberg, Born, Jordan
Fundamental Object	Wave function $\Psi(x,t)$	Infinite matrices X_{mn}, P_{mn}
Dynamics	Schrödinger equation: $i\hbar\frac{\partial\Psi}{\partial t} = \hat{H}\Psi$	Heisenberg equation: $\frac{d\hat{A}}{dt} = \frac{1}{i\hbar}[\hat{A},\hat{H}]$
Physical Interpretation	Probability amplitude	Transition amplitudes
Mathematical Framework	Differential equations	Linear algebra
Year Developed	1926	1925

Table 1: Comparison of the two historical formulations of quantum mechanics

从这些公设出发，玻尔推导出了氢原子的：

$$E_n = -\frac{m_e e^4}{8\epsilon_0^2 h^2}\frac{1}{n^2} = -\frac{13.6 \text{ eV}}{n^2}$$

$$r_n = \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2} n^2 = a_0 n^2 \quad (a_0 = 0.529 \text{ Å})$$



玻尔模型：量子化轨道

成功：解释了氢光谱，预测了里德堡常数。

局限性：仅适用于类氢原子；无法解释谱线强度；经典与量子思想的拼凑。

5.2 量子力学的两种表述

量子力学的两种等效方法

历史上，量子力学通过两种平行的表述形式发展起来，后来证明它们在数学上是等价的：

方面	**波动力学（薛定谔）**	**矩阵力学（海森堡）**
关键人物	德布罗意，薛定谔	海森堡，玻恩，约当
基本对象	波函数 $\Psi(x,t)$	无限矩阵 X_{mn}, P_{mn}
动力学	薛定谔方程：$i\hbar\frac{\partial\Psi}{\partial t} = \hat{H}\Psi$	海森堡方程：$\frac{d\hat{A}}{dt} = \frac{1}{i\hbar}[\hat{A},\hat{H}]$
物理意义	概率幅	跃迁幅
数学框架	微分方程	线性代数
开发年份	1926	1925

表1：量子力学两种历史形式的比较

Historical Unification

The equivalence of wave and matrix mechanics was established through:

- **P.A.M. Dirac** (1927): Transformation theory showing equivalence
- **John von Neumann** (1932): Rigorous Hilbert space formulation
- **Unified framework**: Hilbert spaces with Dirac notation

Remark 5.1 (Historical vs. Modern Presentation)**.** *While historically matrix mechanics came first (Heisenberg, 1925), pedagogically we start with wave mechanics because:*

- ***Wave mechanics*** *is more intuitive for students familiar with differential equations*
- ***Visualizable concepts****: Wave functions, probability densities, interference*
- ***Direct connection*** *to classical wave physics*
- ***Easier calculations*** *for initial problems*

The modern Hilbert space formalism (combining both approaches) will be developed in outline once you have built physical intuition.

Historical Development of Quantum Mechanics

1925: Matrix Mechanics
(Heisenberg, Born, Jordan)

↓

1926: Wave Mechanics
(Schrödinger, de Broglie)

↓

1927-1932: Unification
(Dirac, von Neumann)

↓

Modern Formalism
Hilbert Space + Dirac Notation

Our Pedagogical Approach

In this course, we will follow a **wave mechanics first** approach to build physical intuition before introducing abstract mathematical structures:

- **Part 1 (Lectures 2-6)**: Develop physical intuition through wave mechanics, the Schrödinger equation, and one-dimensional problems.
- **Part 2 (Lectures 7-10)**: Transition to the unified Dirac-Hilbert space formalism, introducing spin and abstract operators.
- **Part 3 (Lectures 11-14)**: Apply the complete quantum toolkit to advanced systems (harmonic oscillator, atoms, entanglement) and modern applications.

This progression allows us to ground the abstract postulates in concrete examples, mirroring the historical development of quantum theory.

历史性统一

波动力学和矩阵力学的等价性是通过以下方式确立的：

- P.A.M. 狄拉克 (1927): 变换理论展示等价性
- 约翰·冯·诺依曼 (1932): 严格的希尔伯特空间形式化
- 统一框架：使用狄拉克符号的希尔伯特空间

备注 5.1（历史与现代呈现）虽然从历史上看矩阵力学（海森堡，*1925*年）是首先出现的，但在教学上我们从波动力学开始，因为：

- 波动力学对熟悉微分方程的学生来说更直观
- 可视化的概念：波函数、概率密度、干涉
- 与经典波物理学的直接联系
- 初始问题的计算更简单

现代希尔伯特空间形式化（结合两种方法）将在你建立起物理直觉后概述发展。

量子力学的历史发展

1925：矩阵力学
(海森堡、玻恩、约当)

↓

1926：波动力学 (薛定谔、
德布罗意)

↓

1927-1932：统一 (狄拉
克、冯·诺依曼)

↓

现代形式化
希尔伯特空间 + 狄拉克符号

我们的教学方法

在本课程中，我们将采用先波动力学的方法来建立物理直觉，然后再介绍抽象的数学结构：

- 第一部分（第2-6讲）：通过波动力学、薛定谔方程和一维问题来培养物理直觉。
方程，和一维问题。
- 第二部分（第7-10讲）：过渡到统一的狄拉克-希尔伯特空间形式主义，介绍
自旋和抽象算子。
- 第三部分（第11-14讲）：将完整的量子工具箱应用于高级系统（谐振子、原子、纠缠）和现代应用。

这一进程使我们能够将抽象的公设落实到具体实例中，反映了量子理论的历史发展历程。

What You Will Learn in This Course

Through our wave-mechanics-first approach, you will develop two complementary but equivalent skill sets:

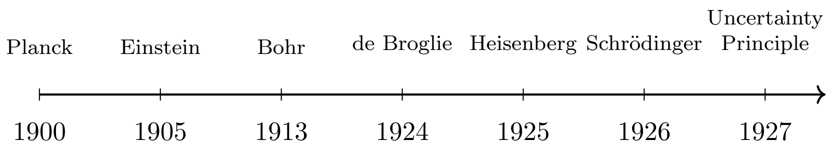
- **Wave Mechanics (Schrödinger Picture – Lectures 2–8)**:
  - Write and solve the Schrödinger equation for key potentials (infinite/finite wells, harmonic oscillator).
  - Physically interpret wavefunctions: $|\Psi(x,t)|^2$ as a probability density.
  - Calculate expectation values, uncertainties, and solve eigenvalue problems in position representation.
  - Understand dynamics through wave packets and the time evolution of probability densities.
- **Abstract Hilbert Space Formalism (Dirac Picture – Lectures 9–14)**:
  - Represent quantum states as abstract vectors $|\psi\rangle$ in Hilbert space.
  - Represent physical observables as operators (e.g., Pauli matrices for spin).
  - Master Dirac notation for calculations: $\langle\phi|\hat{A}|\psi\rangle$, $\sum_n |n\rangle\langle n| = \hat{I}$.
  - Solve problems using operator algebra (e.g., ladder operators for the harmonic oscillator).
  - Describe composite systems and entanglement using tensor products: $\mathcal{H}_A \otimes \mathcal{H}_B$.

Synthesis: By the end of the course, you will be able to translate physical problems into either formalism and appreciate their deep mathematical unity—the same physics expressed in different 'languages'.

Summary: Birth of Modern Quantum Mechanics (1925-1927)

Timeline of Quantum Revolution

Key Developments in Early Quantum Mechanics



Formulation	Creator	Key Idea
Matrix Mechanics	Heisenberg (1925)	Represent observables as matrices
Wave Mechanics	Schrödinger (1926)	Describe particles as waves $\Psi(x,t)$
Transformation Theory	Dirac (1927)	Unified both approaches
Probability Interpretation	Born (1926)	$\|\Psi\|^2$ as probability density
Uncertainty Principle	Heisenberg (1927)	$\Delta x \Delta p \geq \hbar/2$

本课程将学习什么

通过我们的波动力学优先方法，你将发展出两种互补但等价的技能体系：

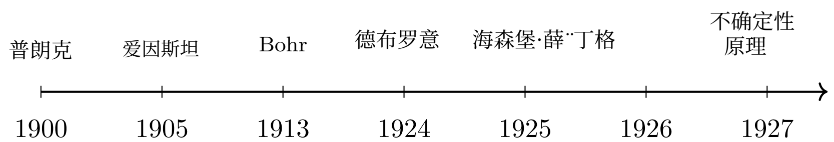
- **波动力学（薛定谔方程——第2-8讲）：**
  - 写出并求解关键势的薛定谔方程（无限/有限深阱、谐振子）。
    。
  - 物理上解释波函数： $|\Psi(x,t)|^2$ 作为概率密度。
  - 计算期望值、不确定度，并求解位置表示中的本征值问题。
    表示。
  - 通过波包和时间演化的概率密度理解动力学。
- **抽象希尔伯特空间形式化（狄拉克图景——第9-14讲）：**
  - 将量子态表示为希尔伯特空间中的抽象向量 $|\psi\rangle$。
  - 将物理可观测量表示为算符（例如，自旋的泡利矩阵）。
  - 掌握狄拉克符号进行计算： $\langle\phi|$ $A|\psi\rangle$ $\sum_{n'}|n\rangle\langle n| = \hat{I}$.
  - 使用算符代数解决问题（例如，谐振子的升降算符）。
  - 使用张量积描述复合系统与纠缠： $\mathcal{H}_A \otimes \mathcal{H}_B$。

合成：课程结束时，你将能够将物理问题转化为形式主义，并欣赏它们深刻的数学统一性——同样的物理学用不同的“语言”表达。

总结：现代量子力学的诞生（1925-1927）

量子革命时间线

早期量子力学的关键发展



表述	创建者	核心思想
矩阵力学	海森堡 (1925)	将可观测量表示为矩阵
波动力学	薛定谔 (1926)	将粒子描述为波 $\Psi(x,t)$
变换理论	狄拉克 (1927)	统一了两种方法
概率诠释	玻恩 (1926)	$\|\Psi\|^2$ 作为概率密度
不确定性原理	海森堡 (1927)	$\Delta x \Delta p \geq \hbar/2$

6 The Double-Slit Experiment: Conceptual Foundation

Three Foundational Quantum Experiments

In this course, we will examine three pivotal experiments that reveal the core principles of quantum mechanics:

Experiment I: Double-slit experiment: Demonstrates wave-particle duality and quantum interference

Experiment II: Stern-Gerlach experiment: Confirms quantization of angular momentum (spin)

Experiment III: EPR correlation experiment: Reveals quantum nonlocality and entanglement

These experiments provide the empirical foundation upon which quantum theory is built.

6.1 Classical Wave Phenomena: Precursor to the Double-Slit Experiment

Before we analyze the quantum double-slit experiment, we must first understand classical wave phenomena. The most direct intuition can be developed through water waves, light waves, and their interference patterns. Richard Feynman's *Lectures on Physics* discusses this topic exceptionally well (Volume I, Chapters 51-52); here we review the essentials needed for quantum mechanics.

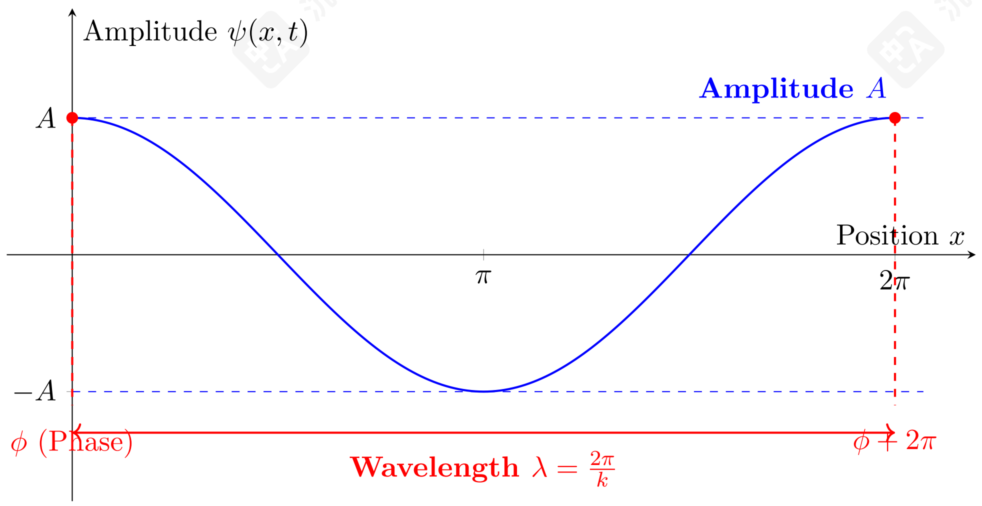
What is a Wave?

A wave is a disturbance that propagates through space and time, transferring energy without permanent displacement of the medium. In water waves, the medium is water molecules vibrating vertically; in light waves, oscillating electric and magnetic fields propagate even through vacuum.

Mathematical Description of Waves

Consider a simple one-dimensional wave propagating along the x-axis. The most general traveling wave can be written as:

$$\boxed{\psi(x,t) = A\cos(\omega t - kx + \phi)}$$



6 双缝实验：概念基础

三个基础量子实验

在本课程中，我们将考察三个关键实验，揭示量子核心原理。
力学:

Experiment I: Double-slit experiment: Demonstrates wave-particle duality and quantum interference

实验II：斯特恩-盖拉赫实验：证实角动量的量子化
(spin)

实验三：EPR关联实验：揭示量子非定域性和纠缠-
ment

这些实验为量子理论奠定了经验基础。

6.1 经典波现象：双缝实验的先驱

在分析量子双缝实验之前，我们必须首先理解经典波现象。最直观的理解可以通过水波、光波及其干涉图样来发展。理查德·费曼的《物理学讲义》对此主题进行了极为出色的讨论（第一卷，第51-52章）；这里我们回顾量子力学所需的要点。

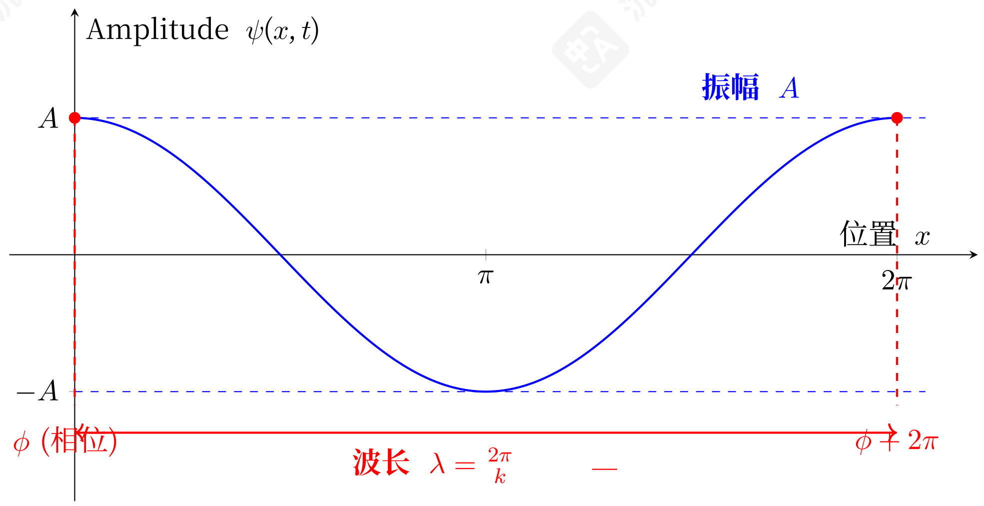
什么是波?

波是一种通过空间和时间传播的扰动，它传递能量而不引起介质的永久位移。在水波中，介质是垂直振动的水分子；在光波中，振荡的电场和磁场即使在真空中也能传播。

波的数学描述

考虑一个沿 x 轴传播的一维简单波。最一般的行波可以表示为:

$$\boxed{\psi(x,t) \ = A\cos(\omega t - kx + \phi)}$$



- **Amplitude** A: Maximum displacement from equilibrium (measured in meters for water waves, volts/meter for electric fields)
- **Angular frequency** ω: Rate of phase change with time ($\omega = 2\pi f$, units: rad/s)
- **Wave number** k: Rate of phase change with position ($k = 2\pi/\lambda$, units: rad/m)
- **Phase constant** ϕ: Initial phase at $x = 0, t = 0$

Phase Velocity and Dispersion Relation

The **phase velocity** v_p is the speed at which points of constant phase move. Setting the phase to constant:

$$\omega t - kx + \phi = \text{constant} \quad \Rightarrow \quad \omega\, dt - k\, dx = 0$$

$$\boxed{v_p = \frac{dx}{dt} = \frac{\omega}{k}}$$

The relationship $\omega = \omega(k)$ is called the **dispersion relation**:

- **Non-dispersive waves**: $\omega = vk$ (constant v), all frequencies travel at same speed (e.g., light in vacuum, sound in air)
- **Dispersive waves**: $\omega(k)$ non-linear, different frequencies travel at different speeds (e.g., water waves, light in materials)

Waves as Fields

Waves are Fields

A wave is an example of a **field**: a physical quantity defined at every point in space and time. Other important fields include:

- Electric field $\mathbf{E}(x, y, z, t)$ and magnetic field $\mathbf{B}(x, y, z, t)$
- Gravitational field $g(x, y, z, t)$
- Temperature field $T(x, y, z, t)$ (heat distribution)
- Quantum wave function $\Psi(x, y, z, t)$ (central to this course!)

This field concept will be crucial throughout quantum mechanics.

Interference: The Hallmark of Wave Behavior

When two or more waves overlap, they **interfere**. Consider two waves with the same frequency but different phases:

$$\psi_{\text{total}}(x, t) = \psi_1(x, t) + \psi_2(x, t) = A\cos(\omega t - kx + \phi_1) + A\cos(\omega t - kx + \phi_2)$$

Using trigonometric identities, this simplifies to:

$$\psi_{\text{total}}(x, t) = 2A\cos\left(\frac{\phi_1 - \phi_2}{2}\right)\cos\left(\omega t - kx + \frac{\phi_1 + \phi_2}{2}\right)$$

- 振幅 A: 偏离平衡位置的最大值（对于水波以米为单位，对于电场以伏/米为单位）
- 角频率 ω: 随时间 ($\omega = 2\pi f$，单位变化：rad/s)
- 波数 k: 相位随位置变化的速率 ($k = 2\pi/\lambda$, 单位：弧度/米)
- 相位常数 ϕ: 在 $x = 0$ $t = 0$ 处的初始相位，

相速度与色散关系

相速度 v_p 是恒定相位点移动的速度。将相位设为恒定：

$$\omega t - kx + \phi = \text{constant} \quad \Rightarrow \quad \omega\, dt - k\, dx = 0$$

$$\boxed{v_p = \frac{dx}{dt} = \frac{\omega}{k}}$$

这种关系 $\omega = \omega(k)$ 称为色散关系：

- 非色散波： $\omega = vk$ （恒定 v），所有频率都以相同速度传播（例如，真空中的光、空气中的声波）
- 色散波： $\omega(k)$ 非线性，不同频率以不同速度传播（例如，水波、材料中的光）

波作为场

波是场

波是场的例子：一个在空间和时间中的每一点都有定义的物理量。其他重要的场包括：

- 电场 $\mathbf{E}(x, y, z, t)$ 和磁场 $\mathbf{B}(x, y, z, t)$
- 引力场 $g(x, y, z, t)$
- 温度场 $T(x, y, z, t)$ (热量分布)
- 量子波函数 $\Psi(x, y, z, t)$ (本课程的核心！)

这个场概念在整个量子力学中至关重要。

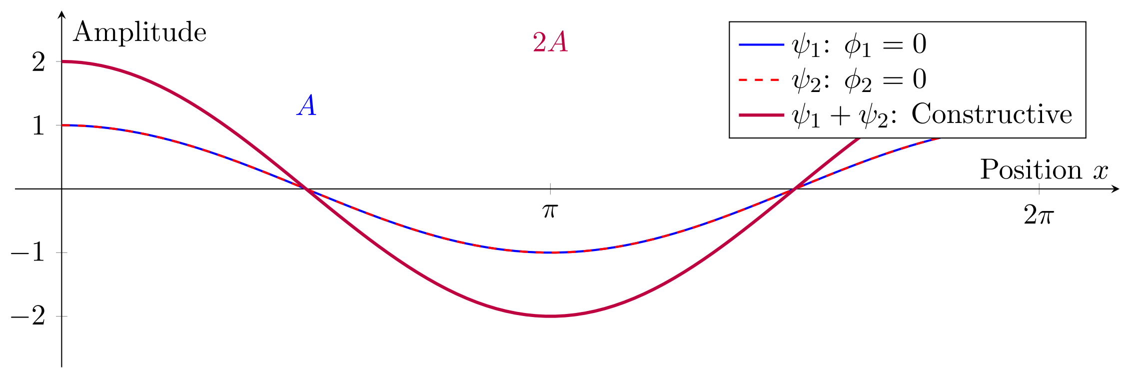
干涉：波的标志性特征

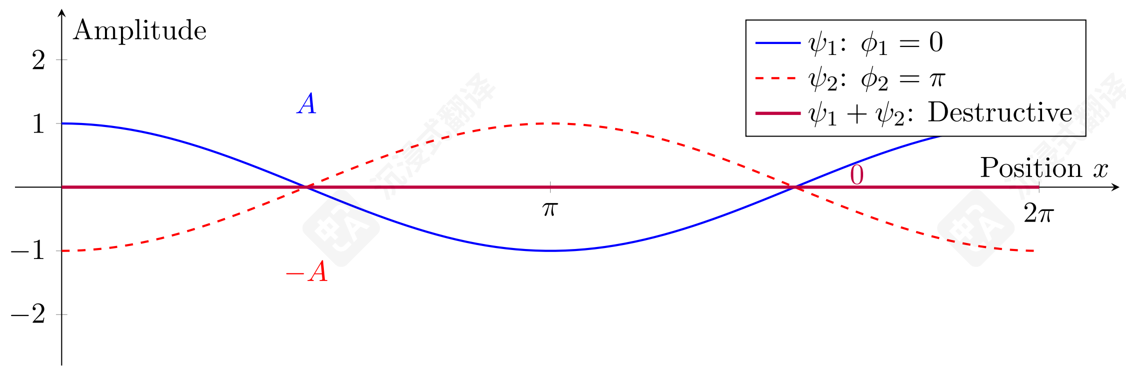
当两个或多个波重叠时，它们会发生干涉。考虑两个频率相同但相位不同的波：

$$\psi_{\text{total}}(x, t) = \psi_1(x, t) + \psi_2(x, t) = A\cos(\omega t - kx + \phi_1) + A\cos(\omega t - kx + \phi_2)$$

利用三角恒等式，这简化为：

$$\psi_{\text{total}}(x, t) = 2A\cos\left(\frac{\phi_1 - \phi_2}{2}\right)\cos\left(\omega t - kx + \frac{\phi_1 + \phi_2}{2}\right)$$





- **Constructive interference**: $\phi_1 - \phi_2 = 2n\pi \Rightarrow$ amplitude doubles ($2A$)
- **Destructive interference**: $\phi_1 - \phi_2 = (2n+1)\pi \Rightarrow$ complete cancellation (0)
- **Partial interference**: Intermediate phase differences yield amplitudes between 0 and $2A$

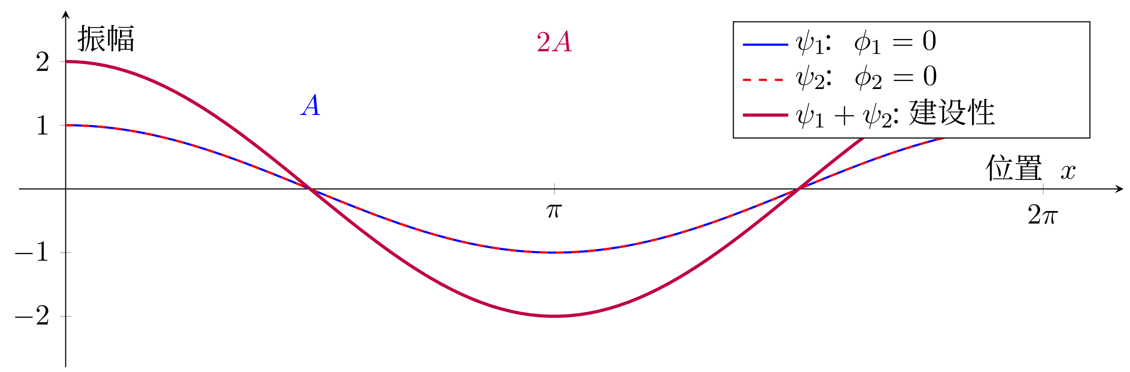
From Water Waves to Light: Young's Double-Slit Experiment (1801)

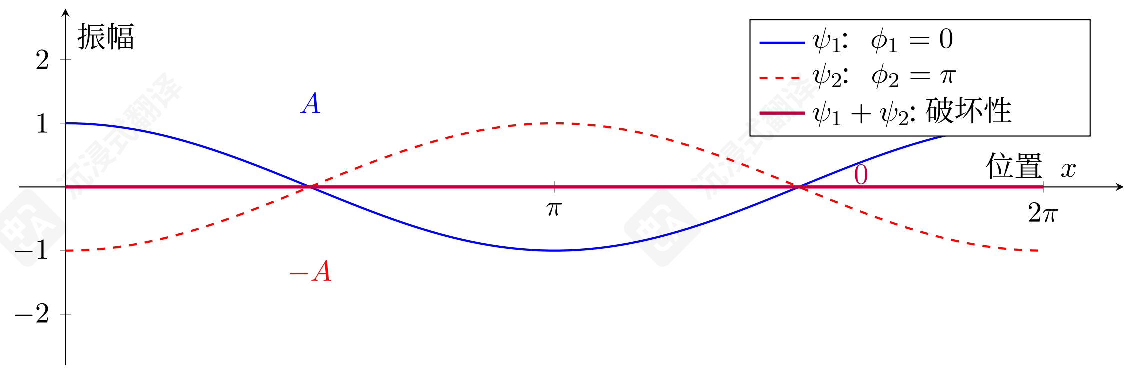
Thomas Young's Experiment

In 1801, Thomas Young performed the definitive experiment demonstrating light's wave nature:

- A monochromatic light source illuminates a barrier with two narrow slits
- Each slit acts as a new source of spherical waves (Huygens' principle)
- Waves from both slits interfere on a screen
- Result: Alternating bright (constructive) and dark (destructive) fringes

This proved light behaves as a wave, contradicting Newton's particle theory.





- **建设性干涉**：$\phi_1 - \phi_2 = 2n\pi \Rightarrow$ **振幅加倍** ($2A$)
- 破坏性干涉：$\phi_1 - \phi_2 = (2n+1)\pi \Rightarrow$ 完全抵消（0）
- 部分干涉：中间相位差产生介于0和 $2A$之间的振幅

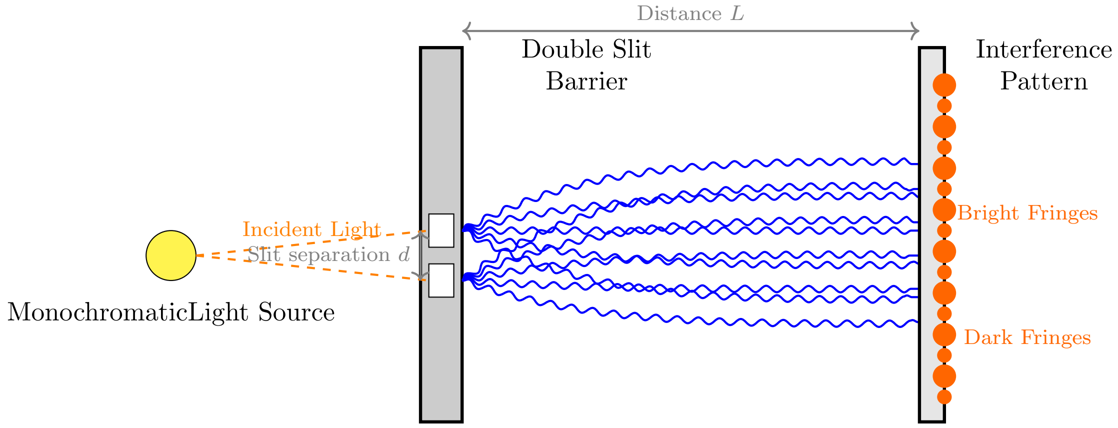
从水波到光：杨氏双缝实验（1801）

托马斯·杨的实验

1801年，托马斯·杨进行了决定性的实验，证明了光的波动性：

- 单色光源照射带有两个狭缝的屏障
- 每个狭缝都充当一个新的球面波源（惠更斯原理）
- 来自两个狭缝的波在屏幕上发生干涉
- 结果：交替出现明（相长）和暗（相消）条纹

这证明了光的行为如同波，与牛顿的粒子理论相悖。



Young's Double-Slit Experiment (1801)

Demonstrated wave nature of light through interference fringes

Why This Matters for Quantum Mechanics

Bridge to Quantum Physics

The classical wave concepts developed here provide the essential foundation for understanding quantum phenomena:

- **Wave-particle duality**: Just as light waves show particle properties (photons), matter particles show wave properties (de Broglie waves)
- **Quantum interference**: The double-slit experiment with electrons produces identical patterns to light waves
- **Probability amplitudes**: Quantum wave functions $\Psi(x, t)$ obey superposition and interference like classical waves
- **Measurement**: The act of observation affects quantum systems analogously to how measuring a wave disturbs it

The mathematical formalism of waves—superposition, interference, and field description—directly carries over to quantum mechanics.

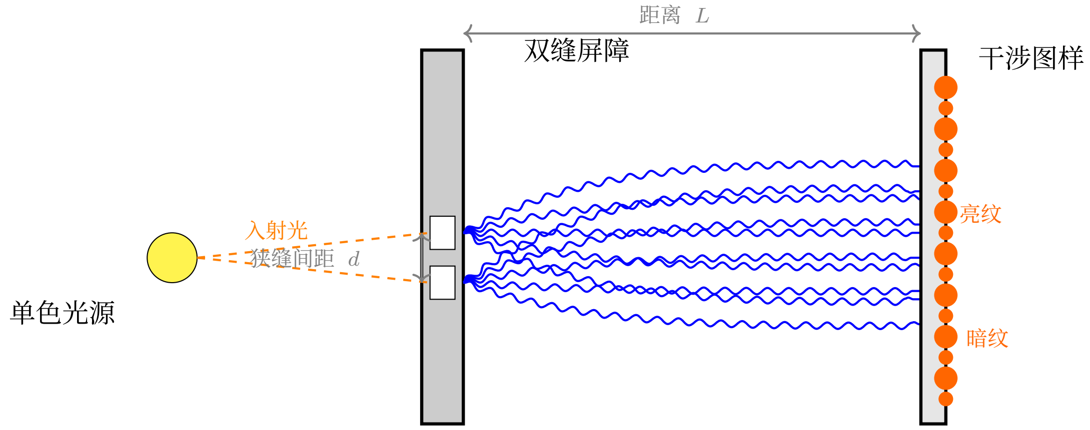
6.2 History of the Electron Double-Slit Experiment

Electrons were discovered in 1897 by J.J. Thomson through his cathode ray tube experiments. Thomson's work established that:

- Cathode rays consist of **negatively charged particles**
- These particles have a **specific charge-to-mass ratio** (e/m) much larger than atoms
- They are **constituents of all atoms**, making them the first identified subatomic particle
- They can be **emitted from metals** via thermionic emission when heated

Key Historical Milestones

The electron double-slit experiment evolved through theoretical proposals and gradual experimental realization:



杨氏双缝实验 (1801)

通过干涉条纹证明了光的波动性

量子力学为何重要

通往量子物理的桥梁

这里发展的经典波概念为理解量子现象提供了基本基础：

- 波粒二象性：正如光波表现出粒子特性（光子），物质粒子也表现出波特性（德布罗意波）cles显示波动性质（德布罗意波）
- 量子干涉：电子的双缝实验产生的图案与光波相同与光波相同
- 概率幅：量子波函数 $\Psi(x, t)$ 遵循叠加和干涉如同经典波
- 测量：观察行为会影响量子系统，类似于测量波会干扰它

描述波的数学形式——叠加、干涉和场描述——可以直接应用于量子力学。

6.2 电子双缝实验的历史

1897年，J.J.汤姆孙通过他的阴极射线管实验发现了电子。汤姆孙的研究确立了：

- **阴极射线由带负电的粒子组成**
- 这些粒子的比荷 (e/m) 比原子大得多
- 它们是所有原子的组成部分，因此是第一个被发现的亚原子粒子
- 当加热时，它们可以通过热电子发射从金属中发射出来

关键历史里程碑

电子双缝实验通过理论提案和逐步的实验实现而发展：

Theoretical Foundation (1920s)

- **1924**: **Louis de Broglie** proposed matter waves with $\lambda = h/p$
- **1927**: **Clinton Davisson and Lester Germer** demonstrated electron diffraction from nickel crystals (Nobel Prize 1937)
- **1928**: **Max Born** first discussed the *thought experiment* of electrons through double slits in his papers on quantum mechanics

First Serious Proposal (1940s-1950s)

- **1949**: **Carlo Ferrari** (Italian physicist) published one of the first detailed theoretical analyses
- **1961**: **Claus Jönsson** (University of Tübingen, Germany) performed the **first actual electron double-slit experiment**

Jönsson's Experiment (1961)

Claus Jönsson achieved the first true electron double-slit interference using:

- 50 keV electrons ($\lambda \approx 0.05$ Å)
- Double slits in copper foil (width ~ 0.3 m, separation ~ 1 m)
- Electron microscope for detection
- Clear interference fringes observed

Published in *Zeitschrift für Physik* 161, 454 (1961).

Refinements and Single-Electron Experiments

- **1974**: **G. Möllenstedt and H. Düker** improved the experiment with biprism technique
- **1976**: **P.G. Merli, G.F. Missiroli, and G. Pozzi** published electron interference patterns
- **1989**: **Akira Tonomura** (Hitachi Labs) performed the **definitive single-electron double-slit experiment**

Tonomura's Single-Electron Experiment (1989)

Akira Tonomura's team demonstrated single-electron interference unambiguously:

- Electrons sent one at a time (spaced minutes apart!)
- Accumulated over time to build interference pattern
- Proved each electron interferes *with itself*
- Used electron biprism instead of physical slits
- Published in *American Journal of Physics* 57, 117 (1989)

This experiment conclusively demonstrated wave-particle duality at the single-particle level.

理论基础（20世纪20年代）

- **1924**：路易·德布罗意提出了具有 $\lambda = h/p$的物质波
- **1927**: 克林顿·戴维森和莱斯特·格默证明了电子在镍晶体上的衍射（1937年诺贝尔奖）
- **1928**: 马克斯·玻恩首次在他的论文中讨论了电子通过双缝的 Gedankenexperiment 关于量子力学

首次严肃提案（1940年代-1950年代）

- **1949**: 卡洛·费拉里（意大利物理学家）发表了首批详细的理论分析之一
- **1961: 克劳斯·约恩松（德国蒂宾根大学）进行了首个实际的电子双缝实验**

约恩松实验（1961年）

克劳斯·约恩松使用以下方法实现了首个真正的电子双缝干涉：

- 50 keV电子 ($\lambda \approx 0.05$ ˚A)
- 铜箔上的双缝（宽度 ~ 0.3 m，间距 ~ 1 m）
- 用于检测的电子显微镜
- 观察到清晰的干涉条纹

发表于《物理杂志》161, 454 (1961)。

改进与单电子实验

- **1974**: G. M¨ollenstedt 和 H. D¨uker 使用双棱镜技术改进了实验
- **1976**: P.G. Merli、G.F. Missiroli 和 G. Pozzi 发表了电子干涉图样
- **1989: Akira Tonomura（日立实验室）进行了决定性的单电子双缝实验**

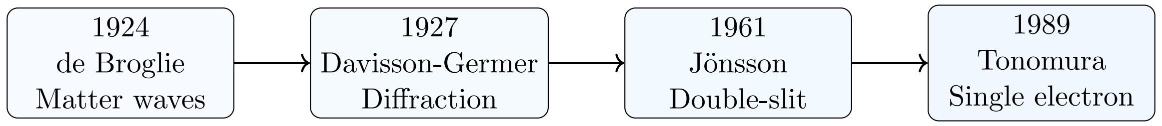
Tonomura 的单电子实验 (1989)

Akira Tonomura 的团队明确地展示了单电子干涉：

- 一次发送一个电子（间隔几分钟！）
- 随时间积累形成干涉图样
- 已证明每个电子会与自己发生干涉
- 使用电子双棱镜代替物理狭缝
- 发表于《美国物理杂志》第*57*卷第*117*页（*1989*年）

这项实验明确证明了在单粒子层面上存在波粒二象性。

Evolution of Electron Interference Experiments



Key Scientific Implications

- **Jönsson (1961)**: First demonstration of electron interference through physical double slits
- **Tonomura (1989)**: Definitive proof of single-particle interference and wave-particle duality
- The experiments confirmed that:
  1. Electrons exhibit wave-like behavior (interference)
  2. Each electron interferes with *itself*, not with other electrons
  3. The wave function describes probability amplitudes
  4. Measurement collapses the wave function

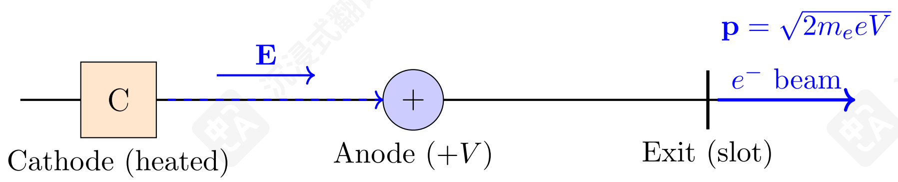
Remark 6.1 (Pedagogical Note)**.** *In textbooks, the "double-slit experiment" is often presented as:*

- *A **thought experiment** proposed by early quantum theorists (Born, Feynman)*
- *Actually performed in stages: Jönsson (1961) → improvements → Tonomura (1989)*
- *Richard Feynman called it "a phenomenon which is impossible [...] to explain in any classical way, and which has in it the heart of quantum mechanics"*

6.3 Double-Slit Experiment with Electrons: Phenomenological Description

The double-slit experiment represents perhaps the most fundamental demonstration of quantum mechanical principles. We will develop its understanding through several progressive stages.

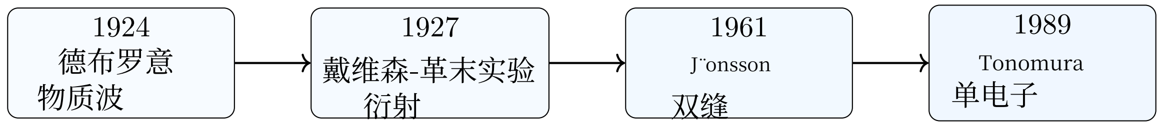
Stage 1: Electron Source Preparation



We begin with an **electron gun** constructed from a cathode ray tube (CRT):

- **Thermionic emission**: Electrons are detached from material by heating
- **Acceleration**: Electrons are accelerated by an electric field within the tube
- **Collimation**: Electrons exit through a narrow slot, producing a beam with controlled momentum $\mathbf{p}$
- **Result**: A monoenergetic electron beam with well-defined initial conditions

电子干涉实验的演变



关键科学意义

- J¨onsson (1961)：首次通过物理双缝实验演示了电子干涉。
- Tonomura (1989)：单粒子干涉和波粒二象性的最终证明
- 实验证实了：
  1. 电子表现出波动行为（干涉）
  2. 每个电子与自己发生干涉，而不是与其他电子发生干涉
  3. 波函数描述概率幅
  4. 测量会坍缩波函数

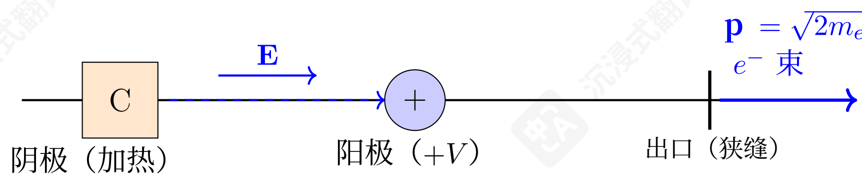
备注 *6.1*（教学提示）。在教科书中，"双缝实验"通常被呈现为：

- 由早期量子理论家（玻恩、费曼）提出的思想实验
- 实际上分阶段进行：*J¨onsson*（*1961*） → 改进 → *Tonomura*（*1989*）
- 理查德·费曼称其为"一个在经典方式上无法解释的现象，并且包含了量子力学核心的" *[...]*

6.3 电子双缝实验：现象学描述

双缝实验或许是量子力学原理最根本的演示。我们将通过几个逐步的阶段来发展对其的理解。

第一阶段：电子源制备

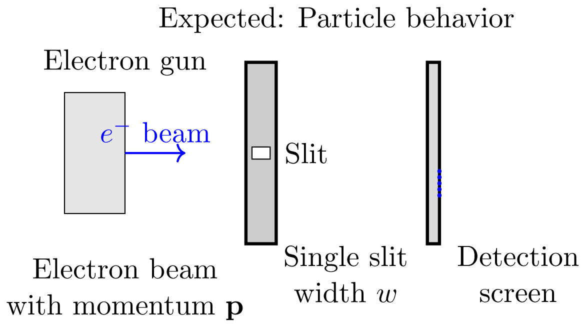


我们从阴极射线管（CRT）制成的电子枪开始：

- 热电子发射：通过加热使电子从材料中脱离
- 加速：电子在管内的电场中被加速
- 准直：电子通过一个狭缝射出，产生具有可控动量p的束流
- 结果：具有明确初始条件的单能电子束

Page 54 of 61

Stage 2: Single-Slit Preliminary Experiment



- **Setup**: The electron beam impinges on a barrier containing a single slit of width w
- **Detection**: A photographic screen records electron impacts as visible dots
- **Initial observation**: For sufficiently large slit width w, electrons form a distribution centered opposite the slit
- **Classical expectation**: Electrons as particles should pass through the slit and strike the screen directly opposite

Stage 3: The Quantum Surprise: Single-Slit Diffraction

Classical vs. Quantum Prediction

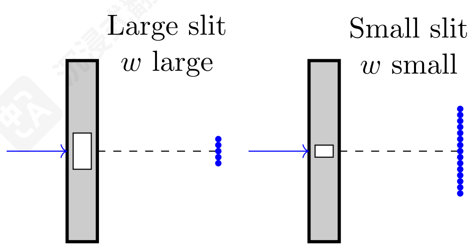
- **Classical particle prediction**: Decreasing slit width $w \to 0$ should produce *sharper* localization on screen

$$\Delta y_{\text{screen}} \propto w \quad \text{(smaller slit} \Rightarrow \text{tighter focus)}$$

- **Experimental observation**: Decreasing slit width produces *broader* distribution

$$\Delta y_{\text{screen}} \propto \frac{1}{w} \quad \text{(smaller slit} \Rightarrow \text{greater spread)}$$

- **Paradox**: The distribution *expands* as the slit narrows!

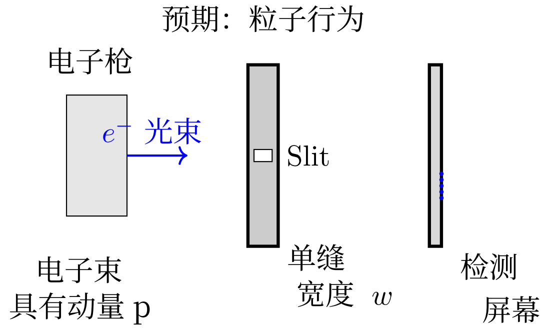


Wave Interpretation of Single-Slit Diffraction

The observed behavior finds natural explanation through wave optics:

- **Wave description**: Electrons exhibit wavelength $\lambda = h/p$ (de Broglie relation)
- **Diffraction principle**: A slit of width w acts as a secondary wave source
- **Angular spread**: $\Delta\theta \approx \lambda/w$ (smaller $w \Rightarrow$ larger $\Delta\theta$)
- **Screen spread**: $\Delta y_{\text{screen}} \approx L\Delta\theta \approx L\lambda/w$

阶段 2：单缝初步实验



- 设置：电子束轰击一个包含单个宽度为 w狭缝的屏障
- 检测：一张照片记录了电子撞击形成的可见点
- 初始观察：对于足够大的狭缝宽度 w，电子会形成以狭缝正对位置为中心的分布
- 经典预期：电子作为粒子应该穿过狭缝并直接撞击狭缝正对位置的屏幕

阶段3：量子惊喜：单缝衍射

经典与量子预测对比

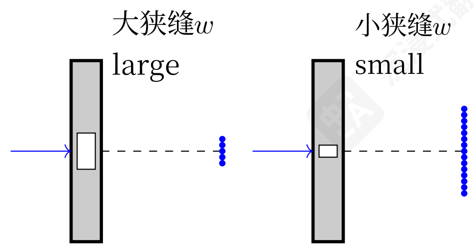
- 经典粒子预测：减小狭缝宽度 $w \to 0$ 应该使屏幕上的定位更加锐利

$$\Delta y_{\text{screen}} \propto w \quad \text{(smaller slit} \Rightarrow \text{tighter focus)}$$

- 实验观察：减小狭缝宽度会产生更宽的分布

$$\Delta y_{\text{screen}} \propto \frac{1}{w} \quad \text{(smaller slit} \Rightarrow \text{greater spread)}$$

- 悖论：当狭缝变窄时，分布反而会扩展！



单缝衍射的波动解释

通过波动光学，可以自然地解释观察到的现象：

- 波描述：电子表现出波长 $\lambda = h/p$ (德布罗意关系)
- 衍射原理：宽度为 w 的狭缝充当次级波源
- 角扩展：$\Delta\theta \approx \lambda/w$ (更小 $w \Rightarrow$ 更大 $\Delta\theta$)
- **屏幕扩散：** $\Delta y_{\text{screen}} \approx L\Delta\theta \approx L\lambda/w$

- **Mathematical consistency**: The observed $\Delta y \propto 1/w$ matches wave prediction

Remark 6.2 (First Evidence of Wave Nature)**.** *The single-slit experiment provides* preliminary evidence *for the wave nature of electrons:*

1. *Electrons exhibit diffraction patterns characteristic of waves*
2. *The spreading obeys the mathematical relationship* $\Delta y \propto 1/w$
3. *This contradicts purely particle-like behavior*

However, this evidence remains ***inconclusive*** *because alternative particle-based explanations (e.g., interaction with slit edges) could potentially account for the observations.*

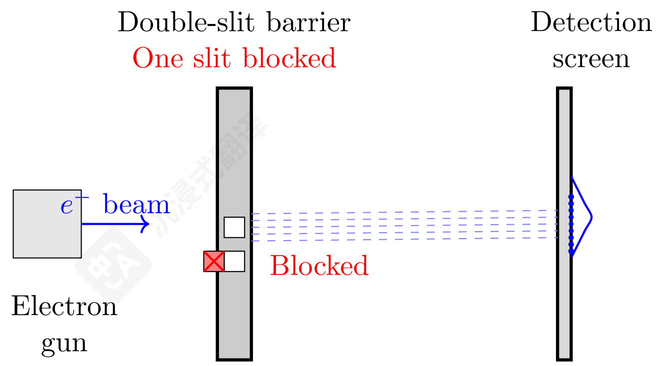
6.4 The Need for the Double-Slit Experiment

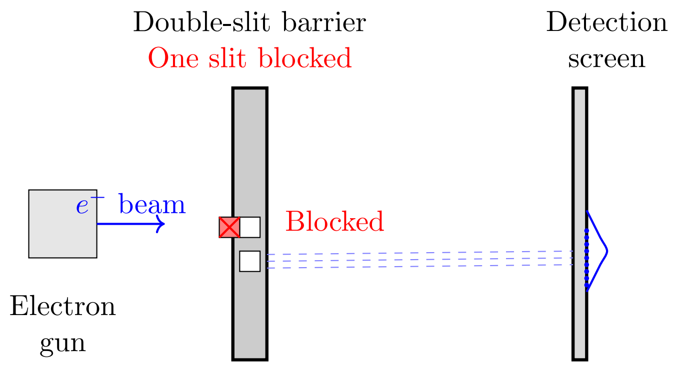
To obtain *definitive evidence* of wave behavior, we require an experiment that demonstrates **interference**—a phenomenon unique to waves:

Why Single-Slit is Not Conclusive

- Single-slit diffraction could potentially be explained by particle-slit interactions
- Particles might be scattered by the slit edges in a width-dependent manner
- Only interference patterns (requiring *two* coherent sources) provide unambiguous wave evidence
- The double-slit experiment eliminates alternative particle explanations

The double-slit configuration, which we will examine next, provides the decisive test by producing **interference fringes** that cannot be explained by any classical particle model. This interference pattern, with its characteristic maxima and minima, constitutes the definitive signature of wave behavior in quantum systems.





- 数学一致性：观测到的 $\Delta y \propto 1/w$ 与波预测相符

备注 *6.2*（波粒二象性的首次证据）。单缝实验为电子的波动性提供了初步证据：

1. 电子表现出与波特征相符的衍射图样
2. 其扩散遵循数学关系 $\Delta y \propto 1/w$
3. 这与纯粹的粒子行为相矛盾

然而，这一证据仍不充分，因为其他基于粒子的解释（例如，与狭缝边缘的相互作用）有可能解释这些观察结果。

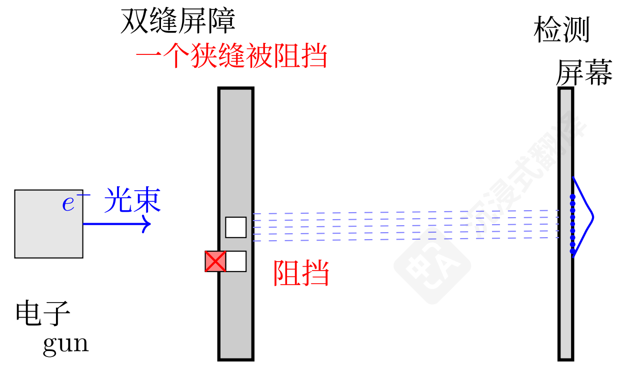
6.4 双缝实验的必要性

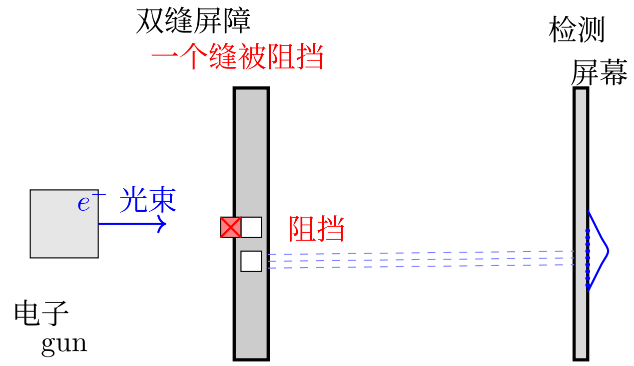
为了获得关于波动行为的决定性证据，我们需要一个能够展示干涉现象的实验——这是波特有的现象：

为什么单缝实验不能得出结论

- 单缝衍射有可能通过粒子与狭缝的相互作用来解释
- 粒子可能以与狭缝宽度相关的 manner 被狭缝边缘散射
- 只有干涉图样（需要两个相干光源）才能提供明确的波动证据
- 双缝实验排除了其他粒子解释的可能性

接下来我们将考察的双缝配置，通过产生无法用任何经典粒子模型解释的干涉条纹，提供了决定性的检验。这种干涉图样，及其特有的最大值和最小值，构成了量子系统中波动行为的明确标志。





Single-Slit Control Experiment

When only one slit is open (the other blocked by a physical barrier), the experiment reduces to the single-slit case:

- **Electron source**: Monoenergetic beam from CRT gun
- **Barrier**: Double-slit apparatus with one slit physically blocked
- **Result**: Only electrons passing through the *unblocked slit* reach the screen
- **Distribution**: Pattern on screen shows **Gaussian-like spread** centered opposite the open slit

Summary of Single-Slit Results

Single-Slit Experiments: Particle-like Behavior

When we repeat the experiment with the **other slit blocked**, we obtain exactly the same type of results:

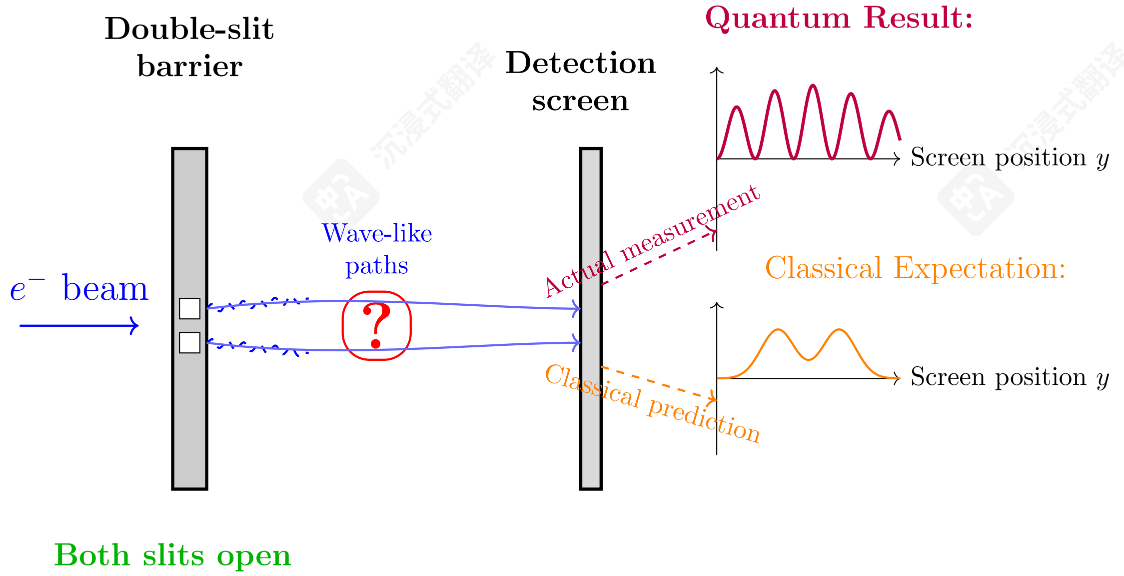
- With only **upper slit open**: Localized probability distribution centered opposite the upper slit
- With only **lower slit open**: Localized probability distribution centered opposite the lower slit

This indicates that electrons behave as **particles** in the sense that:

1. Each electron goes through a *specific slit* when only one is available
2. The arrival pattern follows a predictable probability distribution
3. Individual electrons make localized impacts on the screen

6.5 Transition to Double-Slit Experiment

The Critical Question: Both Slits Open



单缝控制实验

当只有一个缝打开（另一个被物理障碍物阻挡）时，实验简化为单缝情况：

- 电子源：CRT枪产生的单能束
- 障碍物：一个缝物理上被阻挡的双缝装置
- 结果：只有通过未阻挡缝的电子到达屏幕
- 分布：屏幕上的模式显示高斯状分布，中心位于开孔的对侧

单缝实验结果总结

单缝实验：粒子行为

当我们用另一个狭缝进行实验时，我们会得到完全相同的结果：

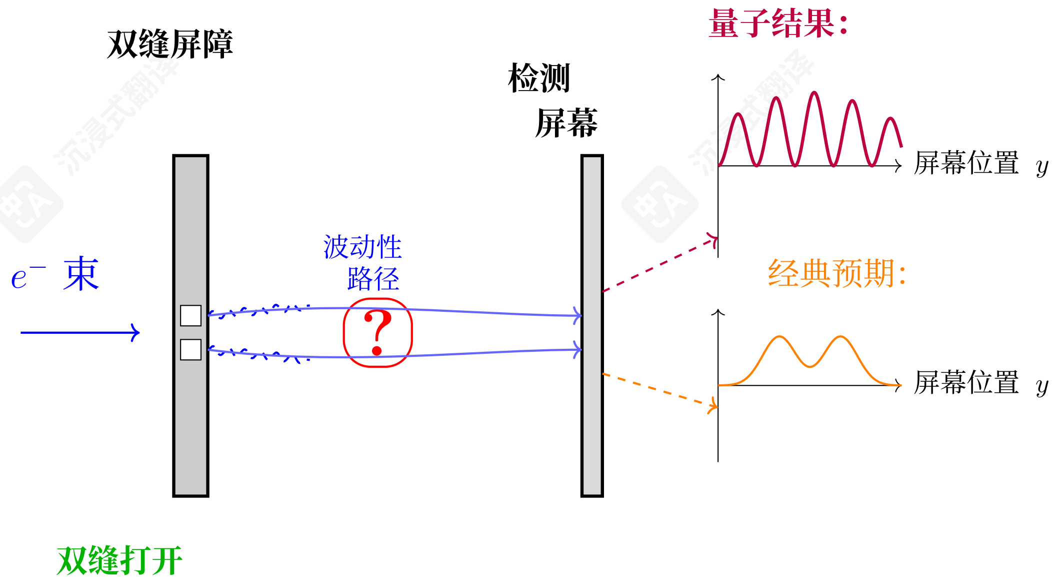
- 仅上缝打开时：局部概率分布中心位于上缝对侧
- 仅下缝打开时：局部概率分布中心位于下缝对侧

这表明电子在某种意义上表现出粒子行为：

1. 当只有一个狭缝可用时，每个电子会通过特定的狭缝
2. 到达模式遵循可预测的概率分布
3. 单个电子对屏幕产生局部影响

6.5 过渡到双缝实验

关键问题：两个狭缝都打开



The Quantum Revelation

The side-by-side comparison reveals the fundamental departure from classical intuition:

- **Classical expectation**: Simple sum of independent probabilities

$$P_{\text{classical}}(y) = P_{\text{upper}}(y) + P_{\text{lower}}(y)$$

- **Quantum reality**: Interference of probability amplitudes

$$P_{\text{quantum}}(y) = |\psi_{\text{upper}}(y) + \psi_{\text{lower}}(y)|^2$$

- **Critical difference**: The cross-term $\psi^*_{\text{upper}}\psi_{\text{lower}} + \psi_{\text{upper}}\psi^*_{\text{lower}}$
  - Can be **positive** (constructive interference → maxima)
  - Can be **negative** (destructive interference → minima)
  - Can be **exactly canceling** (zero probability regions!)

Paradox: Opening an *additional* path (second slit) creates regions where electrons *cannot* go—impossible in classical particle physics!

The Quantum Surprise Revealed

The interference pattern overlay shows what *actually* happens when both slits are open:

- **Not two Gaussians**: The pattern is *not* the sum of single-slit distributions
- **Interference fringes**: Alternating bright (maxima) and dark (minima) bands
- **Zero probability regions**: At minima, probability of detecting electrons is **exactly zero**
- **Quantum signature**: This pattern can only be explained by *wave interference*

$$\boxed{P_{\text{quantum}}(y) = |\psi_{\text{upper}}(y) + \psi_{\text{lower}}(y)|^2 \neq |\psi_{\text{upper}}(y)|^2 + |\psi_{\text{lower}}(y)|^2}$$

The interference terms $\psi^*_{\text{upper}}\psi_{\text{lower}} + \psi_{\text{upper}}\psi^*_{\text{lower}}$ create the characteristic pattern.

The Fundamental Question Introduced

With both slits open, we face profound questions about the nature of quantum reality:

- **Particle perspective**: Does each electron go through *one specific slit*?
- **Wave perspective**: Does the electron pass through *both slits simultaneously* as a wave?
- **Measurement problem**: If we try to *observe* which slit the electron goes through, what happens?
- **Superposition**: Could the electron be in a *superposition* of both paths until measured?

The double-slit experiment with both slits open tests the very foundations of quantum mechanics.

量子启示

并排比较揭示了与经典直觉的根本性偏离：

- 经典预期：独立概率的简单叠加

$$P_{\text{classical}}(y) = P_{\text{upper}}(y) + P_{\text{lower}}(y)$$

- 量子现实：概率幅的干涉

$$P_{\text{quantum}}(y) = |\psi_{\text{upper}}(y) + \psi_{\text{lower}}(y)|^2$$

- **关键差异：交叉项** $\psi^*_{\text{upper}}\psi_{\text{lower}} + \psi_{\text{upper}}\psi^*_{\text{lower}}$
  - 可以是正的（相长干涉 → 极大值）
  - 可以是负的（相消干涉 → 极小值）
  - 可以完全抵消（零概率区域！）

悖论：打开额外路径（第二个狭缝）会形成电子无法到达的区域——经典粒子物理学中不可能发生！

量子惊喜揭晓

干涉图样叠加显示，当两个狭缝都打开时实际发生的情况是：

- 并非两个高斯分布：该模式并非单个狭缝分布的总和
- 干涉条纹：交替出现明暗（最大值和最小值）的条纹
- 零概率区域：在最小值处，检测电子的概率恰好为零
- 量子特征：该模式只能通过波干涉来解释

$$\boxed{P_{\text{quantum}}(y) = |\psi_{\text{upper}}(y) + \psi_{\text{lower}}(y)|^2 \neq |\psi_{\text{upper}}(y)|^2 + |\psi_{\text{lower}}(y)|^2}$$

干涉项 $\psi^*_{\text{upper}}\psi_{\text{lower}} + \psi_{\text{upper}}\psi^*_{\text{lower}}$ 产生了特征模式。

引出的基本问题

当两个狭缝都打开时，我们面临关于量子现实本质的深刻问题：

- 粒子视角：每个电子是否通过特定的一个狭缝?
- 波动视角：电子是否像波一样同时通过两个狭缝?
- 测量问题：如果我们试图观察电子通过哪个狭缝，会发生什么?
- 叠加态：电子在测量前是否处于两条路径的叠加态?

双缝实验中两个缝都打开时，测试了量子力学的基础原理。

Three Possible Interpretations

Classical Particle View — Each electron goes through *either* upper *or* lower slit $\xrightarrow{?}$ **Wave View** — Electron as wave passes through *both slits simultaneously* $\xrightarrow{?}$ **Quantum Superposition** — Electron in state describing *both possibilities at once*

What the Experiment Will Test

- **If classical particle view is correct**: Pattern = Sum of two single-slit distributions

$$P_{\text{classical}}(y) = P_{\text{upper}}(y) + P_{\text{lower}}(y)$$

- **If wave/superposition view is correct**: Pattern = Interference of probability amplitudes

$$P_{\text{quantum}}(y) = |\psi_{\text{upper}}(y) + \psi_{\text{lower}}(y)|^2$$

- **Critical difference**: The quantum prediction includes *interference terms*:

$$|\psi_1 + \psi_2|^2 = |\psi_1|^2 + |\psi_2|^2 + 2\text{Re}(\psi_1^* \psi_2)$$

The last term can be **negative**, creating regions of *zero probability*!

6.6 Fauliure of Classical Logical Analysis

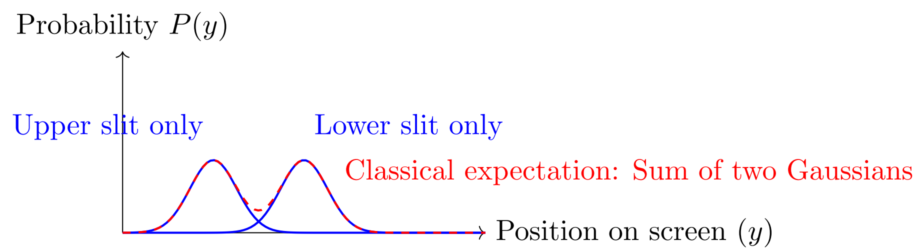
Classical Particle Expectation

Based on classical reasoning about particles, we expect the following:

- Electrons are particles $\Rightarrow$ each electron goes *either* through the upper slit *or* through the lower slit
- The logical operation is OR (union of two mutually exclusive events)
- Classical probability rule: For mutually exclusive events, total probability = sum of individual probabilities

$$P_{\text{classical}}(y) = P_{\text{upper}}(y) + P_{\text{lower}}(y)$$

$$\boxed{P_{\text{classical}}(y) \propto \exp\left[-\frac{(y - y_{\text{upper}})^2}{2\sigma^2}\right] + \exp\left[-\frac{(y - y_{\text{lower}})^2}{2\sigma^2}\right]}$$



三种可能的解释

经典粒子观 每个电子通过上或下狭缝 $\xrightarrow{?}$ **波浪视图** 电子作为波通过两个狭缝同时 $\xrightarrow{?}$ 量子叠加态电子描述两种可能性同时

实验将测试什么

- 如果经典粒子观点是正确的：图案 = 两个单缝分布的总和

$$P_{\text{classical}}(y) = P_{\text{upper}}(y) + P_{\text{lower}}(y)$$

- 如果波/叠加态观点正确：模式 = 概率幅的干涉

$$P_{\text{quantum}}(y) = |\psi_{\text{upper}}(y) + \psi_{\text{lower}}(y)|^2$$

- 关键差异：量子预测包含干涉项：

$$|\psi_1 + \psi_2|^2 = |\psi_1|^2 + |\psi_2|^2 + 2\text{Re}(\psi_1^* \psi_2)$$

最后一项可以是负的，产生零概率区域！

6.6 经典逻辑分析的失效

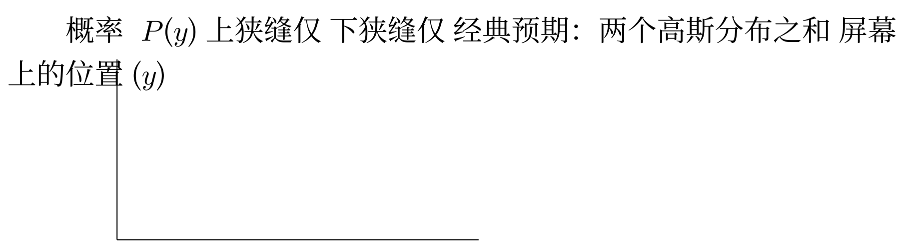
经典粒子预期

基于关于粒子的经典推理，我们预期如下：

- 电子是粒子 $\Rightarrow$ 每个电子要么通过上狭缝，要么通过下狭缝
- 逻辑运算是或（两个互斥事件的并集）
- 经典概率规则：对于互斥事件，总概率 =等于各个概率之和

$$P_{\text{classical}}(y) = P_{\text{upper}}(y) + P_{\text{lower}}(y)$$

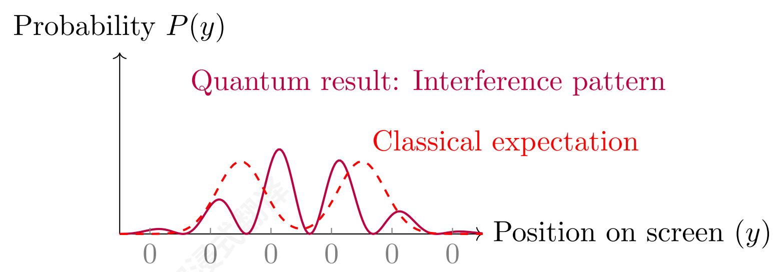
$$\boxed{P_{\text{classical}}(y) \propto \exp\left[-\frac{(y - y_{\text{upper}})^2}{2\sigma^2}\right] + \exp\left[-\frac{(y - y_{\text{lower}})^2}{2\sigma^2}\right]}$$



The Quantum Surprise: Interference Pattern

Experimental Result (Contrary to Classical Expectation)

When the experiment is actually performed with **both slits open**, we do *not* get the classically expected sum of two probability distributions. Instead, we observe a fundamentally different **interference pattern**:



Alternating maxima and **null regions** (zero probability!)

The Profound Implication

The interference pattern contains alternating **maxima** and **null regions**:

- **Null regions**: Probability of detecting electrons at these locations is *exactly zero*
- **Paradox**: When two slits are open, it becomes *impossible* for electrons to reach certain positions
- **Contradiction**: This *cannot* be explained by electrons going through either the upper OR lower slit
- If electrons were classical particles, opening an *additional* path should only *increase* probability, never create regions of *zero* probability!

6.7 Quantum Explanation: Wave-Particle Duality

Wave-Particle Duality and de Broglie Hypothesis (1924)

de Broglie's Insight

If light (waves) can behave as particles, perhaps matter (particles) can behave as waves:

$$\lambda_{\text{de Broglie}} = \frac{h}{p}$$

where p is the particle's momentum.

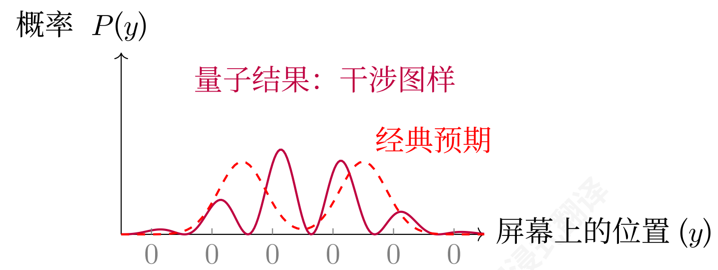
Evidence: Davisson-Germer experiment (1927) showed electron diffraction from crystals, confirming wave nature of electrons.

Significance: Complete symmetry between matter and radiation; foundation for wave mechanics.

量子惊喜：干涉图样

实验结果（与经典预期相反）

当实验实际以两个狭缝都打开时，我们不会得到经典预期的两个概率分布之和。相反，我们观察到一种根本不同的干涉图案：



交替出现的最大值和零概率区域（概率为零！）

深远意义

干涉图样包含交替出现的最大值和零概率区域：

- 零概率区域：在这些位置检测电子的概率正好为零
- 悖论：当两个狭缝都打开时，电子将无法到达某些特定位置
- 矛盾：这无法用电子穿过上狭缝或下狭缝来解释
- 如果电子是经典粒子，那么增加一条路径只会增加概率，绝不会产生零概率区域！

6.7 量子解释：波粒二象性

波粒二象性与德布罗意假设（1924年）

德布罗意的洞见

如果光（波）可以像粒子一样行为，那么物质（粒子）或许也可以像波一样行为：

$$\lambda_{\text{de Broglie}} = \frac{h}{p}$$

其中 p 是粒子的动量。

证据：戴维森-革末实验（1927年）显示了电子的晶体衍射，证实了电子的波动性。

意义：物质与辐射之间存在完全对称性；是波力学的理论基础。

Wave Mechanics Interpretation

The interference pattern can be explained if a **complex-valued wavelike field** is associated with the electron:

- **de Broglie-Schrödinger field**: $\psi(x,t)$ represents the quantum state
- **Wave nature**: The electron behaves as a wave passing through *both slits simultaneously*
- **Interference**: Waves from two slits interfere constructively (maxima) and destructively (nulls)
- **Probability**: $P(y) = |\psi_{\text{upper}}(y) + \psi_{\text{lower}}(y)|^2$ (not simple sum!)

波动力学诠释

如果电子与一个复数值的波状场相关联，则可以解释干涉图样：

- 德布罗意-薛定谔场： $\psi(x,t)$表示量子态
- 波动性：电子表现得像穿过两个狭缝的波一样
- 干涉：来自两个狭缝的波发生相长干涉（最大值）和相消干涉（零点）
- 概率： $P(y) = |\psi上限(y) + \psi下限(y)|^2$ (并非简单相加！)